

3.1. Limit and Colimit as a universal cone.

Recall: diagram (or shape) J in a category C

$\Rightarrow F: J \rightarrow C$ a functor.

Objective: Introduce $\lim$ and colim of diagram
as universal cones over and under the diagram

Def 3.1.1 [Cone]

$C \in C$, J : category,

$c: J \rightarrow C$ is called constant functor.

$$\begin{array}{ccc} x \mapsto c \\ f \downarrow \mapsto \downarrow 1_c \\ y \mapsto c \end{array}$$

$$\Rightarrow \begin{array}{ccc} \Delta: C & \longrightarrow & C^J \\ c \mapsto & c: J \rightarrow C & \\ f \downarrow & \downarrow \alpha_f & \\ d \mapsto & d: J \rightarrow C & \end{array}$$

is an embedding.

α_f : natural trans

$$(\alpha_f)_c = f$$

$$\begin{array}{ccc} & & 1_c \\ & & \downarrow \\ C & \xrightarrow{\quad} & C \\ f \downarrow & \alpha_f & \downarrow f \\ d \xrightarrow{\quad} & d & \\ & \downarrow & \\ & 1_d & \end{array}$$

Def 3.1.2. A cone over $F: J \rightarrow C$ with summit (apex) $C \in C$.

= Natural transformation $\lambda: C \Rightarrow F$.

• Data $\lambda = \{\lambda_j : j \in J\}$

• From natural transtf, for $f: j \rightarrow k \in J$

$$\begin{array}{ccc} C & \xrightarrow{1_C} & C \\ \lambda_j \downarrow & \circlearrowleft & \downarrow \lambda_k \\ F_j & \xrightarrow{Ff} & F_k \end{array} \quad (\Leftrightarrow) \quad \begin{array}{ccc} & C & \\ \lambda_j \swarrow & \circlearrowleft & \searrow \lambda_k \\ F_j & \xrightarrow{Ff} & F_k \end{array}$$

• Thus, if we have data satisfying the commuting diagram, then we can define cone.

A cone under F with nadir C

= Natural transtf $\lambda: F \Rightarrow C$ st

$$\forall f: j \rightarrow k, \quad \begin{array}{ccc} F_j & \xrightarrow{Ff} & F_k \\ \lambda_j \searrow & & \swarrow \lambda_k \\ & C & \end{array}$$

Cone under F is called Cocone

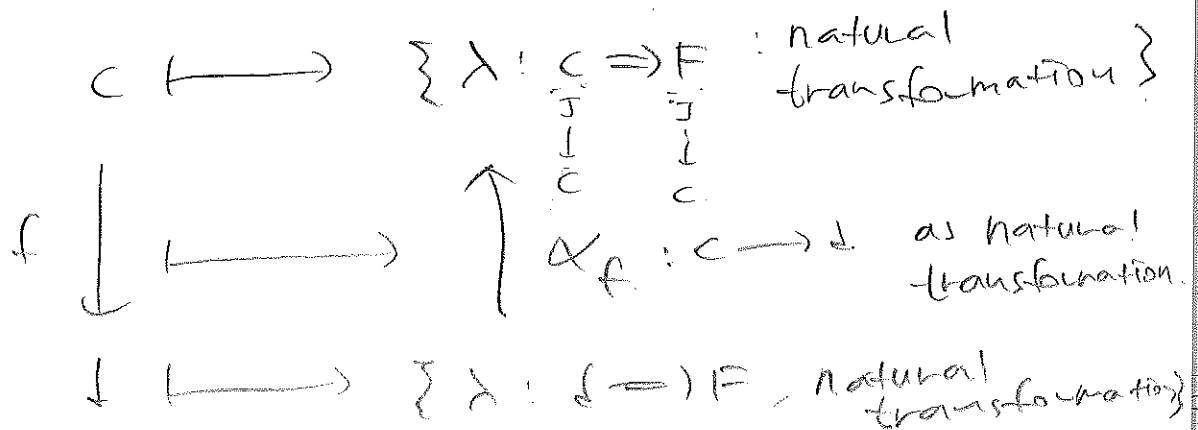
since it is dual of a cone.

(Cone over $F: J^{op} \rightarrow C^{op}$.)

$\lim F$: universal cone over F

Def) $F: J \rightarrow C$ diagram.

$\text{Cone}(-, F): C^{\text{op}} \rightarrow \text{Set}$

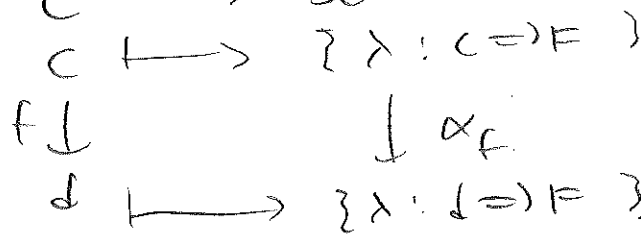


$\lim F$: representation of $\text{Cone}(-, F)$, i.e.

$\text{Cone}(-, F) \cong C(-, \lim F) \Leftrightarrow$ terminal object of $\int \text{Cone}(-, F)$

Similarly,

$\text{Cone}(F, -): C \rightarrow \text{Set}$

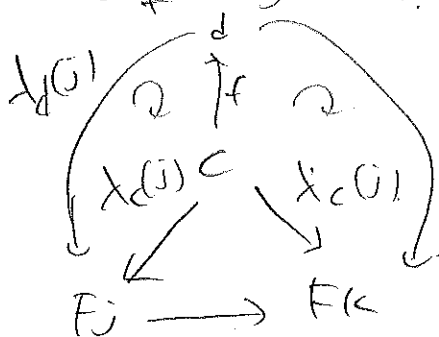


Limit cone:
universal element
in this.

Limit cone:
universal element

$\text{Colim } F$: rep of $\text{Cone}(F, -)$. $\Leftrightarrow$ initial object of $\int \text{Cone}(F, -)$

Notes that if $\lambda_c: c \Rightarrow F$, $\lambda_d: d \Rightarrow F$, $f: c \rightarrow d$.
then α_f sends λ_d to λ_c via comm diagram.



$\lambda_d(x)$

So: $\text{Cone}(-, F)$

is well-defined.

(similar to $\text{Cone}(F, -)$).

Prop 3.1.7. $\lambda: I \Rightarrow F$, $\lambda': I' \Rightarrow F$ are two
limit cones, $\Rightarrow I \cong^{=} I'$. Unique iso. commutes
with legs of the limit cones.

Pf) $(\lim F, \lambda)$, $(\lim F, \lambda')$ are terminal
 $\Rightarrow$ " $\cong$ " . by Corollary 2.3.2.

And isomorphism in $\int \text{Cone}(-, F)$ gives commuting
diagram:

Remark: There may be nontrivial automorphism
between I and I' . but by Corollary 2.3.2
if it commutes with cone, then it is
the iso in prop 3.1.7.

Def 3.1.9. [Product]

Product = limit of a diagram indexed by
 $J :=$ a discrete category with only
identity morphisms.

Thus, $F: J \rightarrow C$ is just collection
 $\{F_j\}_{j \in J}$. (Since J has no nontrivial morphisms)

Thus, cone over J is $\lambda: C \Rightarrow F$
i.e. collection of $\{\lambda_j: C \rightarrow F_j\}_{j \in J}$.

Therefore its limit is $\prod_{j \in J} F_j$

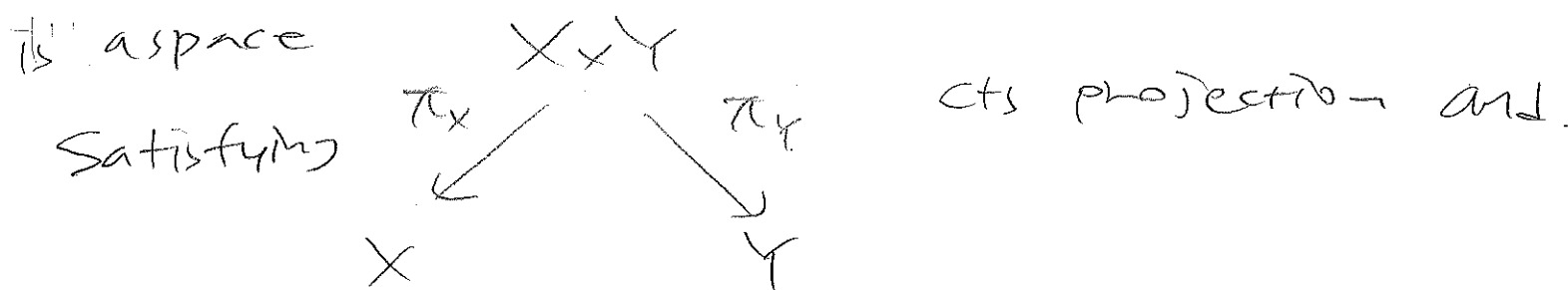
with legs $\pi_k: \prod_{j \in J} F_j \rightarrow F_k$

By the universal property of $\prod_{j \in J} F_j$ as a cone gives natural isomorphism.

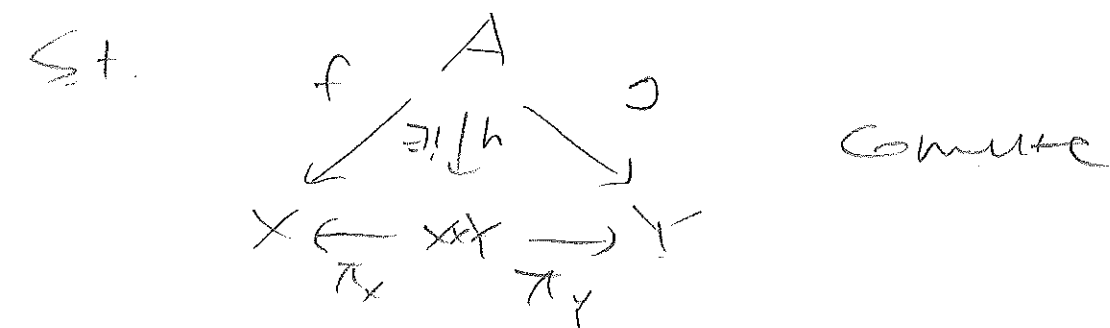
$$\left(C, \prod_{j \in J} F_j \right) \xrightarrow[\cong]{(\pi_k)_*} \text{Cone}(C, F) \cong \prod_{k \in J} (C, F_k)$$

(Yoneda lemma) (just...)

Ex 3.11.0. $X, Y \in \text{Top}$. Product of $X \times Y$



has universal property: For any $A \in \text{Top}$ with $f: A \rightarrow X$, $g: A \rightarrow Y$, $\exists! h: A \rightarrow X \times Y$



To construct $X \times Y$ as usual sense (Cartesian Product)

By taking $A = \{*\}$, A as a constant functor

says that $X \times Y \cong X \times_{\text{Cartesian}} Y$ as a set.

(By similar argument, $X \times Y = X \times_{\text{Cartesian}} Y$ as a set)

This is because $\text{Top}(A, X \times Y) \cong \text{Top}(A, X) \times \text{Top}(A, Y)$
as a set.

To show $X \times Y = X \times Y$ as a top space
Cartesian

take $A = X \times Y$ as a set with
Various topology.

By the universal property,

$X \times Y$ forced to be defined as the coarsest
topology on Cartesian product s.t.

π_X, π_Y are cts. (These are ^{case} when $J = \{\sigma, \tau\}$)

(Do the same thing for any index J .)

Def 3.1.11. Terminal object = product
when the indexing category is empty.

i.e., if $J = \{\}$, cone over J with summit C
is just C . $\Rightarrow \int \text{Cone}(-, J \rightarrow C)$

$$\cong C.$$

By def of limit, $\lim(J \rightarrow C)$ is terminal obj
of $\int \text{Cone}(-, J \rightarrow C) \cong C$.

Ex 3.1.12 $\mathbb{1} : \{ \cdot \} \rightarrow \{ \cdot \}$ called terminal category

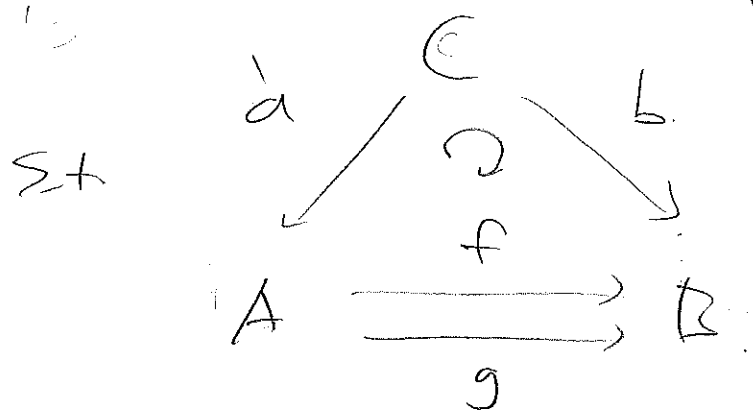
Since it is terminal obj of Cat or AT

Def 3.1.13 [Equalizer]

Equalizer = Limit of $F: J \rightarrow \mathcal{C}$

Where $J = \begin{matrix} \bullet & \xrightarrow{f} & \bullet \\ & \searrow g & \\ & \bullet & \end{matrix}$ parallel pair.

$\Rightarrow$ Let $F(J) = A \begin{matrix} \xrightarrow{f} \\ \xrightarrow{g} \end{matrix} B$. A cone over F with summit C is $(a: C \rightarrow A, b: C \rightarrow B)$

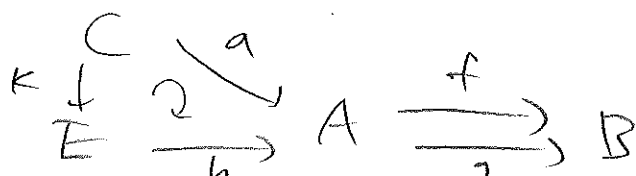


$$(\Leftrightarrow) fa = b = ga$$

Thus cone over parallel pair $A \begin{matrix} \xrightarrow{f} \\ \xrightarrow{g} \end{matrix} B$ (with summit C) is represented by a morphism $a: C \rightarrow A$ st $fa = ga$.

Hence Equalizer is the universal arrow with this property, $h: E \rightarrow A$. In particular

$\forall$ cone $a: C \rightarrow A$, $\exists ! k: C \rightarrow E$ st



Commutative.

Ex 3.1.14. $\phi, \psi: G \rightarrow H \in \text{Group}$

$$\Rightarrow \text{Eq}(\phi, \psi) = (\ker \phi \rightarrow G)$$

Since

$$\begin{array}{ccc} & C & \\ \exists ! \downarrow & \searrow a & \\ \ker \phi & \hookrightarrow G & \xrightarrow[\phi]{\psi} H \end{array}$$

$$\text{Eq}(\phi, \psi) = \left(\{ g \in G : \phi(g) = \psi(g) \} \hookrightarrow G \right)$$

If H is abelian, $(= \ker(\phi - \psi))$

Actually, $\text{Eq}(A \xrightarrow{f} B) \hookrightarrow C$ is monomorphic
 $h: E \rightarrow A \mapsto E$

Def 3.1.15 [Pullback]

Pullback = limit of $(F: \bullet \rightarrow \bullet \leftarrow \bullet \mapsto C)$

$$\text{Let } F(J) = B \xrightarrow{f} A \xleftarrow{g} C$$

Then cone over F with summit D is

$$\begin{array}{ccc} D & \xrightarrow{c} & C \\ b \downarrow & \searrow a & \downarrow g \\ B & \xrightarrow{f} & A \end{array}$$

triple morphism

Satisfies left comm. diagram

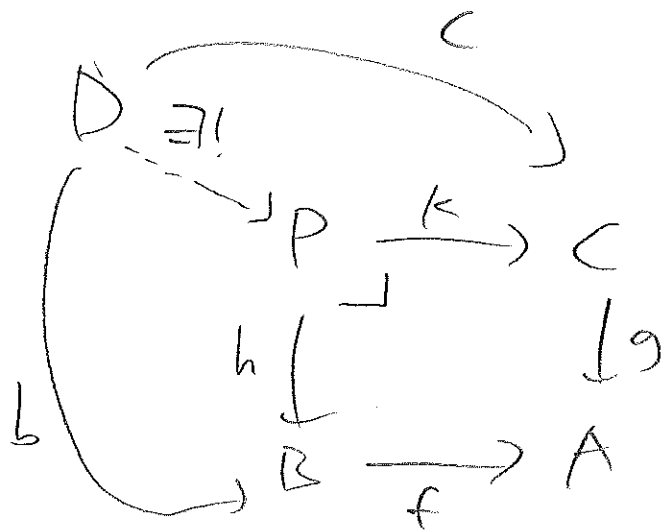
$$\Rightarrow g \circ a = f \circ b$$

Thus we can represent this core by
two morphisms $B \xleftarrow{b} D \xrightarrow{c} C$
satisfying rectangular.

Hence pull back: $B \xleftarrow{h} P \xrightarrow{k} C$

is universal core over $(B \xrightarrow{f} A \xleftarrow{g} C)$

i.e. for any core $B \xleftarrow{b} D \xrightarrow{c} C$,



" $\lrcorner$ " denote P is pullback, i.e. limit diagram
not just commutative square.

P is called "fiber product", denoted by $B \times_A C$.

If C is concrete category

thus $B = \{*\} \subseteq A$ as a set

$$\begin{array}{ccc} P & \longrightarrow & C \\ \downarrow & \lrcorner & \downarrow g \\ \{*\} & \hookrightarrow & A \end{array}$$

implies $P = \{c \in C : g(c) = *\}$
 $= g^{-1}(*)$

this fiber of map g
 over $*$ (or $f(A)$)

Ex 3.1.18. $\rho: \mathbb{R} \rightarrow S^1$ cts.
 $t \mapsto e^{2\pi i t}$

$$\begin{array}{ccc} P & \longrightarrow & \mathbb{R} \\ \vdots & & \downarrow \rho \\ \vdots & & \\ 1 & \hookrightarrow & S^1 \end{array}$$

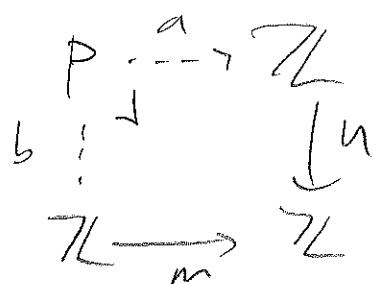
What is P ?

$$\begin{aligned} &= \rho^{-1}(1) \\ &= \{t \in \mathbb{R} : e^{2\pi i t} = 1\} \\ &= \mathbb{Z} \end{aligned}$$

Similar as equalizer

Pullback $(B \xrightarrow{\text{mono}} A \leftarrow C)$ defines $(B \times_A C \xrightarrow{\text{mono}} C)$

Ex 3.1.19.



What is P ? in Ab .

must $na = mb$.

$P : ab \text{ gp s.t. } \forall p \in P$

$$na(p) = mb(p)$$

More over, $\Rightarrow$ it commutes with any other cores, satisfy $na(p) = mb(p)$.

$\Rightarrow na(p) = mb(p)$ should be l.c.m of n, m ,

Thus, we may take $P = \mathbb{Z}$ with

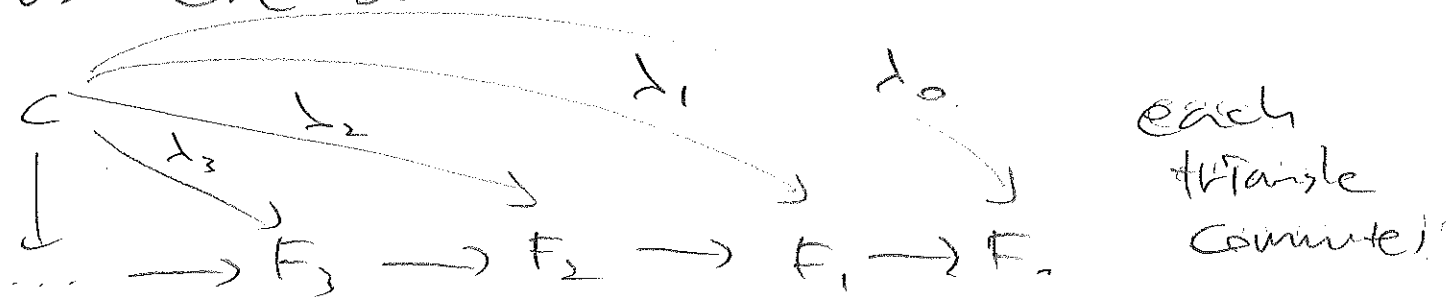
$$a(1) = \frac{\text{lcm}(n, m)}{m}, \quad b(1) = \frac{\text{lcm}(n, m)}{n}$$

Def 3.1.2.1 [Inverse Limit]

Limit of $F: J \rightarrow C$ when $J = \omega^{op}$

$\dots \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0 : \omega^{op}$

Then core over F with summit C is



$\varprojlim F_n$: Is the terminal core.

(Thus direct limit is when $J = W$.)
and Colimit of $F: J \rightarrow C$

Ex 3.1.22

$$\mathbb{Z}_p = \varprojlim \mathbb{Z}/p^n$$

Their dual notions are

Coproduct : Colimit of discrete category

Initial object : " empty "

Coequalizer : " $\cdot \rightrightarrows \cdot$ "

Pushout : " $\cdot \leftarrow \cdot \rightarrow \cdot$ "

Direct limit : " W "

(Sequential Colimit)

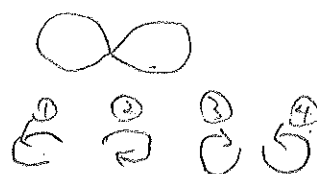
Ex 3.1.24 Pushout of $S' \leftarrow * \rightarrow S'$

$$\begin{array}{ccc} S' \cup S' & & * \xrightarrow{+} S' \\ & \circ \downarrow \quad \downarrow r & \\ & S' \rightarrow S' \cup S' & \end{array}$$

Moreover,

$$\begin{array}{ccc} S' \xrightarrow{aba^{-1}b^{-1}} S' \cup S' \\ i \downarrow \quad \quad \downarrow r \\ D^2 \dashrightarrow T \end{array}$$

T : torus.
 $a, ba^{-1}b^{-1}$ loop



Ex). 1.25 Cokernel in Group

$$= \text{Coequalizer} (\phi: G \rightarrow H, \psi: G \rightarrow H)$$

Ex). 1.26. (n-skeleta)

$$X_0 \hookrightarrow X_1 \hookrightarrow X_2 \hookrightarrow \dots$$

$$\text{Colimit of sequence} = \bigcup_{n \geq 0} X_n$$

ex) CW complex = colimit of its n-skeleta.

$$\text{Ex). 1.27) } f: \coprod_{\bar{\alpha} \in I} A_{\bar{\alpha}} \rightarrow X \iff \{f_{\bar{\alpha}}: A_{\bar{\alpha}} \rightarrow X \mid \bar{\alpha} \in I\}$$

$$\text{Thus, } C(\coprod_{\bar{\alpha} \in I} A_{\bar{\alpha}}, X) \xrightarrow{\cong} \prod_{\bar{\alpha} \in I} C(A_{\bar{\alpha}}, X)$$

$$f \longmapsto (f_{\bar{\alpha}})_{\bar{\alpha} \in I}$$

$$\text{by letting } \bar{c}_{\bar{\alpha}}: A_{\bar{\alpha}} \rightarrow \coprod A_{\bar{\alpha}} \text{ and } f_{\bar{\alpha}} = f \circ \bar{c}_{\bar{\alpha}}.$$

$$\text{Dually, } g: X \rightarrow \prod_{j \in J} B_j \iff \{g_j: X \rightarrow B_j \mid j \in J\}$$

$$\Rightarrow C(X, \prod_{j \in J} B_j) \xrightarrow{\cong} \prod_{j \in J} C(X, B_j)$$

$$g \longmapsto g_{\bar{j}}$$

$$\text{by letting } \pi_j: \prod_{j \in J} B_j \rightarrow B_j \text{ with } g_{\bar{j}} = \pi_j \circ g.$$

$$\text{Then, if } \coprod_{\bar{\alpha} \in I} A_{\bar{\alpha}} \xrightarrow{f} \prod_{j \in J} B_j, \text{ it gives}$$

$$\begin{array}{ccc} \bar{c}_{\bar{\alpha}'} \uparrow & \circlearrowright & \downarrow \pi_{j'} \\ A_{\bar{\alpha}'} & \xrightarrow{f(\bar{\alpha}', j')} & B_{j'} \end{array} \quad \text{left diagram.}$$

i.e., a map from coproduct to product is determined by a matrix of component maps.

In certain categories, Ab , Mod_R , Ch_R , have a notion of zero homomorphism between any pair of objects.

In any finite collection I ,

$$\coprod_{i \in I} A_i \xrightarrow{\begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix}} \prod_{i \in I} A_i \quad \text{do make sense;}$$

This induces isomorphism in "Abelian Category".

$\Rightarrow \coprod_{i \in I} A_i \cong \prod_{i \in I} A_i$. So use notation $\bigoplus_{i \in I} A_i$ a direct sum.

Then any ~~com~~ composition of maps between direct sums is matrix multiplication.

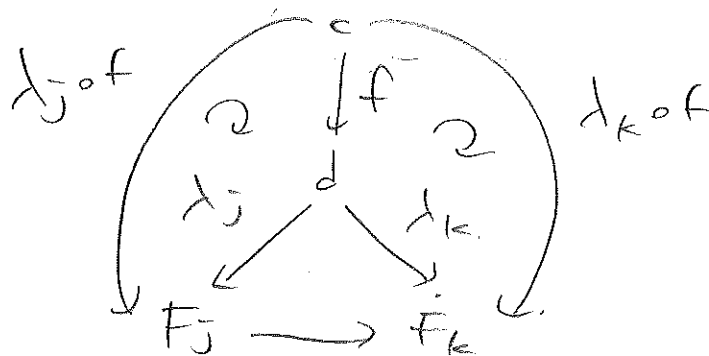
Ex 3.1. i) Let $f: c \rightarrow d \in \text{Mor } C$. Then

$$\text{Cone}(c, F) \xrightarrow{\alpha_f} \text{Cone}(d, F) \quad \text{st. } \alpha_f = \text{Cone}(f, F)$$

$$\alpha_f \uparrow$$

$\text{Cone}(d, F)$

If $\lambda \in \text{Cone}(d, F)$ then



implies that

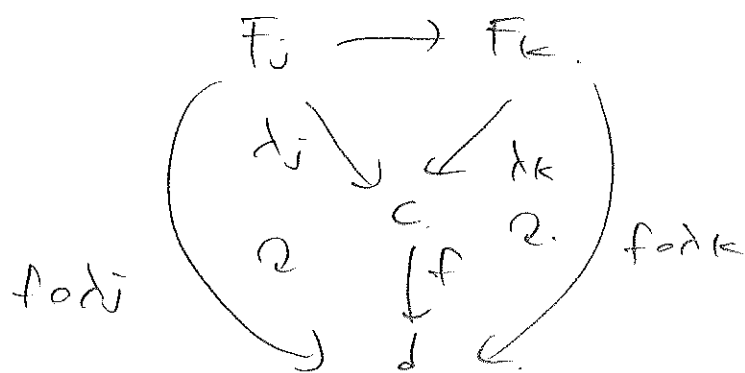
then,

$$\alpha_f(\lambda) = \lambda \circ f \Rightarrow \lambda \in \text{Cone}(c, F)$$

Dually,

$$\text{Cone}(F, f): \text{Cone}(F, c) \xrightarrow{\alpha_f} \text{Cone}(F, d)$$

st. if $\lambda \in \text{Cone}(F, c)$ for λ is a cone in $\text{Cone}(F, d)$



$$\lambda \circ f \Rightarrow c \Rightarrow d \in \text{Cone}(d, F)$$

Ex 3.1. ii) Notes that for any $\lambda \in \text{Cone}(c, F)$
 $\lambda: c \Rightarrow F$ natural transformation, thus $\lambda \in \text{Hom}(\Delta(c), F)$
 $\Rightarrow \text{Cone}(c, F) \subseteq \text{Hom}(\Delta(c), F)$

Conversely, let $\lambda \in \text{Hom}(\Delta(c), F)$. By definition, λ is a cone over F with summit $c \Rightarrow \text{Cone}(c, F) = \text{Hom}(\Delta(c), F)$

Also, $\text{Hom}(\Delta(f), F) \cong \alpha_f = \text{Cone}(f, F)$

Thus, $\text{Cone}(-, F) \cong \text{Hom}(\Delta(-), F)$

Ex. 3.1. iii) WTS, $\int \text{Cone}(-, F) \cong \Delta \downarrow F$

	$\int \text{Cone}(-, F)$	$\Delta \downarrow F$
Obj	$(C, \lambda), C \in \mathcal{C}$ $\lambda \in \text{Cone}(C, F)$	$(\Delta(C), F, \lambda: \Delta(C) \Rightarrow F)$ So $\lambda \in \text{Cone}(C, F)$
Mor.	$(C, \lambda) \xrightarrow{f} (C', \lambda')$ if $f: C \rightarrow C'$ and $\alpha_f \circ \lambda' = \lambda$	$(\Delta(C), F, \lambda) \xrightarrow{(f, 1_F)} (\Delta(C'), F, \lambda')$ So $\alpha_f \circ \lambda' = \lambda$
		$\Rightarrow \alpha_f \circ \lambda' = \lambda$

Hence, there is bijection between them.

Now, for any $F \in (\mathcal{C}^{\text{op}})^{\mathcal{J}}$,

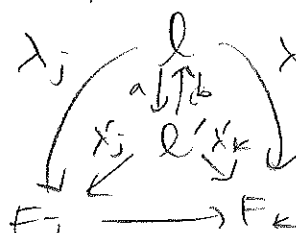
$$\int \text{Cone}^{\text{op}}(-, F) \cong \Delta^{\text{op}} \downarrow F^{\text{op}}$$

Where $\Delta^{\text{op}}: \mathcal{C}^{\text{op}} \rightarrow (\mathcal{C}^{\text{op}})^{\mathcal{J}}$, $F^{\text{op}}: 1 \rightarrow (\mathcal{C}^{\text{op}})^{\mathcal{J}}$

And observe $\int \text{Cone}^{\text{op}}(-, F) = \int \text{Cone}(F, -)$

$$\text{and } \Delta^{\text{op}} \downarrow F^{\text{op}} = F \downarrow \Delta$$

Ex 3.1. iv) From universal property,

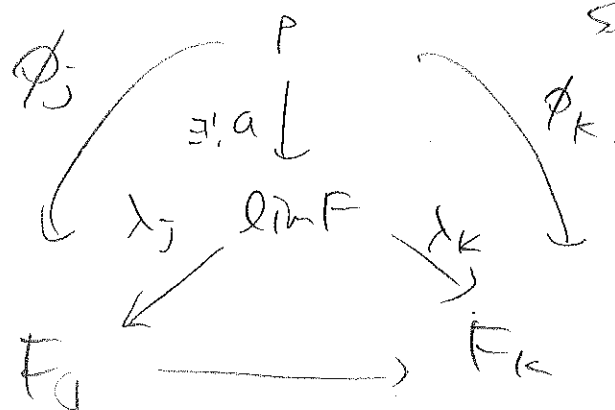
λ_j  λ_k below diagram commutes for any of triangle. Then, $\lambda_j \circ a = \lambda_j$
 $\lambda_j \circ b = \lambda'_j \Rightarrow \lambda_j \circ b \circ a = \lambda_j \Rightarrow b \circ a = 1_Q$
Since $\text{Hom}(Q, L) = \{1_{(Q, L)}\}$.

and I_{ℓ} induces $I_{(\ell, N)}$.

By the similar argument, $a \circ b = I_{\ell'} \Rightarrow a, b$ are isomorphisms $\Rightarrow \ell \cong \ell'$.

Ex 3 (1) If $\lim F$ exists with limit cone λ ;

then for any cone $\phi: P \Rightarrow F$, $\exists! a: P \rightarrow \lim F$ s.t. left diagram commutes



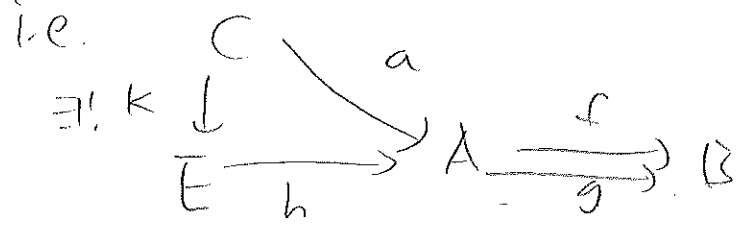
$\Rightarrow$ Let $\text{Summit}(P, \leq)$ be a subcategory of $(P, \leq)$ spanned by elements m in $(P, \leq)$ which is a summit of a cone.

Then, $\lim F$ is "a" maximum of $\text{Summit}(P, \leq)$

(Not "the" maximum since pre-order has two distinct element on which both $\geq$ and $\leq$ hold.)

Conversely, colim is a minimum of subcategory $\text{radix}(P, \leq)$, spanned by elements which is radix of some cocone over F .

Ex 3.1, vi) Let $f', g': C \rightarrow E$ s.t. $hf' = hg'$.
 Let $a = hf' = hg'$. Then, $fa = fhf' = (gh)f' = g(a)$
 Thus universal property



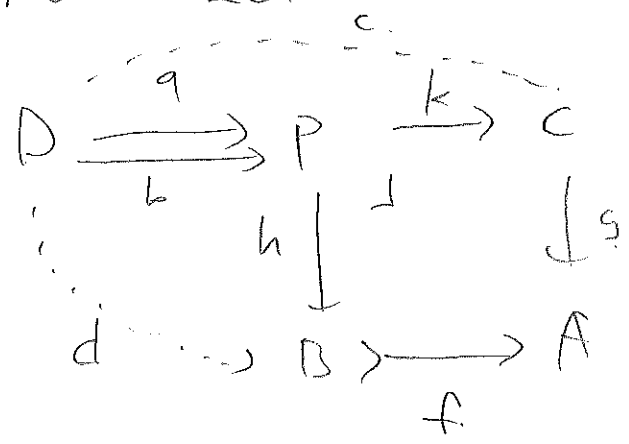
gives Unique $k: C \rightarrow E$
 s.t. $a = hk$

This implies $f' = g'$.

Since uniqueness of k s.t. $a = hk$ implies $f' = k = g'$.

$\Rightarrow h$ is mono.

3.1, vii) Let $a, b: D \rightarrow P$ s.t. $ka = kb$.

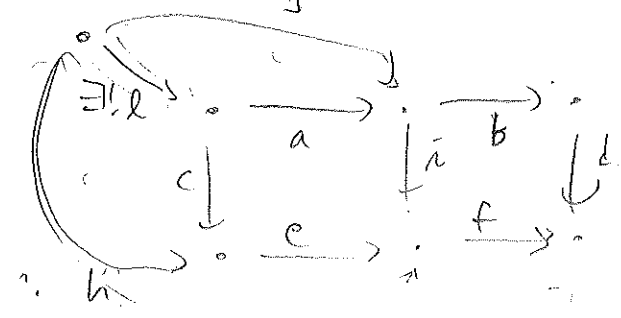


Then, $gka = gkb$
 $\Rightarrow fha = fhb$
 $\Rightarrow ha = hb$

Now let $c = ka = kb$
 $d = ha = hb$

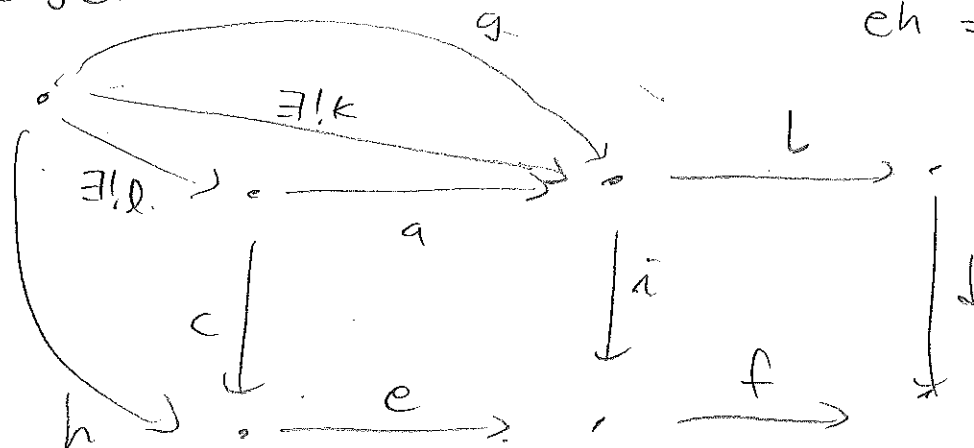
Then, $\exists! l: D \rightarrow P$ s.t. $c = fl$ $d = hl$.
 This implies $a = l = b$ by uniqueness, $\Rightarrow k$ is mono.

3.1 viii) Suppose composite ^(big) rectangle is pull back
 for bg and h , then "big" pull back gives unique l



s.t. $h' = cl$, $bg' = bal$

Apply "small" (right) pull back on bg and eh
 We get unique k s.t. $bk = bg$ and $eh = ik$.



By uniqueness of k ,
 $al = k = g$.

Thus, $h = cl$, $g = al$. This shows the existence of a map l to make $\begin{array}{ccc} & a & \\ c \downarrow & & \downarrow i \\ & e & \end{array}$ (left)

pull back. To see uniqueness, suppose $\exists l'$ s.t.

$g = al'$, $h = cl'$. Then, applying l 's pull back on bg and h gives $h = cl = cl'$.

Since l is unique, $l' = l$. $\square$

Conversely, both left and right one are pullbacks

Then right pull back induce k ,
 for g and eh s.t.

$$g = bk, \quad eh = ik$$

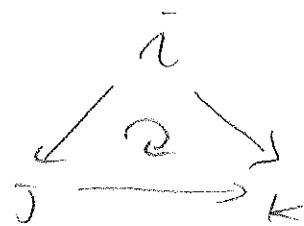
And left pull back for k and h induces l s.t.

$$a k = al \quad h = cl$$

Thus, $g = bal$, $h = cl$. Also if $\exists l'$ s.t. $g = bal'$, $h = cl'$ then, by uniqueness of k and l , $al' = k = al \Rightarrow l = l'$.

$\therefore$ Composite rectangular is pull back.

EX 3.1-IX) Let $\bar{i}$ be initial in J . Then,
for any $j \rightarrow k \in J$.



by uniqueness of $\text{Hom}(\bar{i}, k)$,

Thus, let $\lambda_j = F(\bar{i} \rightarrow j) \quad \forall j \in J$.

Then, $(\lambda, \bar{i})$ is a cone over F with summit $F\bar{i}$.

Let $\phi: C \Rightarrow F$ a cone with summit C .

Then, ϕ_j Commutes since property of cone ϕ make the triangles in each side commute.

Now suppose $f: C \rightarrow F\bar{i}$ st. $\alpha_f(\lambda) = \phi$.

Then, $\Rightarrow f = \phi_{\bar{i}}$. Hence, in $\text{Cone}(C, F)$ $(F\bar{i}, \lambda)$ is terminal element.

$\therefore F\bar{i} = \lim F$.

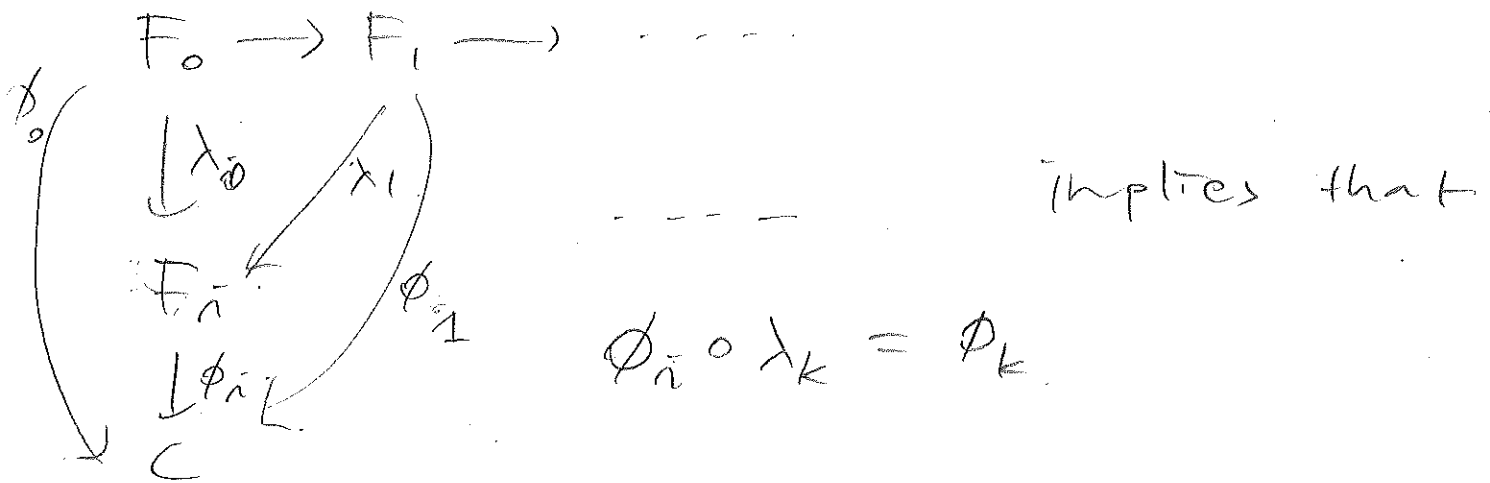
Now suppose J is indexed by w , i.e.

$\exists G: w \rightarrow J$ exists.

If J has terminal $\bar{i}$, then $F: J \rightarrow C$ has colimit

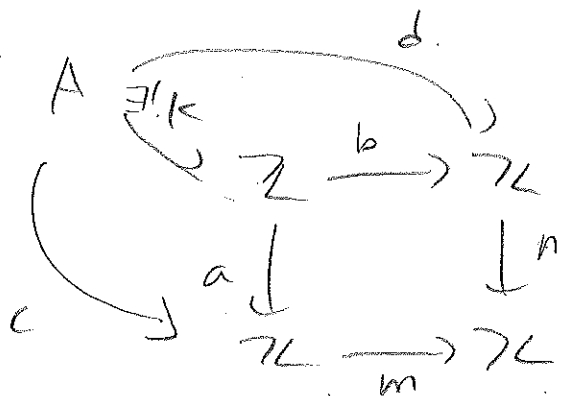
$F\bar{i}$.

Then, if $\mathcal{O}$ is a cocone with radii c and λ is colimit cone,



Since J is indexed by W , this implies that for $k > \bar{n}$, $\lambda_k \circ (F_{\bar{n}} \rightarrow \dots \rightarrow F_k) = 1_{F_{\bar{n}}}$.
 $\Rightarrow F_k = F_{\bar{n}}$. Hence $J = \bar{n}$, a subcategory of W spanned by $G(0), G(1), \dots, G(\bar{n})$.
 And colimit of F is $F_{\bar{n}}$.

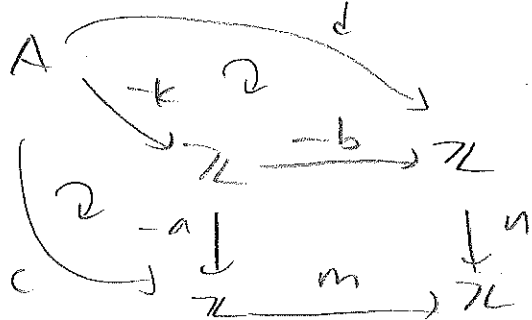
Ex 3.1.x) Let $A \xrightarrow[\lambda]{c} \mathbb{Z}$ be a map s.t.



Then pull back induces

$$k \text{ s.t. } d = bk$$

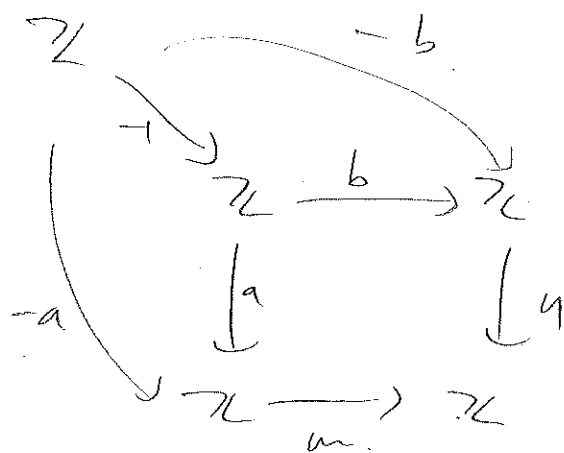
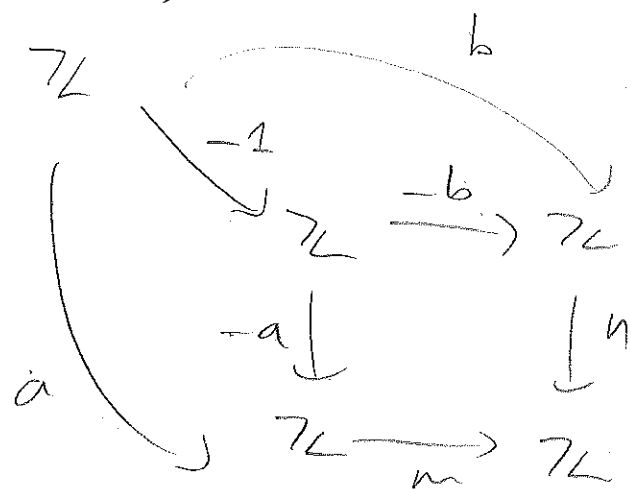
$$c = ak$$



Now, we define $-k$, a composition of k with inversion. Then $(-k) \circ b = k \circ b$ and $-a \circ k = ak$.

Thus, $(-a, -b)$ also has a universal property.

However,



Shows that $(\mathbb{Z}, (a, b)) \cong (\mathbb{Z}, (-a, -b))$

In the category of cores. So, this $(-a, -b)$ is just another limit one naturally isomorphic to (a, b)

Ex 3.1. xi) Note, that each leg is denoted by

$$\bar{c}_j: A_j \longrightarrow \coprod_{j \in I} A_j$$

Since this category has a morphism for each pair of objects, let $f_{\bar{n}}: A_{\bar{n}} \longrightarrow A_j \quad \forall \bar{n} \in I, f_{j\bar{j}} = 1_{A_j}$ we can pick such $f_{j\bar{j}}$ by given assumption.

Then, A_j is coproduct over I , so universal property gives that

Now let $\bar{n} = j$ Then

$$\begin{array}{ccc} A_{\bar{n}} & \xrightarrow{\bar{c}_j} & \coprod_{j \in I} A_j \\ \downarrow f_{j\bar{j}} & \searrow & \downarrow \exists! k_{j\bar{j}} \\ A_j & & A_j \end{array}$$

$$\begin{array}{ccc} A_j & \xrightarrow{\bar{c}_j} & \coprod_{j \in I} A_j \\ \downarrow 1_{A_j} & \searrow & \downarrow k_{j\bar{j}} \\ A_j & & A_j \end{array} \Rightarrow \bar{c}_j \text{ is split mono.}$$

3.1.xii) Let $\lambda: C \Rightarrow F$ is J-shaped cone
 $\eta: C \Rightarrow FE$ is I-shaped cone.

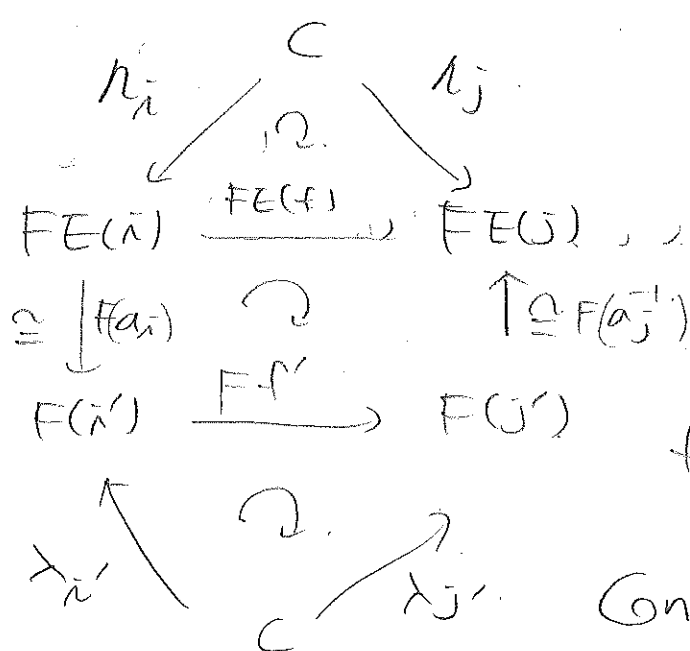
Then, for $f: \bar{\alpha} \rightarrow \bar{\beta} \in J$ essential surjectivity gives $E(\bar{\alpha}) \xrightarrow{a_i} \bar{\alpha}', E(\bar{\beta}) \xrightarrow{a_j} \bar{\beta}'$ for some $\bar{\alpha}, \bar{\beta} \in I$.

Hence, $J(E(\bar{\alpha}), E(\bar{\beta})) \cong J(\bar{\alpha}', \bar{\beta}')$ since composition of isomorphisms induce bijection.

on $J(-, -)$. Thus, $\exists f: \bar{\alpha} \rightarrow \bar{\beta}$ s.t.

$$E(f) = E(\bar{\alpha}) \xrightarrow{\cong} \bar{\alpha}' \xrightarrow{f'} \bar{\beta}' \xrightarrow{\cong} E(\bar{\beta}).$$

Hence



Thus, sending

$$\eta \mapsto \eta a := \{F a_i \circ \eta_i\}$$

sends

$$(C, \eta) \in \text{Cone}(-, FE)$$

$$\text{to } (C, \eta a) \in \text{Cone}(-, F)$$

Conversely, sending

$$\lambda \mapsto \lambda a^{-1} := \{F(a_i^{-1}) \circ \lambda_i\}$$

$$\text{sends } (C, \lambda) \in \text{Cone}(-, F)$$

$$\text{to } (C, \lambda a^{-1}) \in \text{Cone}(-, FE)$$

And this two map is inverse to each other.

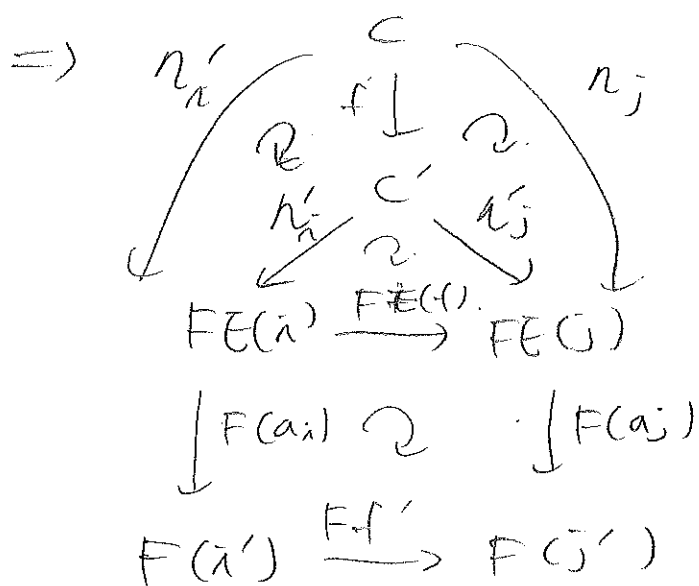
So if we show these are functor, then $\text{Cone}(-, F)$ and $\text{Cone}(-, FE)$ are isomorphism of category.

If $(C, \eta) \xrightarrow{g} (C', \eta') \in \text{Cone}(-, FE)$

then, $g: C \rightarrow C'$ and $\text{Cone}(g, FE)(\eta') = \eta$

And $\int \text{Cone}(g, FE) = \alpha_g: C \Rightarrow C'$.

So, $\eta' \circ \alpha_g = \eta$.



Now sending η to ηa and η' to ηa .

α has

$$\eta a \circ \alpha_g = \eta a$$

So $\eta \mapsto \eta a$.

$$\alpha_g \downarrow \mapsto \alpha_g$$

$$\eta' \mapsto \eta' a$$

is functor

In the similar way, we can conclude that $\text{Cone}(-, FE) \cong \text{Cone}(-, F)$.

Hence if $\lim FE$ exists, $(\lim FE, \lambda)$ is initial in $\text{Cone}(-, FE)$. Thus, $(\lim FE, \lambda a)$ is initial

since they are isomorphic category.

$$\Rightarrow \lim F = \lim FE.$$

Ex 3, 1. xiii). Coproduct of CRings.

Let $\coprod_{j \in J} R_j$ be a coproduct of CRings

Then, $\bar{f}_j: R_j \rightarrow \coprod_{j \in J} R_j$ exists and

for any $f_j: R_j \rightarrow S$ $\forall j$, $\exists! k: \coprod_{j \in J} R_j \rightarrow S$

$$\begin{array}{ccc} R_j & \xrightarrow{\bar{f}_j} & \coprod_{j \in J} R_j \\ & \searrow f_j & \downarrow k \\ & & S \end{array}$$

When $J=2$, then
 Right diagram holds.

$$\begin{array}{ccccc} R_1 & \xrightarrow{\bar{f}_1} & R_1 \coprod R_2 & \xleftarrow{\bar{f}_2} & R_2 \\ & \searrow f_1 & \downarrow k & \swarrow f_2 & \\ & & S & & \end{array}$$

Notes that every comm. ring is $\mathbb{Z}$ -module

$$\begin{array}{ccc} R_1 \times R_2 & \xrightarrow{\otimes} & R_1 \otimes_{\mathbb{Z}} R_2 \\ & \searrow f_1(r_1) \cdot f_2(r_2) & \downarrow \exists! f \\ & & S \end{array}$$

property of tensor product

We claim that $R_1 \otimes_{\mathbb{Z}} R_2 \cong R_1 \coprod R_2$

$\bar{f}_1: R_1 \rightarrow R_1 \coprod R_2$ induces $\text{induces } R_1 \times R_2 \rightarrow R_1 \coprod R_2$

So universal property of tensor product gives

unique $\bar{f}: R_1 \otimes_{\mathbb{Z}} R_2 \rightarrow R_1 \coprod R_2$. Also, universal property

of coproduct on $R_1 \xrightarrow{-\otimes 1} R_1 \otimes R_2, R_2 \xrightarrow{1 \otimes -} R_1 \otimes R_2$

Induces unique $k: R_1 \amalg R_2 \rightarrow R_1 \otimes_{\mathbb{Z}} R_2$

$$\Rightarrow$$

$$\begin{array}{ccccc}
 & \bar{\iota}_1 & \nearrow & R_1 \amalg R_2 & \nwarrow & \bar{\iota}_2 \\
 R_1 & & \searrow & \downarrow \uparrow \bar{\iota} & & \nearrow & R_2 \\
 & \circlearrowleft & k & & \circlearrowright & g & \\
 & - \otimes 1 & \searrow & R_1 \otimes_{\mathbb{Z}} R_2 & \swarrow & 1 \otimes - &
 \end{array}$$

Now $\bar{\iota} \circ k \circ \bar{\iota}_1 = \bar{\iota}(- \otimes 1) = \bar{\iota}_1$

$\Rightarrow$ Thus $R_1 \xrightarrow{\bar{\iota} \circ k \circ \bar{\iota}_1} R_1 \amalg R_2$ with universal property
 $R_2 \xrightarrow{\bar{\iota} \circ k \circ \bar{\iota}_2} R_1 \amalg R_2$

of coproduct induces $\exists! R_1 \amalg R_2 \rightarrow R_1 \amalg R_2$

which is identity. $\Rightarrow \bar{\iota} \circ k = 1_{R_1 \amalg R_2}$

Conversely, $k \circ \bar{\iota} \circ (1 \otimes -) = k \circ \bar{\iota}_2 = 1 \otimes -$

Hence, $R_1 \times R_2 \xrightarrow{\otimes} R_1 \otimes R_2$
 $\circlearrowleft \downarrow \exists! \overline{k \circ \bar{\iota} \circ (1 \otimes -)} \Rightarrow k \circ \bar{\iota} \circ (1 \otimes -) = 1_{R \otimes R}$
 ~~$k \circ \bar{\iota} \circ (- \otimes 1) = 1_{R \otimes R}$~~

Hence, k and $\bar{\iota}$ are isomorphism.

3.2 Limits in the Category of sets.

Def 3.2.1. $F: J \rightarrow C$ diagram is small if J is small category.

C is complete $\Leftrightarrow \forall F: J \rightarrow C$ small diagram $\exists \lim F$.

" Co complete $\Leftrightarrow \forall F: J \rightarrow C$ " $\exists \operatorname{colim} F$.

Claim: $F: J \rightarrow \operatorname{Set}$ small diagram. If $\lim F$ exists, then $\lim F \cong \operatorname{Cone}(1, F)$.

pf) $\operatorname{Set}(X, \lim F) \cong \operatorname{Cone}(X, F) \quad \forall X \in \operatorname{Set}$.

Take $X = 1$, singleton. Then.

$$\lim F \cong \operatorname{Set}(1, \lim F) \cong \operatorname{Cone}(1, F)$$

Def 3.2.3. $\forall F: J \rightarrow \operatorname{Set}$ small diagram, $\lim F := \operatorname{Cone}(1, F)$ and define $\lambda: \lim F \Rightarrow F$ by $(\lambda)_j: \lim F \rightarrow F_j$.

$$\mu: 1 \Rightarrow F \mapsto \mu_j: 1 \rightarrow F_j.$$

(Notes that μ_j can be identified as element of F_j)

Thm 3.2.6. 3.2.3 is well-defined i.e, set is complete.

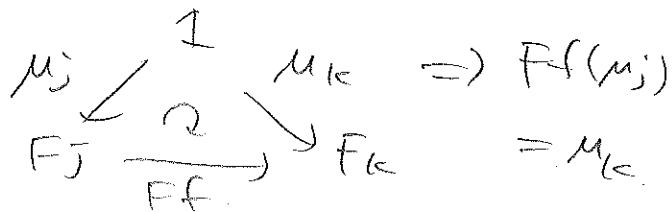
pf) We need to show that λ is a cone and limit cone.

Let $f: J \rightarrow K \in J. \Rightarrow$



induces $Ff(\lambda_j(\mu)) = Ff(\mu_j)$

Now, $\mu: I \Rightarrow F$. Hence



$\Rightarrow Ff(\lambda_j(\mu)) = Ff(\mu_j) = \mu_k = \lambda_k(\mu)$.

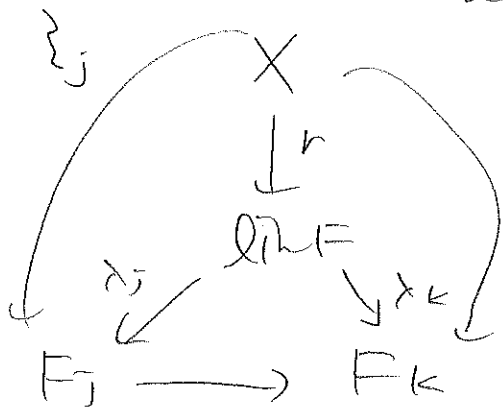
So the triangle commutes. $\Rightarrow \lambda$ is a cone.

To see λ is universal cone, let $\xi: X \Rightarrow F$

be a cone. For each $x \in X$, $\xi x: I \Rightarrow F$

in the sense that $(\xi x)_j = \xi_j|_{\{x\}}$.

Hence let $r: X \rightarrow \lim F$. This induces $x \mapsto \xi x$.



$$\begin{aligned} \text{s.t. } \lambda_j(r(x)) &= \lambda_j(\xi x) \\ &= (\xi x)_j = \xi_j|_{\{x\}} = \xi_j(x) \end{aligned}$$

Moreover, any map $X \rightarrow \lim F$ should send x to ξx , otherwise the above diagram doesn't commute.

$\Rightarrow r$ is unique. $\Rightarrow \lim F$ is well-defined.

Ex 3.2.7) Let $F: J \rightarrow C$ with J is discrete.

Then $\lim F = \prod_{j \in J} A_j$. By thm 3.2.6, $\lim F \cong \text{Cone}(1, F)$

Notes that $\lambda: 1 \Rightarrow F$ consists of $\lambda_j: 1 \rightarrow A_j$, i.e., elements of A_j for each $j \in J$.

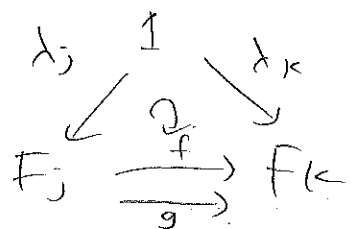
$\Rightarrow \prod_{j \in J} A_j := \{ (a_j \in A_j)_{j \in J} \}$ a Cartesian product.

Ex 3.2.8) Terminal object of $\text{Set} = \text{Cone}$ with $\text{Sum} 1$ over empty diagram.

$= 1$.

Ex 3.2.9.) $F: (\cdot \rightrightarrows \cdot) \rightarrow C$: equalizer.

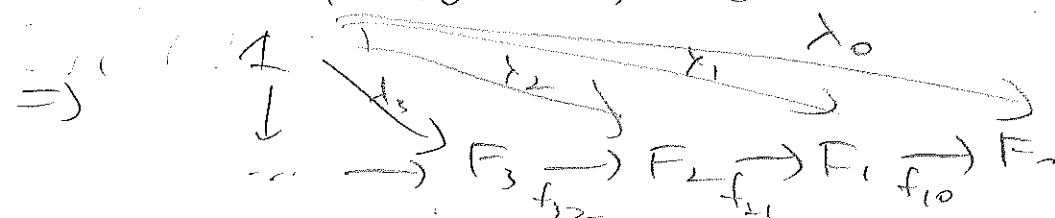
$\text{Cone}(1, F) := \{ \lambda: 1 \Rightarrow F \}$



$\Rightarrow$ So identify λ_j as an element of F_j ,

$\text{Cone}(1, F) = \{ x \in F_j : fx = gx \}$

Ex 3.2.10. $F: W^{op} \rightarrow \text{Set}$. If $\lambda \in \text{Cone}(1, F)$,



So identify λ_i as element of F_i ,

$\text{Cone}(1, F) = \{ (x_n)_{n \in \mathbb{N}} \in \prod_n F_n \mid f_{n,n+1}(x_n) = x_{n+1} \}$

Ex. 3.2.11. For pull back, $\lambda \in \text{Core}(1, F)$

implies

$$\begin{array}{ccc} 1 & \xrightarrow{\lambda_1} & C \\ \lambda_2 \downarrow & \circlearrowleft & \downarrow g \\ B & \xrightarrow[f]{} & A \end{array}$$

So by identifying λ_1 and λ_2 as element of C and B resp.,

$$\text{Core}(1, F) =: B \times_A C := \{(b, c) \in B \times C : f(b) = g(c)\}$$

Ex 3.2.12 $X: BG \rightarrow \text{Set}$. Then $\lambda \in \text{Core}(1, X)$

satisfy

$$\begin{array}{ccc} 1 & & \\ \lambda \swarrow & \circlearrowleft & \searrow \lambda \\ X & \xrightarrow{g_*} & X \end{array}$$

for all $g \in G$.

By identifying $\lambda \in X$, $g\lambda = \lambda$. $\therefore \lambda$ is fixed by g

$$\Rightarrow \lim F = \text{Core}(1, X) = X^G$$

Set of G -fixed pt of X .

Rmk) Let

$$E \longrightarrow B \times C \begin{array}{c} \xrightarrow{(b, c) \mapsto f(b)} \\ \xrightarrow{(b, c) \mapsto g(c)} \end{array} A$$

Then, its equalizer is

$$\begin{aligned} E &= \{ (x, y) \in B \times C \\ &\quad \text{s.t. } f(x) = g(y) \} \\ &= B \times_A C. \end{aligned}$$

This is not a coincidence.

Thm. 3.2.13. $\forall F: J \rightarrow \text{Set}$ small diagram,

$$\lim_J F \longrightarrow \prod_{j \in \text{Obj } J} F_j \begin{array}{c} \xrightarrow{c} \\ \xrightarrow{d} \end{array} \prod_{f \in \text{Mor } J} F(\text{cod } f)$$

is an equalizer diagram, i.e., $\lim_J F$ is an equalizer of c and d .

2f) $\lim_J F = \text{Cone}(I, F)$. Pick $\lambda : I \Rightarrow F$.

Then for any $f \in \text{Mor } J$,

$$\begin{array}{ccc} & \lambda_{\text{dom } f} & \\ & \swarrow & \searrow \\ & 2 & \\ & \swarrow & \searrow \\ F(\text{dom } f) & \xrightarrow{ff} & F(\text{cod } f) \end{array}$$

Hence we may identify λ_j be element of F_j .

$\Rightarrow \lambda$ is identified by $(\lambda_j)_{j \in \text{obj } J} \in \prod_{j \in \text{obj } J} F_j$

Now, define

$$c : (\lambda_j)_{j \in \text{obj } J} \longmapsto (\lambda_{\text{cod } f})_f \in \prod_{f \in \text{Mor } J} F(\text{cod } f)$$

by copying existing components of $(\lambda_j)_{j \in \text{obj } J}$.

$$d : (\lambda_j)_{j \in \text{obj } J} \longmapsto (Ff(\lambda_{\text{dom } f}))_f \in \prod_{f \in \text{Mor } J} F(\text{cod } f)$$

Then, If $c(\lambda) = d(\lambda)$, then for each $f \in \text{Mor } J$,

$$Ff(\lambda_{\text{dom } f}) = \lambda_{\text{cod } f} \Rightarrow \lambda \in \text{Cone}(I, F)$$

$\Rightarrow \text{Equalizer} \subseteq \lim_J F$.

And every element in $\lim_J F$ satisfy $c(\lambda) = d(\lambda)$

$\Rightarrow \lim_J F$ is Equalizer of c and d . 

(mk) c, d can be obtained by universal property of

$$\begin{array}{ccc} \lim_J F & \xrightarrow{\quad} & \prod_{j \in \text{obj } J} F_j \\ & & \uparrow \pi_f \\ & & F(\text{cod } f) \\ & \nearrow \pi_{\text{cod } f} & \\ & \xrightarrow{\exists! c} & \prod_{f \in \text{Mor } J} F(\text{cod } f) \end{array} \quad \left. \begin{array}{l} \prod_{j \in \text{obj } J} F_j \\ \prod_{f \in \text{Mor } J} F(\text{cod } f) \end{array} \right\} \prod_{j \in \text{obj } J} F_j$$

$$\begin{array}{ccc} \lim_{\leftarrow} F \longrightarrow \prod_{j \in \mathcal{O}(\mathcal{I})} F_j & \xrightarrow{\exists! \downarrow} & F(\text{cod } f) \\ \pi_{\text{out}} \downarrow & & \downarrow \pi_f \\ F(\text{dom } f) & \xrightarrow{ff} & F(\text{cod } f) \end{array}$$

Then, we can use the same argument showing that $\lim_{\leftarrow} F$ is equalizer.

Ex. 3.2.16 (Idempotent) $e: A \rightarrow A$ is idempotent if $e \cdot e = e$. So $F = (\mathcal{C} \cdot \mathcal{C}e) \rightarrow \text{Set}$ s.t. $e \cdot e = e$.

If $\lambda: 1 \Rightarrow F$, then,

$$\begin{array}{ccccc} & & 1 & & \\ & \swarrow \lambda & & \searrow \lambda & \\ A & \xrightarrow[e]{} & A & \xrightarrow[e]{} & A \end{array}$$

$e\lambda = \lambda$ and $e \cdot e\lambda = e\lambda$.

Thus $\lim F = \{a \in A : ea = a\} = A^e$: fixed pt of idempotent.

Is this Exercise 3.2.ii., $A^e \xrightarrow{s} A \xrightleftharpoons[e]{1} A$.

as an equalizer. Then, $\exists! h$ s.t. below diagram commutes.

$$\begin{array}{ccccc} A & & & & \\ \downarrow r & \searrow e & & & \\ A^e & \xrightarrow{s} & A & \xrightleftharpoons[e]{1} & A \end{array}$$

$\Rightarrow sh = e$
i.e. sh splits e .

Now $shs = es = s$. Thus

$$\begin{array}{ccc} A^e & & \\ \downarrow r_s & \searrow s & \\ A^e & & \end{array}$$

$$\Rightarrow rs = 1_{A^e}$$

by uniqueness

$$A^e \xrightarrow{s} A \xrightarrow[e]{1} A$$

Also notes that $r|_B$ is surjective: $\forall a \in A^e$,

$$ra = a \Rightarrow sr(a) = a \quad \text{and } s \text{ is canonical.}$$

injection, thus $r(a) = a$.

Conversely, $B \xrightarrow{s} A \xrightarrow{r} B$ with $rs = 1_B$ gives an idempotent sr on A .

($sr sr = s 1_B r = sr$) which is split by B .

Ex 32.1) Take a pullback P of below diagram.

$$\begin{array}{ccc} P & \xrightarrow{\quad} & \text{mor } C \\ \vdots & & \downarrow \text{dom} \end{array}$$

$$\text{mor } C \xrightarrow{\text{cod}} \text{ob } C$$

$$\text{Then } P = \text{mor } C \times_{\text{ob } C} \text{mor } C$$

$$= \{ (g, f) \in \text{mor } C \times \text{mor } C \mid \text{dom } g = \text{cod } (f) \}$$

Thus if $(g, f) \in P$, then $g \circ f$ is possible composition.

And we have $m: P \rightarrow \text{mor } C$ It is well-defined function.
 $(g, f) \mapsto gf$

Now we have

$$P \xrightarrow{m} \text{mor } C \begin{array}{c} \xrightarrow{\text{dom}} \\ \xleftarrow{\text{id}} \\ \xrightarrow{\text{cod}} \end{array} \text{ob } C$$

Then $m(gf)$ has dom $\text{cod } g$ and cod $\text{cod } f$.

We claim that $B \times_A B \xrightarrow{m} B \xrightarrow[\begin{smallmatrix} \xrightarrow{f} \\ \xleftarrow{g} \\ \xrightarrow{h} \end{smallmatrix}]{h} A$

with ① $f(m(b_1, b_2)) = f(b_2)$

② $h(m(b_1, b_2)) = h(b_1)$ has a category structure

③ $m(b_3, m(b_2, b_1)) = m(m(b_3, b_2), b_1)$ ④ $m(g(a), b) = b = m(b, g(a))$

⑤ $m(b, g(a)) = b$

① Object = A

② Morphism = B i.e. by identifying $b \in B$

as $f(b) \rightarrow h(b)$.

③ Composition, m . Suppose (b_1, b_2, b_3) are

composable, i.e. $(b_2, b_1), (b_3, b_2) \in P$.

Then, $f \circ m(b_2, b_1) = f(b_1)$ $f \circ m(b_3, b_2) = f(b_2)$

$h \circ m(b_2, b_1) = h(b_2)$ $h \circ m(b_3, b_2) = h(b_3)$

Hence, $f(b_3) = h(b_2) = h \circ m(b_2, b_1)$

implies $(b_3, m(b_2, b_1)) \in B \times_A B$.

Also, $f \circ m(b_3, b_2) = f(b_2) = h(b_1) \Rightarrow (m(b_3, b_2), b_1) \in B \times_A B$.

Then,

$m(b_3, m(b_2, b_1)) = m(m(b_3, b_2), b_1)$

shows associativity.

Also, if $g(a)$ has a role of identity morphism.

Thus, the diagram with ①, ②, ③, ④, ⑤

gives a small category structure J .

with $Ob J = A$

$Mor J = B$.

Ex 3.2 ii) Change $\prod_{f \in \text{Hom}(J)} F(\text{cod } f)$, with $\prod_{f \in \text{Hom}(J_{\text{ob}})} F(\text{cod } f)$

in the diagram p. 88. In this case, $\text{Ifun } J$ is not a "discrete category" then we still have c and d which is the same as above.

So equalizer must be same.

In case of J is discrete, then $c = d$.

thus $\text{colim}_J F = \prod_{j \in \text{Ob } J} F_j$, a product, as we expected.

Ex 3.2 iii) If $\begin{array}{ccc} & a & \xrightarrow{h} c \\ f \downarrow & \circlearrowleft & \downarrow g \\ & b & \xrightarrow{l} d \end{array}$, then $lf = gh$.

$\Rightarrow P \rightarrow \text{Hom}(a, c)$ let P be a pull back, then,

$$\begin{array}{ccc} & & \downarrow g^* \\ \downarrow & & \downarrow \\ \text{Hom}(b, d) & \xrightarrow{f_*} & \text{Hom}(a, d) \end{array}$$

$$P = \text{Hom}(a, c) \times_{\text{Hom}(a, d)} \text{Hom}(b, d)$$

$$\text{Set } \{ (h, l) \in \text{Hom}(a, c) \times \text{Hom}(b, d) \mid g^*(h) = f_* l \}$$

$$\text{i.e. } gh = lf$$

$$\text{Thus, } Sg(f, g) := \text{Hom}(a, c) \times_{\text{Hom}(a, d)} \text{Hom}(b, d)$$

Ex 3.2 iv Let $J^\S$ be a category st.

$Ob J^\S = Mor J$. $Mor J^\S$: for each $f: x \rightarrow y \in J$

$$\begin{array}{c} 1_x \rightarrow f \\ 1_y \rightarrow f \end{array} \in Mor J^\S$$

Let $\alpha \in Hom(F, G)$. Then, for any $f: x \rightarrow y \in J$

$$\begin{array}{ccc} Fx & \xrightarrow{\alpha_x} & Gx \\ Ff \downarrow & \alpha & \downarrow Gf \\ Fy & \xrightarrow{\alpha_y} & Gy \end{array}$$

Now construct $H: J^\S \rightarrow Set$

$$\begin{array}{ccc} 1_x & \xrightarrow{\quad} & Hom(Fx, Gx) \\ \downarrow & \xrightarrow{\quad} & \downarrow (Gf)_* \\ f: x \rightarrow y & \xrightarrow{\quad} & Hom(Fx, Gy) \\ \uparrow & \xrightarrow{\quad} & \uparrow (Ff)^* \\ 1_y & \xrightarrow{\quad} & Hom(Fy, Gy) \end{array}$$

If $\lambda \in Core(1, H)$. then,

$$\begin{array}{ccc} & 1 & \\ \lambda_x \swarrow & & \searrow \lambda_y \\ & \alpha & \end{array}$$

$$Hom(Fx, Gy) \xrightarrow{(Gf)_*} Hom(Fx, Gy) \xleftarrow{(Ff)^*} Hom(Fy, Gy)$$

By identification, this shows that

$$(Gf)_*(\lambda_x): Fx \rightarrow Gy = \lambda_f = Ff^*(\lambda_y)$$

ie.

$$\begin{array}{ccc} x & \xrightarrow{Ff} & y \\ \lambda_x \downarrow & & \downarrow \lambda_y \\ & \xrightarrow{Gf} & \end{array}$$

Hence we identify λ as
 $\lambda \in Hom(F, G)$

Conversely, if $\lambda \in \text{Hom}(F, G)$

then define $\lambda_f = Gf \circ \lambda_x = \lambda_y \circ Ff$

for any $f: x \rightarrow y \in J$ nontrivial morphism

Then, by identifying λ_x and λ_y as a map

$1 \rightarrow \text{Hom}(F_x, G_x)$ and $1 \rightarrow \text{Hom}(F_y, G_y)$

we have a diagram

$$\begin{array}{ccc} & 1 & \\ \lambda_x \swarrow & & \searrow \lambda_y \\ \circlearrowleft & \downarrow \lambda_x & \circlearrowright \\ \sim & \xrightarrow{(Gf)^*} & \sim \\ & \circlearrowleft & \circlearrowright \\ & \xleftarrow{(Ff)_*} & \end{array}$$

which commutes every triangle.

Hence such λ is a core.

So, $\text{Hom}(F, G) \cong \text{Lim} H$ as a set.

Ex. 3.2.4) It suffices to show that equalizer is $\text{Hom}(F, G)$. Let E be equalizer of diagram. Then,

$$E = \{ (\lambda_j)_{j \in O_L J} \in \prod_{j \in O_L J} \text{Hom}(F_j, G_j) \mid$$

$$(Ff)^* \circ \pi_{j'}((\lambda_j)_{j \in O_L J})$$

$$= Gf_* \circ \pi_j((\lambda_j)_{j \in O_L J})$$

i.e., $\lambda_{j'} \circ Ff = Gf \circ \lambda_j$ i.e.

$$\begin{array}{ccc} F_j & \xrightarrow{\lambda_j} & G_j \\ Ff \downarrow & \circlearrowleft & \downarrow \\ F_{j'} & \xrightarrow{\lambda_{j'}} & G_{j'} \end{array}$$

Thus $E \subseteq \text{Hom}(F, G)$. Conversely,
 any $\lambda \in \text{Hom}(F, G)$ as identified by
 $(\lambda_j)_{j \in \text{Obj } J} \in \prod_{j \in \text{Obj } J} (F_j, G_j)$ satisfy the
 diagram $\Rightarrow \lambda \in E. \Rightarrow E = \text{Hom}(F, G)$.

Ex 32. vi).

$\lim F \cong \text{Cone}(1, F)$. Let $\lambda \in \lim F$.
 Then, $\lambda_c: 1 \rightarrow F_c$ can be identified as an element of F_c .

Thus, for each $\lambda \in \lim F$,

let $G: \text{Cone}(1, F) \rightarrow \text{Section}(\pi)$.
 $\lambda \longmapsto F_\lambda: C \rightarrow \int F$
 $c \longmapsto (c, \lambda_c)$.

Conversely, let $H \in \text{Section}(\pi)$.

Then, $\forall c \in \text{Obj } C, H_c = (c, \lambda_c)$

for some $\lambda_c \in F_c$. And, for if $\exists c \xrightarrow{f} d \in C$,

$Hf: (c, \lambda_c) \rightarrow (d, \lambda_d)$

s.t. $f(c) = d$ and $Ff(\lambda_c) = \lambda_d$.

Hence.

$\lambda = \{ \lambda_c : c \in \text{Obj } C \}$
 $\Rightarrow$ can be identified as
 a cone $1 \Rightarrow F$.

And this is image of G .

Recall limit cone λ is $\lambda: \lim F \Rightarrow F$ s.t.

$\forall \mu: X \Rightarrow F$ a cone, $\exists! f: X \rightarrow \lim F \in \text{Set}$.

s.t. $\lambda \circ \alpha_f = \mu$.

Then, $\forall x \in X$, $\mu(x): I \Rightarrow F$. Thus,
 $\lambda \circ \alpha_f(x): I \Rightarrow F$ " "

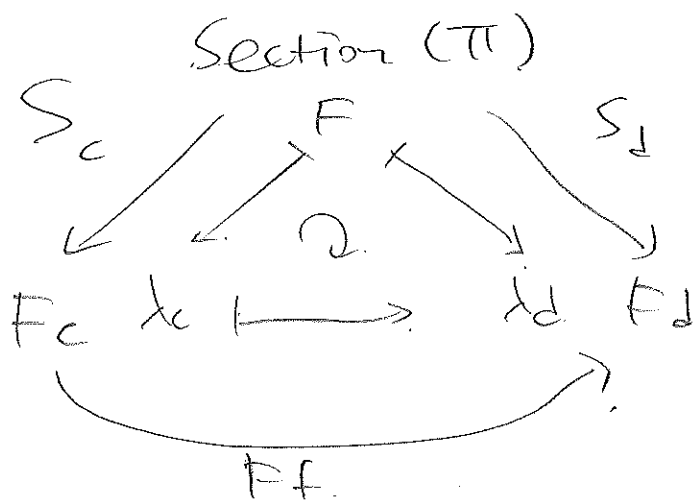
Since $f|_{\alpha_f}$ is an iso, we know that.

$\mu(x) = \lambda(f(x))$ as a natural transformation.

Thus, every cone $\mu: X \Rightarrow F$ is just set of
 $\mu(x): I \Rightarrow F$ s.t. $\mu(x)$ is a section.

Thus limit cone is just a cone induced by
 Section (π) s.t. for $f: c \rightarrow d \in \text{Mor } C$,

when $F: C \rightarrow \text{Set}$
 $c \mapsto (c, \lambda_c)$



left diagram
 is induced,
 thus:

$S: \text{Section}(\pi) \Rightarrow F$

Can be regarded as a limit. For any $\mu: X \Rightarrow F$,

$\mu(x): I \Rightarrow F$ so $\mu(x) \in \text{Section}(\pi)$. Thus, the map.

$X \rightarrow \text{Section}(\pi)$ by $x \mapsto \mu(x)$ induces desired diagram.

3.3. Preservation, reflection, creation.

Def 3.3.1. For any class $\mathcal{K}$ of diagrams $K: J \rightarrow C$
and $F: C \rightarrow D$

① F preserves those limits:

$\forall K: J \rightarrow C \in \mathcal{K}$ and $K: \lim K \Rightarrow K$ is a
limit cone $\Rightarrow FK$ is a limit cone of FK .
i.e. $F(\lim K) = \lim FK$ and $FK: \lim FK \Rightarrow FK$.

② F reflects " "

$\forall K: X \Rightarrow K$ in C , and FK is a limit cone
of $FK \Rightarrow K$ is a limit cone, i.e.

$X = \lim K$ and K has universal
property

③ F creates " "

If $\lim FK$ exists, $\exists K: \lim FK \Rightarrow FK$ a
limit cone s.t. it can be lifted to a

limit cone over K and F reflects the limits.

(For colimit, dualize these.)

In other words, if $K: \lim K \Rightarrow K$ is a limit cone,
 $K': X \Rightarrow K$ is a cone,

① preserve: " FK is a limit cone" (Thus $F(\lim K) = \lim FK$)

② reflects: " FK is a limit cone $\Rightarrow K$ is limit cone

③ creates FK has a limit cone $\Rightarrow K$ has a limit cone
and F reflects.

Prop 3.3.3. ① $F: C \rightarrow D$ creates limit. ② D has a limit.
 $\Rightarrow C$ has a limit and F preserves it.

3f) Let $K: J \rightarrow C$ be in the class.

$\mu: J \Rightarrow FK$ is a limit cone.

Since F creates these limits, $\exists \lambda: C \Rightarrow K$
 a limit cone s.t. $F\lambda$ is a limit cone.

By Ex. 3.1. iv, $F\lambda \cong \mu$.

To see F preserves then, let $\lambda': C' \Rightarrow K$ be
 another limit cone. By prop 3.1. ii, $\lambda' \cong \lambda$

$\Rightarrow F\lambda' \cong F\lambda \cong \mu$. $\Rightarrow F\lambda'$ is a limit cone.

Def 3.3.4 (Sheaf) $\mathcal{O}(X)$: poset of open subsets of X
 ordered by inclusion. Let I be a discrete ^{small} category.

$\{U_i\}_{i \in I}$ cover U . If U_i 's are image of I .

and $\text{colim}_{i \in I} (\{U_i \leftarrow U_i \cap U_j \rightarrow U_j\}) = U$.

Notes that colimit in this diagram is an open set

J' with $U_i \rightarrow U'$ and $U_i \cap U_j \rightarrow U'$.

$\Rightarrow U' = \bigcup_{i \in I} U_i$. So, this definition is consistent with
 usual definition of open coverings.

Let $F: \mathcal{O}(X)^{\text{op}} \rightarrow \text{Set}$. By def in 1.3. ii (v) U

since $\mathcal{O}(X)$ is small. F is presheaf. Sheaf is a presheaf
 preserving these colimits $\forall$ pair i, j and I index I
 covering U .

Now let I' be a small category st $I' \supseteq I$ st
 ① Indexes $\{U_i\}_{i \in I}$ a cover of U .

② I' has objects (i, j) corresponds to

$$U_i \cap U_j \quad \forall i, j \in I$$

Let $K: (I')^{op} \hookrightarrow \mathcal{O}(X)^{op}$ and $F: \mathcal{O}(X)^{op} \rightarrow \text{Set}$

as a sheaf. Notes that any morphism $U_i \rightarrow U_j$ actually induces $U_i \cap U_j \hookrightarrow U_i$

Then, Thm 3.2.13 yields

$$\lim_{I' \text{ op}} FK \xrightarrow{\quad} \prod_{(i,j) \in I'} FK(i,j) \xrightarrow[\downarrow]{\quad} \prod_{f \in \text{Mor}(I')^{op}} F(\text{cod } f^{op})$$

Let f be an arbitrary morphism in $(I')^{op}$.

We may regard it as $U_i \cap U_j \hookrightarrow U_i$ by the above argument.

Then, $\subset$ is just sends $F(U_i \cap U_j)$ component in the middle to the same component in the last, by seeing p. 88 diagram.

Now, $\downarrow$ is a little different. $\pi_{\text{cod } f^{op}} = \pi_{\text{cod } f}$.

So if $f: U_i \cap U_j \hookrightarrow U_i$ then $\downarrow$ sends

$F(U_i)$ in the middle to $F(U_i \cap U_j)$ in the last

$U_i \cap U_j \hookrightarrow U_i$.

Now let $\lambda \in \prod_{(i,j) \in I'} FK(i,j)$ If $c(\lambda) = \downarrow(\lambda)$, then,

$Ff(\lambda|_{U_i}) = \lambda|_{U_i \cap U_j} = Ff(\lambda|_{U_j})$ by applying $\downarrow$ twice and $\subset$ once.

This shows that if $Ff(\lambda|_{U_i}) = Ff(\lambda|_{U_j})$

for all i, j , then $\lambda \in \lim_{I'} FK$.

Thus, instead of C , we may use this condition.

$$\lim_{I'} FK \longrightarrow \prod_{(i,j) \in I'} FK(\bar{i}, j) \xrightarrow[\substack{Ff(\lambda|_{U_i}) \circ \pi_i \\ Ff(\lambda|_{U_j}) \circ \pi_j \cdot f_{ij}^{op} \in \text{Mon}(I')^{op}}]{Ff(\lambda|_{U_i}) \circ \pi_i} \prod F(\text{cod } f^{op})$$

Since these two maps only depend on $FK(\bar{i}, j)$ with $\bar{i} = j$, we may delete components with $\bar{i} \neq j$, so.

$$\lim_{I'} FK \longrightarrow \prod_{\bar{i} \in I} FK(\bar{i}). \xrightarrow[\quad]{\quad} \prod \mathbb{Q} //$$

Also, since the map only came of $FK(\bar{i}, j)$ with $\bar{i} \neq j$. So we may delete $F(U_i)$ parts in the last

$$\lim_{I'} FK \longrightarrow \prod_{\bar{i} \in I} FK(\bar{i}) \xrightarrow[\substack{f \in \text{Mon}(I')^{op} \\ f: U_{\bar{i}} \cap U_j \rightarrow U_j}]{\quad} \prod F(U_{\bar{i}} \cap U_j)$$

then, by change of notation, we may get.

$$\lim_I F \longrightarrow \prod_{i \in I} F(U_i) \xrightarrow[\quad]{\quad} \prod_{i, j \in I} F(U_i \cap U_j)$$

as desired.

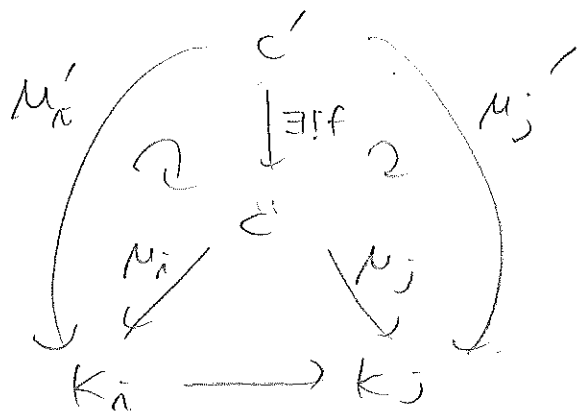
Lemma 33.5 $F! C \rightarrow D$ fully faithful.

$\Rightarrow$ F reflects limits and colimits which are in Codomain .

pf) Let $\mu: C \rightrightarrows K$ be a core of $K \rightarrow C$.

S.t. $F_M: F_C \Rightarrow F_K$ is a limit cone.

Let $\mu': c' \Rightarrow k$, WTS. $\exists! f: c' \rightarrow c$ s.t.



Left diagram commutes.

Now notes that $F \models B$,
still a core; this comes
from functionality of F .

Hence $\exists! \bar{f} : F_C \rightarrow F_C$ st. $\#u' = \bar{f} \circ \#u$.

By fully faithfulness, $\exists f: C' \rightarrow C$ s.t. $\#f = \bar{f}$.

then, $f_{\mu'_i} = f_j(\mu_i \circ f) \Rightarrow \mu'_i = \mu_i \circ f$

Since fully faithful implies $C(c, k_i) \xrightarrow{F(-)} D(Fc, Fk_i)$ is bijection.

is bijective.
Thus, the above diagram commutes. So F reflects
limit.

limit, for the colimit, 'apply this' on contravariant functors and get reflection of colimit by duality.

Lemma 3.3-6. Equivalence of Categories preserves, reflects, and creates any limit and colimits that are present in either its domain or codomain.

5-f) Let $F: C \rightarrow D$ $G: D \rightarrow C$ be equivalence of categories. Then F, G are fully faithful and essentially surjective.

It suffices to show that each F and G creates limit in D or C . Suppose we showed it.

① reflectivity: done by Lemma 3.3.5

② preserves limit/colimit: If λ is a limit cone in C .

then, G creates a limit cone λ' in D .

Is it. $G\lambda' \cong \lambda$. Thus, $G\lambda'$ is a limit cone.

Hence $F(\lambda) \cong FG(\lambda') \cong \lambda'$ implies F

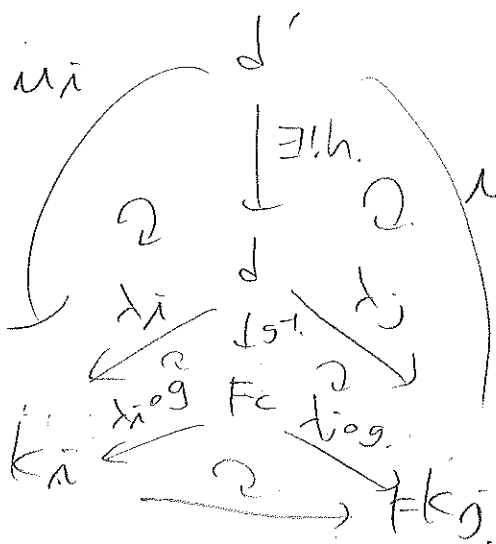
preserves a limit.

Thus, all we have to do is show that F creates limit/colimit.

Let λ be a limit cone in D , i.e. $\lambda: d \Rightarrow Fk$.

By essential surjectivity, $\exists c \in C$ s.t. $Fc \xrightarrow{g} d$.

Now we claim $\lambda_g: Fc \Rightarrow Fk$ is also a limit cone; to see this, let $\mu: d' \Rightarrow Fk$ be a cone. Then,



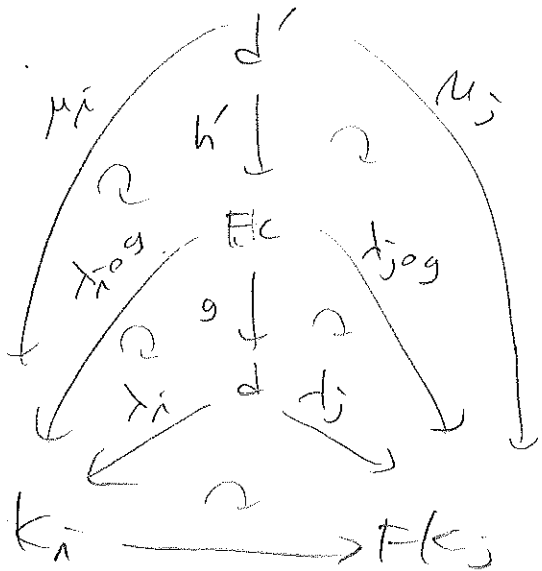
left triangle commutes

Thus, $\exists$ a map $g^!h: d' \rightarrow Fc$
 s.t. $\mu = \alpha_{g^!h} \circ \lambda g$.

Moreover this $g^!h$ is unique;

if $\exists h': d' \rightarrow Fc$ s.t.

the above one commutes, then left triangle



commute, thus, universal property
 of $d: d \Rightarrow Fc$ shows that

$$g h' = h$$

$$\Rightarrow h' = g^!h$$

Now, fully faithfulness of F shows that

any $\lambda_i \circ g$ comes from $\bar{\lambda}_i \circ g \in C(c, k_i)$

via F . Thus, $\exists \bar{\lambda}: c \Rightarrow k$ s.t. $F\bar{\lambda} = \lambda g$.

Since F reflects a limit, $\bar{\lambda}$ is a limit cone.

Hence, (since F is already reflects the limit)

F creates a limit. Since F was chosen

arbitrarily, both F and G creates limits

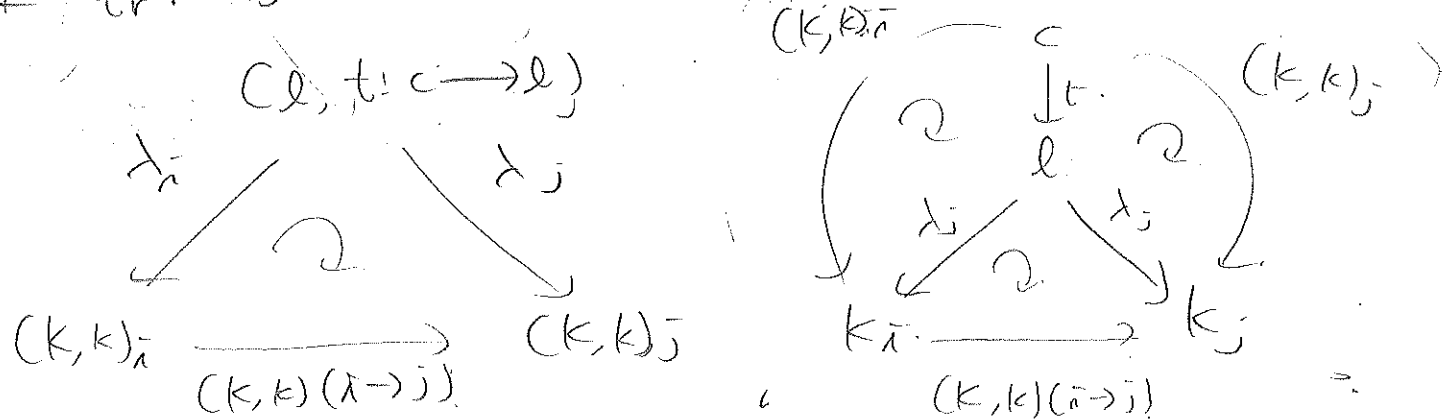
By duality, $\square$

$K := \Pi K'$ and $k: C \Rightarrow K$ is a cone induced by a diagram $K': J \rightarrow \mathcal{C}$, with summit C .

WTS if $K: J \xrightarrow{(K, k)} \mathcal{C}/\mathcal{C} \xrightarrow{\Pi} \mathcal{C}$ admits a limit cone in $\mathcal{C}$, then this cone lifts to define a unique limit cone over (K, k) in $\mathcal{C}/\mathcal{C}$.

Let $\lambda: l \Rightarrow K$ be a limit cone in $\mathcal{C}$. Also we have $k: C \Rightarrow K$. Hence $\exists! t: C \rightarrow l$ st.
 $K = \lambda \circ \alpha_t$.

Now, we can identify $\lambda_{\bar{i}}: l \rightarrow K_{\bar{i}}$ as a morphism from $t: C \rightarrow l$ to $(K, k)_{\bar{i}}: C \rightarrow K_{\bar{i}}$ in $\mathcal{C}/\mathcal{C}$. Thus, below left Δ commutes since right triangle commutes in $\mathcal{C}$.



So, λ is a cone $t \Rightarrow (K, k)$.

To see it is a limit cone, suppose that

$\exists \mu: r \Rightarrow (K, k)$ in $\mathcal{C}/\mathcal{C}$. Then, by letting $r: C \rightarrow r_j$

Def) 3.3.7. $F: C \rightarrow D$ strictly creates limit.

If $\forall K: J \rightarrow C$ in the class and $\varinjlim$ core λ over FK ,

① $\exists!$ lift $\bar{\lambda}$ of λ over K in C as a core

② $\bar{\lambda}$ is limit core.

Def) J : category is connected if $J \neq \emptyset$ and in the underlying directed graph, every pair of objects is connected by a finite zig-zag of morphisms.

A diagram $K: J \rightarrow C$ is connected $\Leftrightarrow J$ is connected.

Prop 3.3.8 $\forall c \in \text{Ob } C$, $\Pi: C/C \rightarrow C$

strictly creates

1) all limits C admit

2) all connected colimits C admit

st. $\left(\begin{array}{ccc} h: d \rightarrow e & \downarrow & \mapsto h \downarrow \\ \begin{array}{c} c \\ \swarrow \downarrow \searrow \\ f \downarrow g \downarrow \\ x \xrightarrow{h} y \end{array} & & (e, g: c \rightarrow e) \mapsto e \end{array} \right)$

st) Let $K: J \rightarrow C$ be a diagram. Then

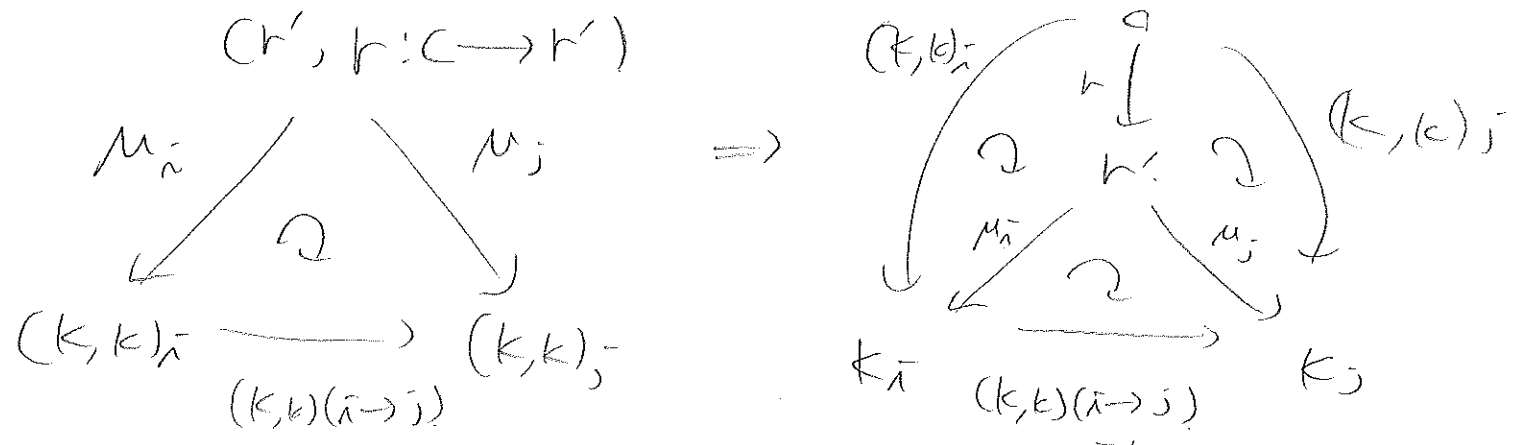
$\forall \bar{i} \rightarrow \bar{j} \in \text{Mor } J$, $K(\bar{i} \rightarrow \bar{j})$ is a morphism

$h: \text{Cod}(K\bar{i}) \rightarrow \text{Cod}(K\bar{j})$ s.t. $\begin{array}{ccc} K\bar{i} & \xrightarrow{c} & K\bar{j} \\ & \searrow \downarrow \swarrow & \\ & \text{Cod}(K\bar{i}) & \xrightarrow{h} \text{Cod}(K\bar{j}) \end{array}$ in C .

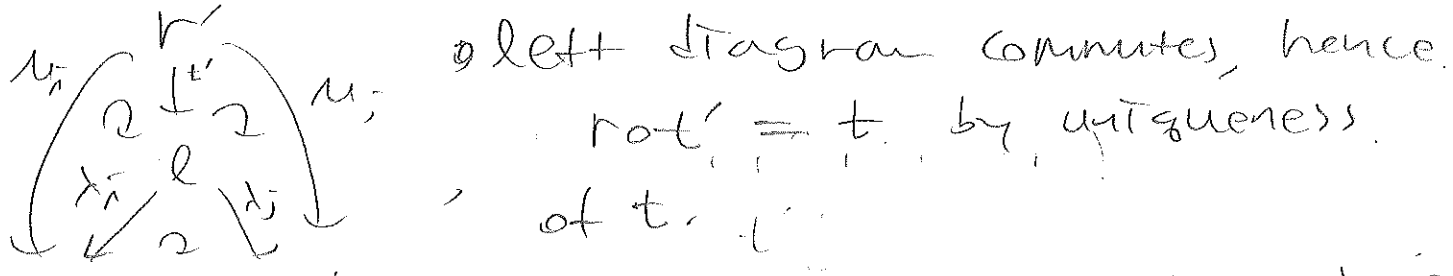
Hence K always induces a cone $K: C \Rightarrow (J \rightarrow C/C \Rightarrow C)$ in C .

Thus, we use notation $K' = (K, K): J \rightarrow C/C$ s.t.

we have



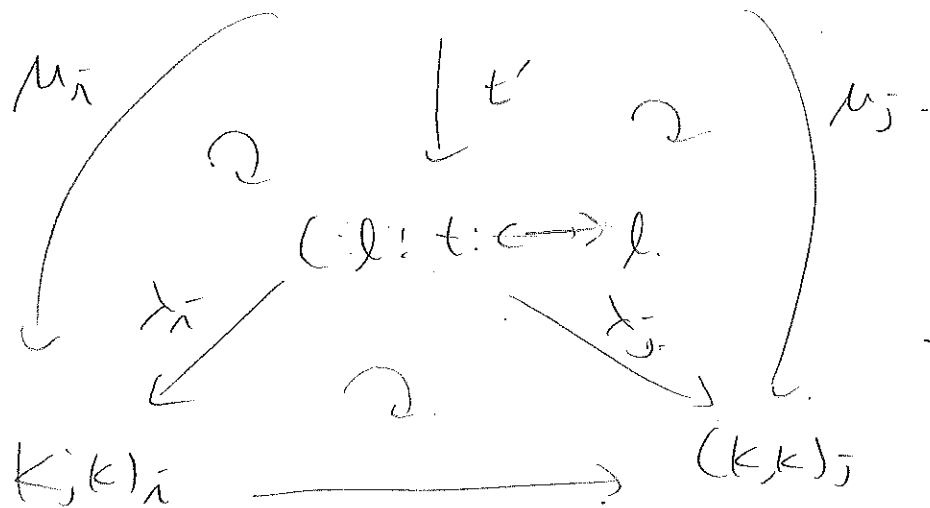
Since λ is a limit cone, $\exists! t: r' \rightarrow l$ s.t.



Let diagram commutes, hence $rot' = t$ by uniqueness of t .

Thus, (t, λ) is a morphism in $\mathcal{C}/\mathcal{C}$.

Hence, $(r', r: C \rightarrow r')$



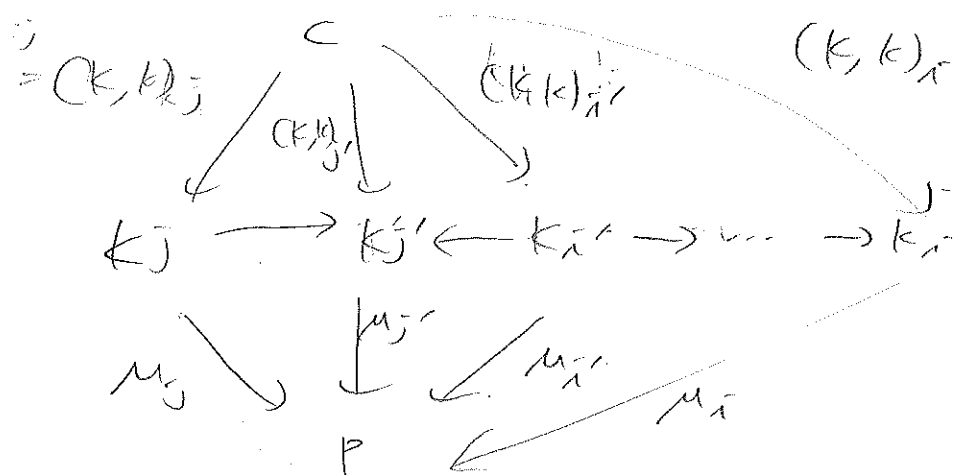
left diagram commutes by Figure 1.

$\Rightarrow \lambda$ is a limit cone in $\mathcal{C}$.

In case of colimit, let $\mu: K \Rightarrow P$ is a colimit and J is connected. Still we have $(c: C \Rightarrow P)$ a cone. Then, let η be a morphism. $\eta_i: C \xrightarrow{k_i} k_j \xrightarrow{\mu_j} P$

We claim that $\lambda: C \rightarrow P$ as an object of $\mathcal{C}/C$ is independent of choice of $j \in J$. To see this

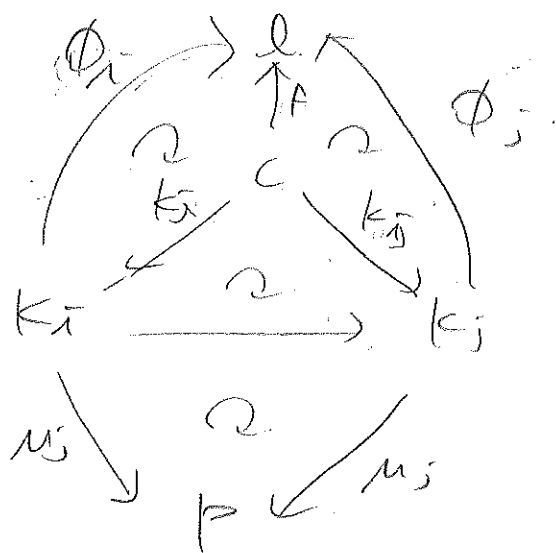
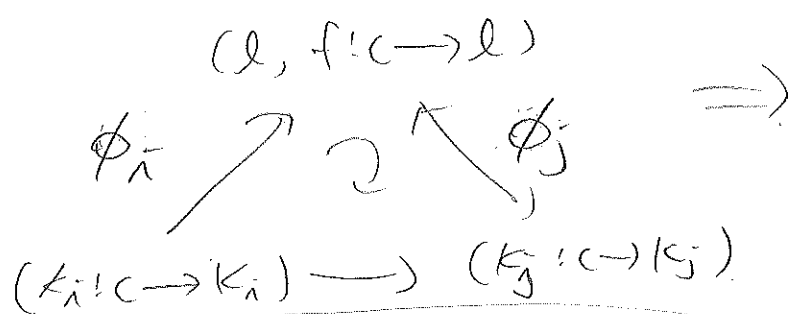
For any $j, \tilde{\lambda} \in \text{obj}$, $\exists z_1 g z_2 g$ w/o μ_B . Thus



left diastole.
with diagram
charm
shows that +.

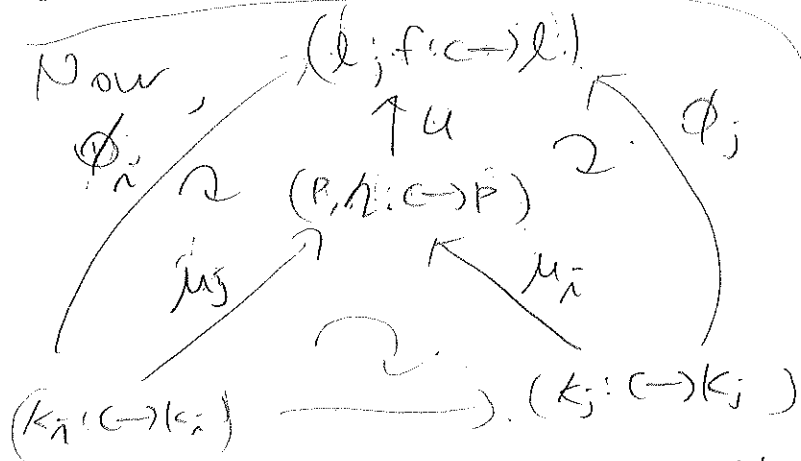
$$\mu_1^*(k_1, \omega), \mu_2^*(k_2, \omega)$$

Now let $\phi: (J \rightarrow \mathcal{C}) \Rightarrow f$ be a cone in $\mathcal{C}$.
under $(\ell, f: C \rightarrow \ell)$ Then,



Since $A B^a$ is a GIMT core,
 $\exists! \eta: P \rightarrow Q$ s.t.

$$u_{0\mu\bar{x}} = \phi_i$$



So it suffices to show that u is a morphism in $\mathcal{C}/\mathcal{C}$.

Since $f = \phi_i \circ k_i$, $\begin{array}{ccc} & P & \xrightarrow{u} L \\ & \nearrow \eta & \nearrow f \\ & C & \end{array}$
 $\eta = \mu_i \circ k_i$

Thus $u \circ \eta = u \circ \mu_i \circ k_i = \phi_i \circ k_i = f$.

$\Rightarrow u$ is a morphism in $\mathcal{C}/C$.

Thus $\mu: \eta \Rightarrow (J \rightarrow \mathcal{C})$ is a colimit cone.

These colimit and l.h construction is unique since it depends only on ~~the~~ cone (a cocone) structure of C . $\Rightarrow \Pi$ strictly creates connected colimits or limits.

Prop: If $K: J \rightarrow C^A$ is a diagram, then $\lim_J K$ is a functor $\text{fun } A \rightarrow C$ s.t. whose evaluation at a is a limit of $J \rightarrow C$ by $j \mapsto F_j(a)$. (This is prop 3.3-9)

Prop 3.3-9.) Let A : small category. $\Pi: C^A \rightarrow C^{ob A}$ be a forgetful functor where $ob A$ is a discrete subcategory of A . Then Π strictly creates all limit and colimit that exist in C .

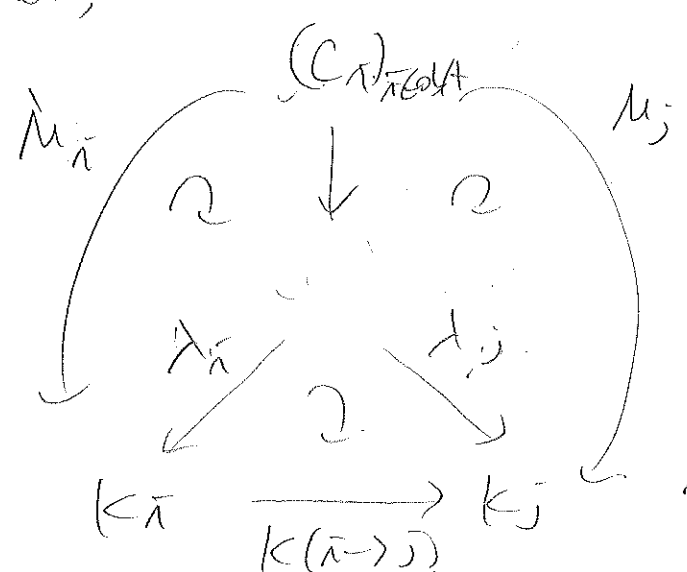
These limits are defined objectwise, meaning that $\forall a \in A$, $ev_a: C^A \rightarrow C$ preserves all limits and colimits existing in C .

pf) C^{obA} is isomorphic to $\prod_{obA} C$; since any functor in C^{obA} is just choosing obA indexed elements in $Ob C$.

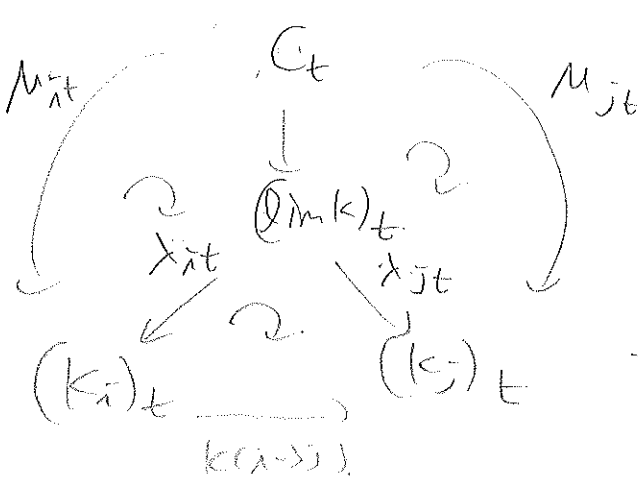
By applying universal property of product in CAT, any diagram $K: J \rightarrow C \cong \prod_{obA} C$ is just obA -indexed family of diagram $J \rightarrow C$.

Let $\lim K$ be a limit in $\prod_{obA} C$, λ is a limit cone.

Then, for any $\mu: (C_i)_{i \in obA} \Rightarrow K$, with $\lambda \rightarrow J \in J$.



but in the product of category, every homomorphism can be decomposed into components indexed by obA . So for any index $t \in obA$,



Hence, $\lim K$ is assemble of limit in each C . and so is its limit cone.

Thus, C admit $\lim$ iff C^{obA} does. (dualy for colimit)

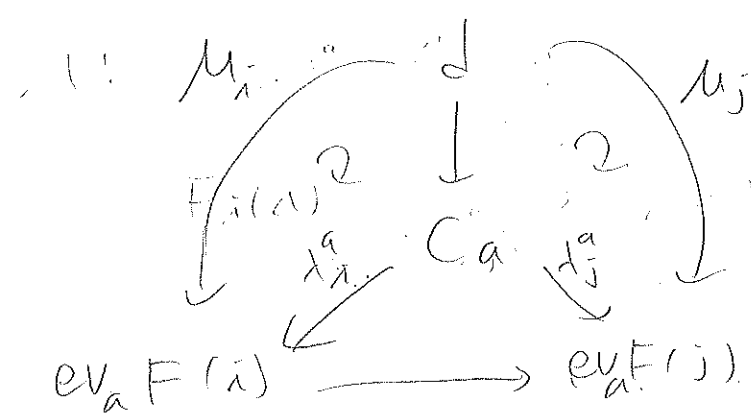
Also, $ev_a: C^{obA} \rightarrow C$ can be viewed as a projection, thus it preserves limit. (dualy colimit)

To show $\Pi : C^A \rightarrow C^{ob A}$ strictly creates $\lim / \text{colim}$,
 Let $F : J \rightarrow C^A$ a diagram s.t. ΠF has a limit
 and limit cone $\lambda : (C_i)_{i \in J} \Rightarrow \Pi F$.

This equivalent to say that C admits a limit.

So, $F : J \rightarrow C^A \xrightarrow{\text{eval}_A} C$ admits a limit for any
 $a \in \text{ob } A$. Let $\lambda^a : C_a \Rightarrow \text{eval}_a F$ be a limit cone.

Then, for any $u : d \Rightarrow \text{eval}_a F$, $i \mapsto J_i \in J$.

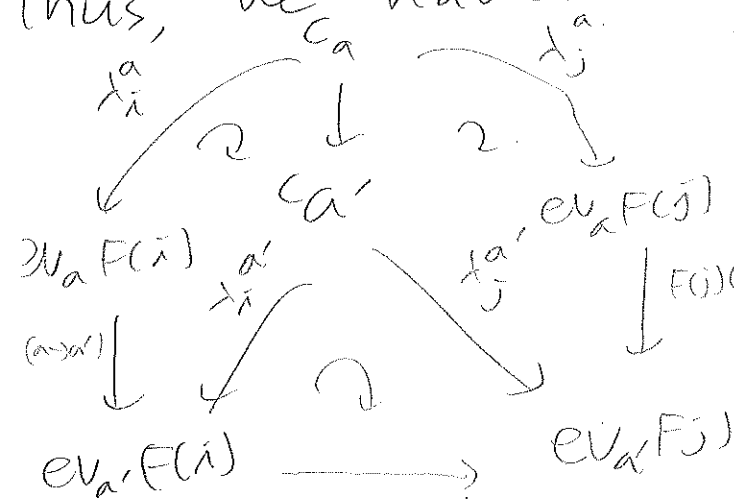


Now let $a \rightarrow a'$ be a morphism in C .

Then, since $F(i)$ is a functor, $\exists$ a morphism

$$F(i)(a \rightarrow a') : \text{eval}_a F(i) \rightarrow \text{eval}_{a'} F(i)$$

Thus, we have the unique map $\lambda^a : C_a \rightarrow C_{a'}$



S.t. λ fits diagram
 Commute.

Now define $G : A \rightarrow C$
 s.t. $G(a) = C_a$.

$G(a \rightarrow a') = (a \rightarrow a') C_{a'}$
 induced by the universal
 property of limit cone $\lambda^{a'}$.

Then functoriality is satisfied by the uniqueness of universal property. I_a goes to I'_{ca} by uniqueness and $a \rightarrow a' \rightarrow a''$ goes to $a \rightarrow a''$ by uniqueness too.

Thus G is a functor. Now we claim

$$G \text{ is } \lim (J \rightarrow C^A)$$

Let $n: H \Rightarrow F$ be a cone. Then,

$$\begin{array}{ccc} n_i & H & n_j \\ & \searrow & \swarrow \\ F(i) & \xrightarrow{F(i \rightarrow j)} & F(j) \end{array}$$

Thus, by ev_a , we set

Then, the universal property gives left diagram with $h_a: H(a) \rightarrow C_a = G(a)$

$$\begin{array}{ccc} H(a) & & \\ \downarrow (n_i)_a & \xrightarrow{h_a} & \downarrow (n_j)_a \\ F(i)(a) & \xrightarrow{ev_a F(i \rightarrow j)} & F(j)(a) \end{array}$$

Now we claim that $h = \{h_a: a \in \text{Ob } A\}$ is a natural transformation because of uniqueness;

$$\begin{array}{ccccc} F(i)(a) & \xleftarrow{\lambda_i^a} & H(a) & \xrightarrow{\lambda_j^a} & F(j)(a) \\ & \searrow & \downarrow h_a & \swarrow & \downarrow \lambda_j^a \\ & & G(a) & & \\ & \searrow & \downarrow h_{a'} & \swarrow & \downarrow \lambda_j^{a'} \\ F(i)(a') & \xleftarrow{\lambda_i^{a'}} & H(a') & \xrightarrow{\lambda_j^{a'}} & F(j)(a') \end{array}$$

$H a \rightarrow G a \rightarrow G a'$
and $H a \rightarrow H a' \rightarrow G a'$
gives the commuting cone structure $\Rightarrow$ uniqueness of limit cone gives the same map.

Finally, this natural transformation make cones commute via left diagram.
 $\Rightarrow G \text{ is } \lim_J F.$

3.3.3.i) Let $\bar{I} \rightarrow \bar{J} \in \bar{\mathcal{J}}$, $\lambda_i (i \in \bar{I}) \Rightarrow \text{Colim } K$

(i)

be a colimit cone.

Then, left diagram

commutes by functoriality

$$\begin{array}{ccc} & \text{Colim } K & \\ F\lambda_i \nearrow & & \nwarrow F\lambda_j \\ FKi & \longrightarrow & F Kj \end{array} \Rightarrow F\lambda \text{ is a cone in } \mathcal{D}$$

Thus, the universal property of $\text{Colim } FK$ gives

$$\exists! t: \text{Colim } FK \rightarrow \text{Colim } K$$

s.t. left diagram commutes

($\mu: FK \Rightarrow \text{Colim } FK$ is a colimit cone)

$$\begin{array}{ccc} & \text{Colim } K & \\ F\lambda_i \nearrow & \uparrow \exists! t & \nwarrow F\lambda_j \\ & \text{Colim } FK & \\ \mu_i \nearrow & & \nwarrow \mu_j \\ FK_i & \longrightarrow & FK_j \end{array}$$

(ii) If t is an isomorphism, then

$F\lambda$ is also a colimit cone since $\mu_i = t^{-1} F\lambda_i \cdot \theta_i$.

So if $\eta: FK \Rightarrow X$ is a cocone then,

$$\exists! t': \eta \rightarrow \text{Colim } FK \text{ s.t. } \eta_i = t' \circ \mu_i \quad \forall i$$

$$\Rightarrow \eta_i = t' \circ t^{-1} \circ F\lambda_i \cdot \theta_i. \text{ And } t' \circ t^{-1} \text{ is uniquely}$$

determined by the universal property of $\text{Colim } FK$.

Thus, F preserves colimit.

3.3.ii), 3.3.iii) already done in lecture note.

Ex 3.3iv). Let $K: J \rightarrow C$ be in class. and
 $\lambda_i: C \rightarrow K_i$ for $i \in I$.

By (i), $\exists \mu: C \Rightarrow K$ is a limit cone and
 $F\mu$ is a limit cone in D . Thus FK has
a limit cone over FK ($F\mu$) which can
be lifted to a limit cone (μ) over K .

Thus, we need to show that F reflects limits.

Let $\lambda: C' \Rightarrow K$ s.t. $F\lambda$ is a limit cone.

Then, $F\lambda \cong F\mu$ by Prop 3.1.7 (Uniqueness of limit cone)

$\Rightarrow FC \cong FC' \Rightarrow C \cong C'$ by (ii) (F reflects iso.)

Then, it suffices to show that λ is a limit cone.

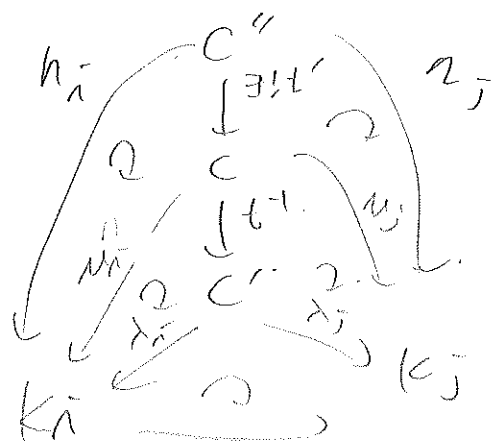
Notes that for any $\eta: C'' \Rightarrow K$,

then left diagram commutes

$\Rightarrow \exists ! t, t': C'' \rightarrow C'$
s.t. cones @ commute,

$\Rightarrow \lambda$ is a limit cone.

□



Ex 3.3.v) Let $(A, *)$, $(B, \#)$ $\in \text{Set}_*$

Then their coproduct is disjoint union of A, B

but picking apt from A or B .

Thus, if point picked from B , then

$(A, \#) \rightarrow (A, *) \sqcup (B, \#)$ is not a function.

$A \rightarrow A \sqcup B$ in Set . Hence it is not a

coproduct, in Set , since it is possible that

in Set , $\#$ and $*$ can be sent to distinct objects.

(This happens in Top_* too.)

If I was connected, then $\exists$ morphism $S \rightarrow I$ sending $*$ to $\#$ or vice versa in limit.

Ex 3.3 vi) done. In lecture note.

3.4. Representable nature of $\lim$ and colim .

Thm 3.4.2 For any $F: J \rightarrow \mathcal{C}$ s.t. $\lim F$ exists.
 $\text{ad } C(-, \lim J F) \cong \lim J C(-, F-)$

pf) First of all, Let $X \in \text{Set}$, Then, we claim
 $\lim J C(X, F-) \cong \text{Cone}(X, F) \left(\cong C(-, \lim J F) \right)$
 by definition of limit.

If $\lambda \in \text{Cone}(X, F)$ then λ satisfies the universal

in Thm 3.2.3 $\Rightarrow \lambda \in \lim J C(X, F-)$

Conversely, any $\lambda \in \lim J C(X, F-)$ is an
 element of $\prod_{j \in J} C(X, F_j)$, and $\forall f: i \rightarrow j \in J$

$\lambda_i \xrightarrow{X} \lambda_j \Rightarrow \lambda$ is a cone over F with summit X
 $F_i \xrightarrow{Ff} F_j \Rightarrow \exists$ isomorphism.

To see this is natural, think when $g: X' \rightarrow X$
 is a set map. Then, for any $\lambda \in \lim J C(X', F)$

$\lambda \in \prod_{j \in J} C(X', F_j)$
 thus $\lambda g \in \prod_{j \in J} C(X, F_j)$
 But as a cone, λg is left trivial which is the same one.

So $\lim J C(X', F-) \xrightarrow{\cong} \text{Cone}(X', F)$
 $\downarrow g_* \quad \downarrow g_*$
 $\lim J C(X, F-) \xrightarrow{\cong} \text{Cone}(X, F)$

Thm 3.4.3', natural iso expresses

"representable universal property of the limit"

$\lim_j F$ is defined representably as $\lim_j C(X, F-)$ for chosen X .

Ex. 3.4.4.) $F: I \rightarrow C$ is an iterated product,

i.e. $\lim F = A^I := \prod_I A$, then the property

$$\text{gives } C(X, A^I) \cong \prod_I C(X, A).$$

i.e. any morphism $h: X \rightarrow A^I$ is determined by $h_i: X \rightarrow A$ for all $i \in I$ with each product projection $\epsilon_i: A^I \rightarrow A$.

In a category of set, this is already known.

Prop 3.4.5: $C(X, -): C \rightarrow \text{Set}$ preserves limit.

if C is locally small.

pf) Thm 3.4.2 shows $C(X, \lim F) \cong \lim (C(X, F-))$

Moreover, remember that the natural isomorphism is constructed via $\prod_{j \in J} C(X, F_j)$. Hence,

$$\lim_j C(X, F-) \cong \text{core}(X, F) \cong C(X, \lim_j F)$$

u: limit core in Set
 λ : limit core in C
 as $\prod_{i \in I} \lambda_i$

And if $\lambda_j: \lim_j F \rightarrow F_j$ is a leg of limit core,

$C(X, \lambda) = \prod_{j \in J} C(X, \lambda_j)$ is RHS diagonal map, and LHS is product of legs of limit core.

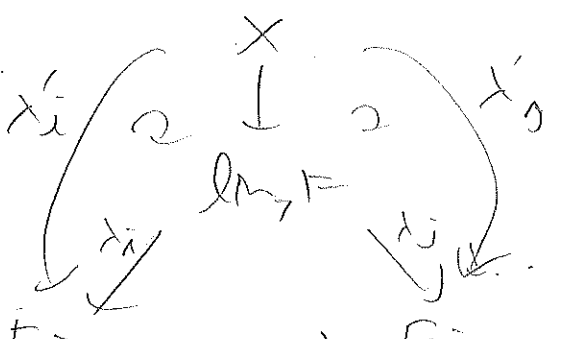
More precisely,

μ is constructed by Thm 3.2.13;

Actually Thm 3.2.13 construction shows that μ is a limit cone.

Then $\text{Cone}(X, F) \xrightarrow{\quad} C(X, \lim_j F)$ is just application
 $\downarrow \quad \swarrow C(X, \lambda)$
 $\prod_{j \in J} (X, F_j)$ of the universal property of limit cone λ ;

any cone $\lambda': X \Rightarrow F$ has a map $\exists! k: X \rightarrow \lim_j F$.

St. $\lambda'_i \circ \eta_i = \eta_i \circ \lambda_i$ commutes. So, $\lambda' \in \text{Cone}(X, F)$
 can be sent to k ,
the unique map.

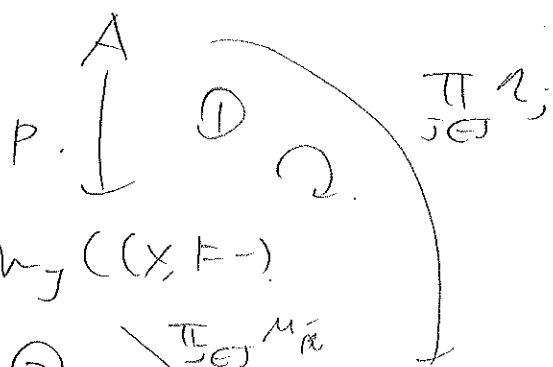
and $k \circ \lambda_i = \lambda'_i \quad \forall i \in J$ shows that
the triangle commutes, $\eta \circ \lambda \in C(X, \lambda)$

Hence, by identifying $C(X, \lambda)$ as a cone (X, F)

the diagram shows that it acts
as a limit cone; i.e. for any
cone $\eta: A \Rightarrow (X, F_-)$ in Set



$\exists!$ unique map $P: A \rightarrow \lim_{j \in J} (X, F_j)$ s.t.



① Commutes

However, ② is also commutes by above



$$\begin{array}{ccc} A & \xrightarrow{P} & \lim_{j \in J} (X, F_j) \\ \downarrow & \searrow & \downarrow \pi_j \\ (X, \lim_{j \in J} F_j) & \xrightarrow{\quad} & (X, \lambda) \end{array}$$

$\Rightarrow \exists!$ unique map KP .

$$\text{s.t. } A \xrightarrow{KP} C(X, \lim_{j \in J} F_j) \xrightarrow{\pi_j} (X, \lambda)$$

$\lim_{j \in J} F_j$ equal to $\lim_{j \in J} F_j$

This implies $\pi_j = C(X, \lambda) \circ KP \Rightarrow C(X, \lambda)$ is a limit cone.

Thm 3.4.6 (Second Interpretation of Thm 3.4.2)

C : locally small

(i) $C(X, -)$ preserves all limits that exist in C

(ii) The covariant Yoneda embedding

$\gamma: C \hookrightarrow \text{Set}^{C^{op}}$ preserves and reflects limits.

pf) (i) is just result of Prop 3.4.5.

(ii) Yoneda embedding is fully faithful $\Rightarrow$ reflects limit.

$$\text{Also, } \gamma(\lim_{j \in J} F_j) = C(-, \lim_{j \in J} F_j)$$

$$\text{And for any } X \rightarrow X' \in C, C(X', \lim_{j \in J} F_j) \cong \lim_{j \in J} C(X', F_j)$$

by Thm 3.4.2.

$$C(X, \lim_{j \in J} F_j) \cong \lim_{j \in J} C(X, F_j)$$

This shows. $\gamma(\lim_J F) = C(-, \lim_J F) \cong \lim_J C(-, F)$
 as we've constructed in Prop 3.3.9. $\square$

Dually, for any $F: J \rightarrow C$, $X \in C$, let

$$C(F-, X) := J^{op} \xrightarrow{F} C^{op} \xrightarrow{C(-, X)} \text{Set}.$$

Thm 3.2.6 shows that a limit exists.

Thm 3.2.13 shows construction of limit:

By the very similar argument of pf of

$$\text{thm 3.4.2, } \lim_{J^{op}} C(F-, X) \cong \text{Cone}(F, X).$$

Thm 3.4.7) For any $F: J \rightarrow C$,

$$\exists \text{ natural iso } C(\text{colim}_J F, X) \cong \lim_{J^{op}} C(F-, X)$$

"Representable uni. prop. of colimit"

Ex 3.4.8. $X, Y, Z \in \text{Ob } C$, thm 3.4.7 implies

$$C(X \sqcup Y, Z) \cong C(X, Z) \times C(Y, Z).$$

Ex 3.4.9. Iterated coproduct of $A \in C$: "copowers" or "tensors".

$$\Rightarrow C(\bigsqcup_I A, X) \cong C(A, X)^I.$$

i.e. $h: \bigsqcup_I A \rightarrow X$ is determined by $h_i: A \rightarrow X$.

with coproduct inclusion $\tilde{c}_i: A \rightarrow \bigsqcup_I A$.

Ex 3.4.10.

By rep universal property.

$$\begin{array}{ccc} U & \xrightarrow{f} & V \\ \downarrow g & & \downarrow h \\ W & \xrightarrow{k} & P \end{array}$$

ish. out.

$$\begin{array}{ccc} C(h, X) & & \\ C(p, X) \searrow & \xrightarrow{\quad} & C(u, X) \\ C(k, X) \downarrow & & \downarrow C(f, X) \\ C(w, X) & \xrightarrow{C(g, X)} & C(v, X) \end{array}$$

pull back.

Thm 3.4.11. $\mathcal{C}$: locally small

(i) $\mathcal{C}(-, X)$ carries colimits to limit

(ii) $\gamma : \mathcal{C}^{op} \hookrightarrow \text{Set}^{\mathcal{C}}$ Yoneda embedding
preserves and reflects limits in $\mathcal{C}^{op}$.

pf) Dualize the argument for covariant case.

Thm 3.4.12 (Generalization of 3.2.13)

$F : J \rightarrow \mathcal{C}$ small diagram, suppose $\mathcal{C}$ has coproduct and coequalizer.

$$\coprod_{f \in \text{Mor } J} F(\text{dom } f) \xrightarrow[\mathcal{C}]{\downarrow} \coprod_{j \in \text{obj } J} F_j \twoheadrightarrow \text{colim}_J F$$

In particular, if $\mathcal{C}$ admits coproduct and coequalizer then it is co complete. Dually,

$$\lim_J F \twoheadrightarrow \prod_{j \in \text{obj } J} F_j \xrightarrow[\mathcal{C}]{d} \prod_{f \in \text{Mor } J} F(\text{cod } f)$$

and if $\mathcal{C}$ admits product and equalizer, then $\mathcal{C}$ is complete.

pf) Using Remark 3.2.15, we have below diagram.

$$\begin{array}{ccccc} & & F(\text{dom } f) & & \\ & \swarrow \bar{c}_{\text{dom } f} & \downarrow \bar{r}_f & & \\ \mathcal{C} & \leftarrow \coprod_{j \in \text{obj } J} F_j & \xleftarrow[\mathcal{C}]{d} \coprod_{f \in \text{Mor } J} F(\text{dom } f) & \xrightarrow{F_h} & \\ & \uparrow \bar{c}_{\text{cod } f} & \uparrow \bar{c}_f & & \\ & F(\text{cod } f) & \leftarrow F(\text{dom } f) & & \end{array}$$

where $(d)_f = \bar{c}_{\text{dom } f}$
 $(c)_f = \bar{c}_{\text{cod } f} \circ Ff$

By hypothesis $\mathcal{C}$ (a coequalizer of c and d) exists.

By Thm. 3.4.11 (ii), Yoneda embedding carries coequalizer diagram to an equalizer diagram.
 Thus for any $X \in \mathcal{C}$,

$$\begin{array}{ccc}
 & & C(F(\text{dnt}), X) \\
 & \nearrow C(\tilde{\lambda}_{\text{dnt}}, X) & \uparrow C(\tilde{\lambda}_f, X) \\
 C(Y) \longrightarrow C\left(\coprod_{j \in \text{Ob}} F_j, X\right) & \xrightarrow[C(C, X)]{C(d, X)} & C\left(\coprod_{f \in \text{Ob}} F(\text{dnt}), X\right) \\
 \downarrow C(\tilde{\lambda}_{\text{codf}}, X) & & \downarrow C(\tilde{\lambda}_f, X) \\
 C(F(\text{codf}), X) & \longrightarrow & C(F(\text{dnt}), X)
 \end{array}$$

Applying 3.4.11 (i) carries coequalizer to a limit,
 we have

$$\begin{array}{ccc}
 & & C(F(\text{codf}), X) \\
 & \nearrow \pi_{\text{codf}} & \uparrow \pi_f \\
 C(C, X) \longrightarrow \prod_{j \in \text{Ob}^{\text{op}}} C(F_j, X) & \xrightarrow[\prod]{C} & \prod_{f \in \text{Ob}^{\text{op}}} C(F(\text{codf}), X) \\
 \downarrow \pi_{\text{dnt}} & & \downarrow \pi_f \\
 C(F(\text{dnt}), X) & \xrightarrow[C(F, X)]{} & C(F(\text{codf}), X)
 \end{array}$$

(change from $F(\text{dnt})$ to $F(\text{codf})$ occur by
 indexation with j^{op}) where $c = C(C, X)$
 $d = C(d, X)$

If X is fixed, by Prop 3.3.9,
 with $\text{ev}_X: \text{Set}^{\mathcal{C}} \rightarrow \text{Set}$ the above one is
 an equalizer diagram in Set .

By Th 3.2.13, The diagram's equalizer

is equal to $\lim_{J^{op}} (F_{-}, X)$.

$$\Rightarrow \lim_{J^{op}} (F_{-}, X) \cong C(C, X) \quad \forall X \in C.$$

Now these are assembled into a functor
in $\text{Set}^{ob C}$. Using forgetful functor $\text{Set}^C \rightarrow \text{Set}^{ob C}$
with prop 3.3.9, $C(C, -) \cong \lim_{j \in J^{op}} (F_j, -)$

$$\text{By Thm 3.4.7, } C(C, X) \cong \lim_{j \in J^{op}} (F_j, -) \cong C(\text{colim}_J F, X)$$

$$\forall X \in C. \Rightarrow C \cong \text{colim}_J F.$$



Rmk 3.4.15. (Generalized element)

Direct pt of above Thm is just $\checkmark$ ^{checks} C and d
has the universal property that defines $\text{colim}_J F$.
This can be viewed as the same thing of what we actually
did for getting proof, via 'philosophy of generalized
elements =

Ex) $\forall a \in A$, ^{set} we can identify a as $a: 1 \rightarrow A$.
In this case, $\forall f: A \rightarrow B$, $f(a)$ is identified as $f \circ a$.

In any category C , a "generalized element" of A

is a morphism $a: X \rightarrow A$ with $\text{codom } a = A$

Then, $f: A \rightarrow B$ act on X -shaped element by composition

$\Rightarrow \text{Hom}(-, A): C^{op} \rightarrow \text{Set}$ send object X to the
set of X -shaped generalized element of A .

Since $C \hookrightarrow \text{Set}^{C^{op}}$ Yoneda embedding
 is fully faithful, no information about A
 is lost by considering not A but $\text{Hom}(-, A)$.

For example, if we want to show $f, g: A \Rightarrow B$
 are equal in this philosophy, it suffices to show that

$$fa = ga \quad \forall a: X \rightarrow A, \quad A \in \text{Hom}(X, A)$$

if we want to show a core is limit core,
 then show that induced function in Set is a
 core in Set , etc...

Lemma 3.4.16 Let C : category with terminal
 object 1 . $\Rightarrow$ $A \times B \xrightarrow{\pi_B} B$ pull back
 $\downarrow \pi_A$ defines
 $A \xrightarrow{!} 1$ product $A \times B$
 of $A, B \in C$.

p.f.) Let $A \xleftarrow{\pi'_A} C \xrightarrow{\pi'_B} B$ a diagram.

Then, universal property of pull back gives
 a map $\exists! k: C \rightarrow A \times B$ st.

$$\begin{array}{ccc} \pi'_A & C & \pi'_B \\ \swarrow & \downarrow k & \searrow \\ A & A \times B & B \end{array}$$

left diagram commutes.

$$\begin{array}{ccc} A & \xleftarrow{\pi_A} & A \times B & \xrightarrow{\pi_B} & B \\ & \searrow & & \swarrow & \\ & 1 & & 1 & \end{array}$$

By deleting $!$ and $!$,

this is just universal property
 of product of two elements.

Lemma 3.4.17. $\mathcal{C}$: category with binary products.

$\Rightarrow$ pullback defines an equalizer, i.e.

$$\begin{array}{ccc} E & \xrightarrow{e} & A \\ \downarrow \lrcorner & & \downarrow (f, g) \\ B & \xrightarrow{(1_B, 1_B)} & B \times B \end{array} \rightsquigarrow E \xrightarrow{e} A \xrightarrow[f]{g} B$$

pf) The representable universal property of the product

$$\Rightarrow \text{Hom}(X, B \times B) \cong \text{Hom}(X, B) \times \text{Hom}(X, B)$$

Also, for pullback we have (by Thm 3.4.6 (i))

$$\begin{array}{ccc} \text{Hom}(X, E) & \xrightarrow{e_*} & \text{Hom}(X, A) \\ \downarrow 1_X & & \downarrow (f_*, g_*) \\ \text{Hom}(X, B) & \xrightarrow{(1_*, 1_*)} & \text{Hom}(X, B) \times \text{Hom}(X, B) \end{array}$$

By concrete construction of pullback in Set as we've seen in Ex 3.2.11,

$$\begin{aligned} \text{Hom}(X, E) &\cong \{ (h_1, h_2) \in \text{Hom}(X, A, X \times B) \\ &\quad \text{s.t. } (1_X, 1_X)(h_2) = (f_*, g_*)(h_1) \} \\ &\Leftrightarrow fh_1 = h_2 = gh_1 \end{aligned}$$

Thus, $\text{Hom}(X, E) \xrightarrow{e_*} \text{Hom}(X, A) \xrightarrow[f_*]{g_*} \text{Hom}(X, B)$
 which is equalizer of f_* and g_*
 Thus, $\text{Hom}(X, E) \xrightarrow{e_*} \text{Hom}(X, A) \xrightarrow[f_*]{g_*} \text{Hom}(X, B)$
 is an equalizer diagram. ~~Now Thm 3.4.6 (ii) shows E~~

Now Thm 3.4.6 (ii) shows that E is an equalizer of f, g . $\square$

Ex 3.4.1) done in lecture note.

Ex 3.4.ii) Yoneda embedding preserves limit

via Thm 3.4.2

" reflects limit since

it is fully faithful.

But it does not create limit.

since limit may be not representable;

Ex 3.4.iii) $F: \mathcal{C} \rightarrow \text{Set}$, $\Pi: \int F \rightarrow \mathcal{C}$.

(i) Π strictly creates all limits that $\mathcal{C}$ admit f, g and F preserves.

Let $k': J \rightarrow \int F$ a small diagram, s.t. $k'_i = (k_i, x_i)$

By this notation, we may define $k: \text{ob } J \rightarrow \mathcal{C}$

by $i \mapsto k_i$.

Also, if $i \rightarrow j \in \text{Mor } J$, $k'(i \rightarrow j): (k_i, x_i) \xrightarrow{f} (k_j, x_j)$

and f comes from a morphism $k_i \rightarrow k_j$ s.t.

$Ff(x_i) = x_j$. Hence by defining $k(i \rightarrow j) = f$,

we may define $k: J \rightarrow \mathcal{C}$ a functor.

Now, let $\lambda: (c, x) \Rightarrow k'$ be a limit cone.

Then $\begin{array}{ccc} \lambda_i & \xrightarrow{(c, x)} & \lambda_j \\ \downarrow & \circlearrowleft & \downarrow \\ (k_i, x_i) & \rightarrow & (k_j, x_j) \end{array} \Rightarrow \begin{array}{ccc} \lambda_i & \xrightarrow{c} & \lambda_j \\ \downarrow & \circlearrowleft & \downarrow \\ k_i & \rightarrow & k_j \end{array}$ thus this λ

induces a cone λ in $\mathcal{C}$, which is a unique lift,
 since other cone in $\mathcal{C}$ must have at least
 one distinct λ_i , which induces distinct legs M_i .
 $\square$

Now it suffices to show that λ is a limit cone in $\mathcal{C}$.
 Let $M_i d \Rightarrow k$ be a cone. Then,

$$\begin{array}{ccc} M_i & \xrightarrow{d} & M_j \\ \downarrow \mu_i & & \downarrow \mu_j \\ k_i & \xrightarrow{\quad} & k_j \end{array} \Rightarrow \begin{array}{ccc} (d, \gamma) & & \\ \downarrow \mu_i & \searrow & \downarrow \mu_j \\ (k_i, \mu_i(\gamma)) & \xrightarrow{\quad} & (k_j, \mu_j(\gamma)) \end{array} \quad \forall \gamma \in F_d$$

By universal property of λ , $\exists! f : (d, \gamma) \rightarrow (c, x)$
 Σ left diagram commutes.

$$\begin{array}{ccc} M_i & \xrightarrow{d} & M_j \\ \downarrow \mu_i & \searrow f & \downarrow \mu_j \\ \lambda_i & \xrightarrow{c} & \lambda_j \\ \downarrow \mu_i & & \downarrow \mu_j \\ (k_i, \mu_i(\gamma)) & \xrightarrow{\quad} & (k_j, \mu_j(\gamma)) \end{array}$$

$$\Rightarrow \begin{array}{ccc} M_i & \xrightarrow{d} & M_j \\ \downarrow \mu_i & \searrow f & \downarrow \mu_j \\ \lambda_i & \xrightarrow{c} & \lambda_j \\ \downarrow \mu_i & & \downarrow \mu_j \\ k_i & \xrightarrow{\quad} & k_j \end{array}$$

$\therefore \lambda$ is a limit cone in $\mathcal{C}$.

ii). Π strictly creates connected colimit.

Let λ be a colimit cone. Then

$$\begin{array}{ccc} (k_i, \gamma_i) & \xrightarrow{\quad} & (k_j, \gamma_j) \\ \downarrow \lambda_i & \searrow & \downarrow \lambda_j \\ (c, x) & & \end{array} \Rightarrow \begin{array}{ccc} k_i & \xrightarrow{\quad} & k_j \\ \downarrow & \searrow & \downarrow \\ c & & \end{array}$$

So λ induces a cocone in $\mathcal{C}$.

To see $\lambda: M \subset C$ is a colimit in C ,

let $M: K \Rightarrow d$ a cocone. Then, $\text{Idem } i \rightarrow j \in J$

$$\begin{array}{ccc} k_i & \longrightarrow & k_j \\ \mu_i \searrow & & \swarrow \mu_j \\ & d & \end{array} \quad \Rightarrow \quad \begin{array}{ccc} (k_i, x_i) & \longrightarrow & (k_j, x_j) \\ \mu_i \searrow & & \swarrow \mu_j \\ & (d, \mu_i(x_i)) & \\ & \parallel & \\ & \mu_j(x_j) & \end{array}$$

Notes that this M is unique cone in $\int F$

Since connectedness fixes γ regardless of choice of i .
precisely, if we fix x_i for some $i \in J$, x_j is fixed via

Zig Zag map over the cone $M_i \Rightarrow \gamma$ is fixed.

Now universal property of d shows that.

$$\begin{array}{ccc} (k_i, x_i) & \longrightarrow & (k_j, x_j) \\ \mu_i \searrow & & \swarrow \mu_j \\ & (c, x_c) & \\ \mu_i \searrow & & \swarrow \mu_j \\ & (d, \mu_i(x_i)) & \\ & \parallel & \\ & \mu_j(x_j) & \end{array} \quad \Rightarrow \quad \begin{array}{ccc} k_i & \longrightarrow & k_j \\ \mu_i \searrow & & \swarrow \mu_j \\ & c & \\ \mu_i \searrow & & \swarrow \mu_j \\ & d & \end{array}$$

Thus d is a colimit cone in C .

Ex. 3.4.14) done in lecture note.

3.5. Completes and CoCompletes Categories.

A function is continuous / cocontinuous if it preserves all small limits or colimits.

Prop 3.5.1. Set is Gcomplete.

pf) By thm 3.4.12, it suffices to show that Set has coproducts and coequalizers.

Coproduct = disj. union.

Coequalizer of $A \xrightarrow{f} B \xrightarrow{g}$ is $B / \sim_{f \circ g}$

Prop 3.5.2. Top is co/complete.

pf) $U: \text{Top} \rightarrow \text{Set} \cong \text{Top}(*, -)$

By prop 3.4.5, if Top has any limits, then its underlying set is limit in the category Set.

① Product: $\prod_{\alpha} X_{\alpha}$: Cartesian product with product topology.

(by definition, it is the coarsest topology s.t. $\prod_{\alpha} X_{\alpha} \rightarrow X_{\alpha}$ is cts.)

② Equalizer: the equalizer $E \rightrightarrows X \rightrightarrows Y$ with subspace topology inherited from X i.e. The coarsest topology s.t. $E \hookrightarrow X$ is cts.

$\Rightarrow$ Thm 3.4.12 shows that Top is complete.

Dually, ① Coproduct: disj union as Top space. the finest topology s.t. $X_0 \rightarrow \coprod X_{\alpha}$ is cts.

② Coequalizer of $X \Rightarrow Y \Rightarrow Y/\sim$ with quotient topology.

Hence Thm 3.4.12 shows that Top is co-complete.

Ex 3.5.3. Any limit/colimit can be constructed

via diagram in Prop 3.5.2 by

constructing $\left(\begin{array}{l} \text{the coarsest (limit)} \\ \text{the finest (colimit)} \end{array} \right)$ topology

on the limit/colimit in Set.

ex) $J: \omega^{\text{op}} \rightarrow \text{Top}$

$n \rightarrow S^1$

$\downarrow \hookrightarrow \downarrow P$

$n+1 \rightarrow S^1$

$p: S^1 \rightarrow S^1$ is p th power map.
 $e^{2\pi i x} \mapsto (e^{2\pi i x})^p$

So we have $\cdots \rightarrow S^1 \xrightarrow{p} S^1 \xrightarrow{p} S^1$

$\Rightarrow \lim J = p\text{-adic solenoid.}$

Ex 3.5.4

Space of ordered configurations of n points

$: \text{PConf}_n(X) := \{ (x_1, \dots, x_n) \in X^n : x_i \neq x_j \forall i \neq j \}$
 with subspace topology $\subseteq X^n$.

Let Σ_n : symmetric group.

$\text{Conf}_n(X): \Sigma_n \rightarrow \text{Top.}$

$\hookrightarrow \text{PConf}_n(X)$

$\downarrow \hookrightarrow \downarrow$ permutation $\hookrightarrow g$

$\hookrightarrow$

$\Rightarrow \text{Conf}_n(X)$

$:= \text{colim} (\Sigma_n \xrightarrow{\text{PConf}_n(X)} \text{Top})$

Prop. 3.3.5. $\mathcal{C}$ is complete/co complete

$\Rightarrow \mathcal{C}/\mathcal{C}$ and $\mathcal{C}/\mathcal{C}$ are complete/co complete
 $\forall \mathcal{C} \in \mathcal{C}$.

Pf) Prop 3.3.8. shows that $\mathcal{C}$ complete/co complete

$\Rightarrow \mathcal{C}/\mathcal{C}$ has all limits and all connected limit.

created by $\Pi: \mathcal{C}/\mathcal{C} \rightarrow \mathcal{C}$.

Now $1_{\mathcal{C}}$ is initial in $\mathcal{C}/\mathcal{C}$. Dual statement

of Lem 3.4.16 creates coproduct in $\mathcal{C}/\mathcal{C}$.

(Since pushout is connected colimit.)

And dual of Lem 3.4.17 generates coequalization.

So, by thm 3.4.12, $\mathcal{C}/\mathcal{C}$ is co complete $\square$

In this proof, coproduct of $(\mathcal{C} \rightarrow \mathcal{X}_a)$ is a

map $\mathcal{C} \rightarrow (\mathcal{X}_a \sqcup \mathcal{X}_{a+1}) \sqcup \dots$

defined by $\mathcal{C} \rightarrow \mathcal{X}_a$ s. $\square$

Rmk) $\text{ob}, \text{mor}(\text{Cat} \Rightarrow \text{Set})$ are representable by

Ex 2.1.5 (ix) and (x).

By Thm 3.4.6(i), Both preserve limits when

limit of $K: \mathcal{J} \rightarrow \text{Cat}$ exists.

Furthermore, domain, codomain identities are
natural transformations between ob and mor .

$\Rightarrow$ They are preserved. $\forall \text{ ob}$ and mor .

Also Composition as a $\text{Cat}(\mathcal{J}, -) \rightarrow \text{Cat}(\mathcal{J}, -)$

preserved. This gives a candidate of product and
 Thus, $\textcircled{2}$ Equalizer in Cat or CAT.
 Prop 3.5.6. Cat and CAT are complete.

Pf) $\textcircled{1}$ Product: product of category

$\textcircled{2}$ Equalizer: $E \longrightarrow C \begin{matrix} \xrightarrow{F} \\ \xleftarrow{G} \end{matrix} D$

where E is a subcategory of C

$$\text{s.t. } \text{ob } E = \{ c \in \text{ob } C : Fc = Gc \}$$

$$\text{Mor } E = \{ f \in \text{Mor } C : Ff = Gf \}$$

Apply Thm 3.4.12. $\square$

Ex 3.5.7. Let $F: C \rightarrow \text{Set}$. Pull back in CAT
 of $C \xrightarrow{F} \text{Set} \xleftarrow{U} \text{Set}_*$

is a category Set_* .

Object: $(C, (X, x))$ Set_* . $Fc = X$.

Mor: $(C, (X, x)) \xrightarrow{(f, g)} (C', (X', x'))$

s.t. $\begin{cases} f: C \rightarrow C' \in C \\ g: (X, x) \rightarrow (X', x') \in \text{Set}_* \end{cases}$

$$Ff = Ug$$

Note that $(C, (X, x))$ can be denoted as (C, x)
 with $x \in Fc$.

And morphism can be denoted as $(C, x) \xrightarrow{f} (C', x')$
 s.t. $Ff(x) = x'$.

$\Rightarrow$ This pullback is $\int F$.

(Claim) Cat and $\mathcal{A}l$ are co-complete.

But proof is a little bit delicate;

① Coproduct: Just disjoint union of category

(Notes that any category is C-product of its connected components)

To use

② Lemma 3.4.17, we need pushout, But it is difficult.

Ex 3.5.8: $1 \sqcup 1 \longrightarrow 2$ ~~We~~ we want to have pushout P .

$$\begin{array}{ccc} 1 \sqcup 1 & \longrightarrow & 2 \\ \downarrow & \searrow f & \downarrow \\ 1 & \longrightarrow & P \end{array}$$

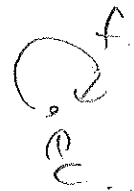
Let $P \rightarrow C$ a functor.

Then $1 \sqcup 1 \longrightarrow 2$ gives a cocone over diagram $1 \leftarrow 1 \sqcup 1 \rightarrow 2$ with vertex C .

$$\begin{array}{ccc} 1 & & 1 \\ \downarrow & & \downarrow \\ 1 & \longrightarrow & P \longrightarrow C \end{array}$$

Thus, $1 \rightarrow P \rightarrow C = 1 \sqcup 1 \rightarrow 1 \rightarrow P \rightarrow C$
 $= 2 \rightarrow P \rightarrow C$

implies that all these maps denote



Thus, P should have an endomorphism without any relation on the composite $f \circ f$.

$\Rightarrow P =$ free category with one object and one non-identity morphism.

By identifying $\text{Mor}(\cdot, \cdot) = \{1_c, f, f^2, \dots\}$

as a monoid, we know that $P = \mathbb{B}\mathbb{N}$

Where $\mathbb{N}$ is ^{monoid of} natural number under addition

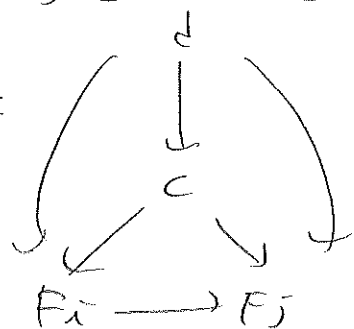
Similarly, $1 \sqcup 1 \rightarrow 1$ since 2 is changed
 $\downarrow \quad \downarrow$ to working iso 1
 $1 \rightarrow \mathbb{B}\mathbb{Z}$ thus f should be
 iso.

Co-completeness proved at chapter 4

Notes that equivalence of categories preserve
 Complete / Co-complete

Prop. 3.5.9. P : poset $\mathbb{B}$ Complete and co-complete
 as a category $\Leftrightarrow P$ is a complete lattice
 i.e. every subset $A \subset P$ has both infimum and
 a supremum.

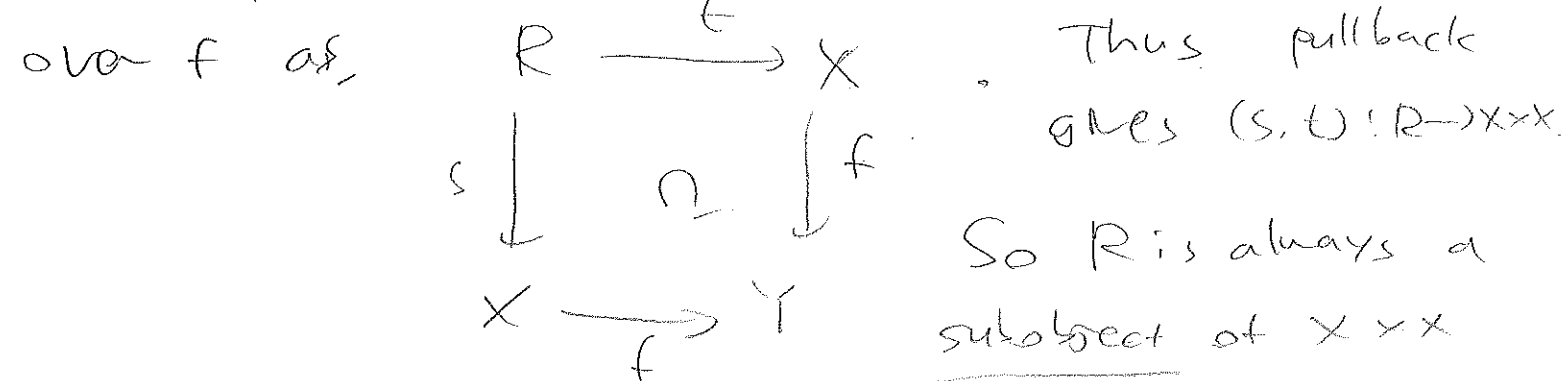
Pt) In any poset or preorder, if limit
 (or colimit) exists, it should be infimum of
 its object (or supremum of its object)
 Since $\downarrow$ implies that any $d \leq f_i \forall i$
 must $d \leq c$, as a term of
 order.



(Similar to colimit)

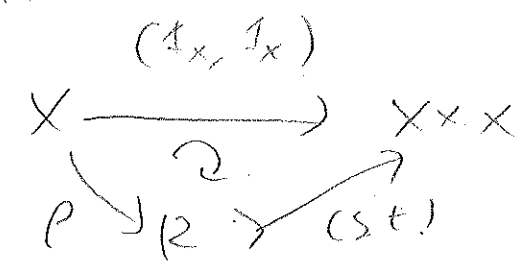
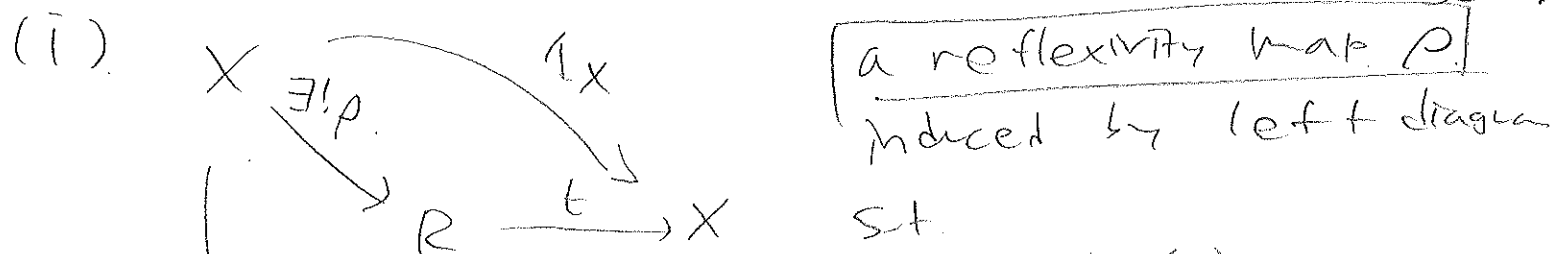
It doesn't matter when J is discrete and

Ex 3.5.10. Let C : category with finite limits
 kernel pair of $f: X \rightarrow Y$ = pullback



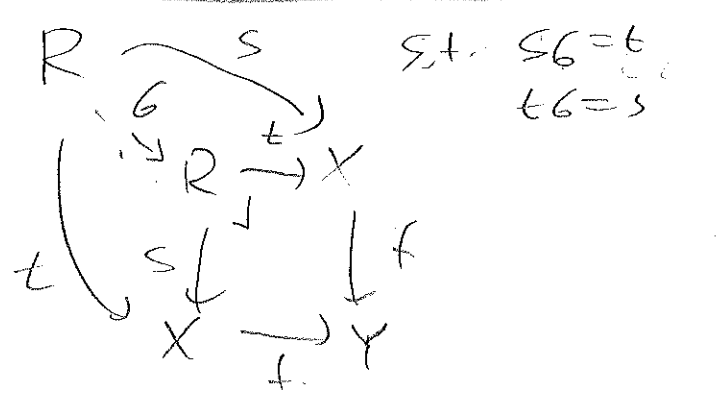
(= monomorphism with codomain $X \times X$)

In Set , $R \subset X \times X$ defines relation on X .
 This is always "equivalence relation" in the below sense.

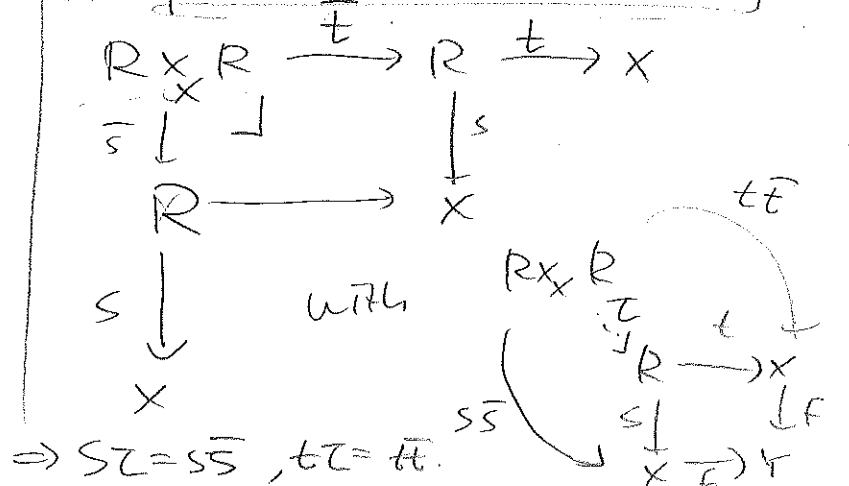


ρ factors $(1_X, 1_X)$

i.e. ρ is a section of
 (ii) Symmetry map $\bar{g}$



(iii) Transitivity map $\bar{t}$



An equivalence relation in a category C with finite limits

$$= (s, t): R \rightrightarrows X \times X$$

with p, q, z in a manner of (i), (ii), (iii).

In this case, Coequalizer of $s, t: R \rightrightarrows X \xrightarrow{c} X/R$
 gives $e: X \rightarrow X/R$ a quotient object.

In Set, X/R is the set of R -equivalence classes of elements of X . Moreover, $\exists!$ factorization

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ e \searrow & \nearrow m & \\ & X/R & \end{array}$$

In a good situation (Grothendieck Topos)
 m is monomorphism. In this case, the factorization is called image factorization and m is

identified as the image X/R as a subobject of Y .

Ex 3.5.1) BG : 1-object category

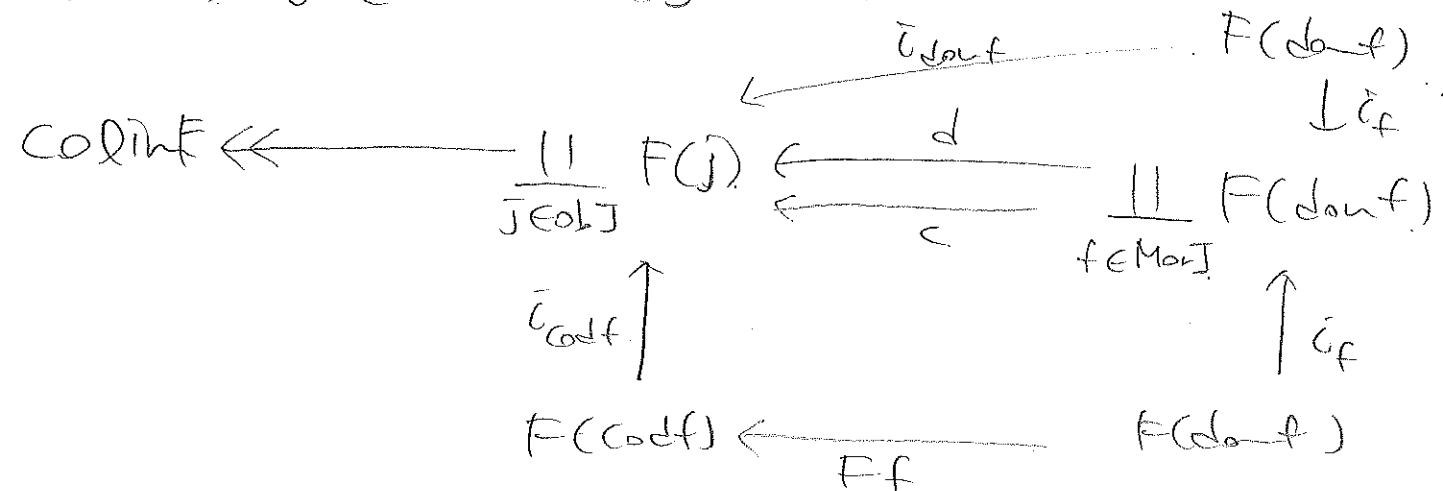
Let $X: BG \rightarrow \text{Set}$. (Colimit of X can be decomposed as

$$\begin{array}{ccccc} & & X & & \\ & \nearrow 1_X & \downarrow i_g & & \\ \text{Colim } X & \xleftarrow{d} & X & \xleftarrow{c} & X \\ & \nwarrow 1_X & \uparrow i_g & & \\ & & X & & \end{array}$$

Thus $d = 1_X$
 $c = g_X$

Hence,
 $\text{Colim } X = X / \sim_{g(X)}$
 $= \{G \cdot x : x \in X\}$
 a set of orbits of G -action.

Ex 3.5.ii) Use the diagram in (i) again;



Then $\text{Colim } F = \{ [\lambda] : \lambda \in \coprod_{j \in \text{Obj } J} F(j) \}$

where $[\lambda] := \{ \lambda' \in \coprod_{j \in \text{Obj } J} F(j) : \exists \lambda'' \in \coprod_{f \in \text{Mor } J} F(\text{dof } f) \text{ s.t. } \lambda' = c(\lambda''), \lambda = d(\lambda'') \text{ or vice versa} \}$

Since d is just inclusion, c is Ff for each component $F(\text{dof } f)$, this means that

$$[\lambda] := \left\{ \lambda' \in \coprod_{j \in \text{Obj } J} F(j) : \exists \lambda'' \in \coprod_{f \in \text{Mor } J} F(\text{dof } f) \text{ s.t. } \lambda' = \lambda'', \lambda = Ff(\lambda'') \text{ or vice versa} \right\}$$

Now we claim that (j, λ) and (i, λ') is connected in SF

$\iff [\lambda] = [\lambda']$.

Suppose LHS. Then $\exists$ a finite length zig-zag morphism, $(j, \lambda) \xrightarrow{f_1} (j_1, \lambda_1) \xleftarrow{f_2} (j_2, \lambda_2) \rightarrow \dots \xleftarrow{f_n} (i, \lambda')$

Say $j_n = i$, $\lambda_n = \lambda'$. Then, $[\lambda] = [\lambda_1]$ since

$Ff_1(\lambda) = \lambda_1$, $[\lambda_1] = [\lambda_2]$ since $Ff_2(\lambda_1) = \lambda_2$...

So $[\lambda_{n-1}] = [\lambda_n]$ since either $Ff_n(\lambda_{n-1}) = \lambda_n$ or $Ff_n(\lambda_n) = \lambda_{n-1}$

is well-defined and true.

Conversely, let $[\lambda] = [\mu]$. Then $\exists \lambda' \in [\lambda]$

s.t. $\lambda = \lambda'$ and $\mu = Ff(\lambda')$ for some $f \in \text{Ob } J$
 or $\lambda = \lambda'$ and $\mu = Ff(\lambda')$

This shows $\bar{\lambda} \rightarrow \bar{\mu} \iff \bar{\mu} \rightarrow \bar{\lambda}$, so $(\bar{\lambda}, \bar{\mu})$
 $(\bar{\lambda}, \bar{\mu})$ is connected in Set .

Thus, if $\lambda: F \Rightarrow \text{Colim } F$ is a colimit cone,

$F\bar{\lambda} \xrightarrow{Ff} Fj$ Thus, if $x \in F\bar{\lambda}$, then
 $\lambda_{\bar{\lambda}} \downarrow_{\text{Colim } F} \lambda_j$ $\lambda_{\bar{\lambda}}(x)$ denotes the orbit of x
 under $Ff, \forall f \in \text{Mor } J$.

Thus colimit cone is just set of functions
 sending element in $F\bar{\lambda}$ to its orbit in Set .

Ex. 3.5.11) Let $A = \{ E \xrightarrow{s} V \xrightarrow{t} V \}$ be a category
 with two parallel nontrivial morphisms.

Then,

① Set^A is the category of directed multigraphs.

of). Let $F: A \rightarrow \text{Set}$ as $E_F \xrightarrow{s} V_F \xrightarrow{t} V_F$
 $G: A \rightarrow \text{Set}$ $E_G \xrightarrow{s} V_G \xrightarrow{t} V_G$

By thinking, $e \in E_F$ is an arrow from $s(e)$ to $t(e)$
 $=$ gives a multigraph. Moreover, if $\exists \alpha: F \Rightarrow G$

Then, $\alpha_e: E_F \rightarrow E_G$

$$E_F \xrightarrow{s} V_F$$

Thus, if $V = s(e)$ with $W = t(e)$,

$$\begin{array}{ccc} \alpha_E \downarrow & \curvearrowright & \downarrow \alpha_G \\ E_G & \xrightarrow{s} & V_G \end{array}$$

then α_G maps $s(e)$ as $s(\alpha_E(e))$

and W as $t(\alpha_E(e))$

so that incidence relation is preserved.

② Set^A is complete/cocomplete.

This is direct application of Thm 3.3.9 and the fact that Set is co/complete.

③ Dir Graph is a full subcategory of Set^A .

p.f.) any morphism between digraph is a morphism as multi-digraph.

Thus $\text{Dir Graph} \xhookrightarrow{\tilde{i}} \text{Set}^A$ is a fully faithful functor.

④ There is a functor $G: \text{Set}^A \rightarrow \text{Dir Graph}$ s.t. $\text{Dir Graph} \xhookrightarrow{\tilde{i}} \text{Set}^A \xrightarrow{G} \text{Dir Graph} = \text{Id}_{\text{Dir Graph}}$

p.f.) Define $G: \text{Set}^A \rightarrow \text{Dir Graph}$ as

$$E \rightrightarrows V \longmapsto E/\sim \rightrightarrows V$$

where $e_1 \sim e_2 \iff s(e_1) = s(e_2), t(e_1) = t(e_2)$

Then, $\sim$ is equivalence relation. And for each $[e] \in E/\sim$

$s([e]) := s(e)$ is well-defined. Moreover, $[e] \neq [e']$,

$t([e]) := t(e)$

then $s(e) \neq s(e')$ or $t(e) \neq t(e')$

Thus, $E/\sim \xrightarrow[\quad]{s, t} U$ is a directed graph.

If $E_1 \Rightarrow U_1$ is a morphism.

$\downarrow (\alpha_E, \alpha_G)$ notes that $\alpha_{E/\sim}([e]) := [\alpha_E(e)]$

$E_2 \Rightarrow U_2$ gives

$$\begin{array}{ccc} E_1/\sim & \Rightarrow & U_1 \\ \alpha_{E_1/\sim} \downarrow & \curvearrowright & \downarrow \alpha_{G_1} \\ E_2/\sim & \Rightarrow & U_2 \end{array}$$

Since if $e, e' \in [e]$, then

$$\begin{aligned} s(\alpha_E(e)) &= \alpha_G(s(e)) = \alpha_G(s(e')) = s(\alpha_E(e')) \\ t(\quad) &= \quad (t(e)) = \quad (t(e')) = t(\quad) \end{aligned}$$

$\Rightarrow [\alpha_E(e)]$ and $[\alpha_E(e')]$ are equivalent.

So, $[e] \xrightarrow{\quad} t([e]) = t(e)$ So $\alpha_{E/\sim}$ is well-defined.

$\downarrow \quad \curvearrowright \quad \downarrow$
 $\alpha_G(s(e)) \quad \alpha_G(t(e))$ Commutes.

$$\begin{array}{ccc} [\alpha_E(e)] & \xrightarrow{\quad} & s(\alpha_E(e)) \\ & \xrightarrow{\quad} & t(\alpha_E(e)) \end{array}$$

Hence $\text{Set}^A \xrightarrow{G} \text{DirGraph}$ is a functor.

Notes that $\text{DirGraph} \hookrightarrow \text{Set}^A \rightarrow \text{DirGraph}$
 $= 1_{\text{DirGraph}}$

5) DM Graph has colimit if Set^A has a corresponding colimit.

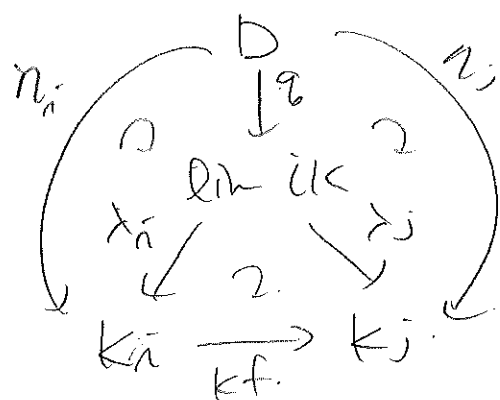
pf) Let $K: J \rightarrow \text{DM Graph}$.

Then, $\bar{K}: J \rightarrow \text{DM Graph} \rightarrow \text{Set}^A$
 is also a small diagram.

Since Set^A is complete / co complete, so
 $\lim \bar{K}$ and $\text{colim } \bar{K}$ exists.

Now we claim that $G(\lim \bar{K}) \cong \lim(G\bar{K})$
 $= \lim K$.

Let $\lambda: \lim \bar{K} \Rightarrow \bar{K}$ be a limit cone. Then,
 for any cone $\eta: D \Rightarrow \bar{K}$, for $i \rightarrow j \in J$, we have



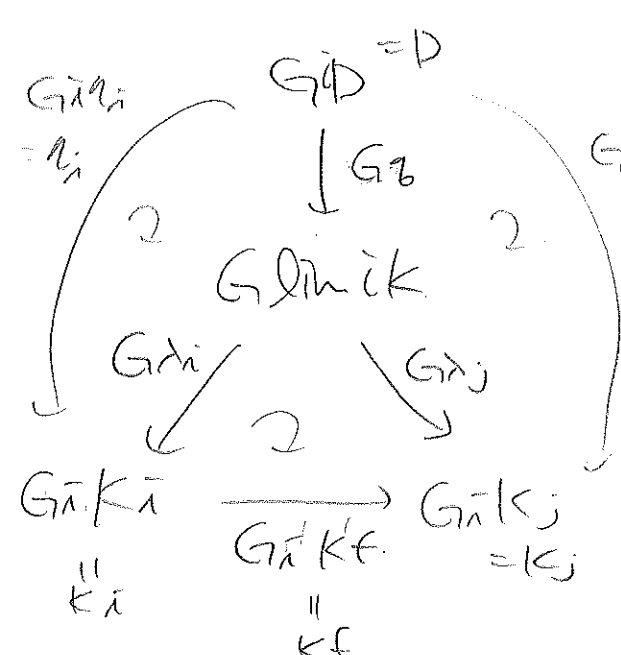
$\exists ! \eta: D \rightarrow \lim \bar{K}$ s.t.
 Left diagram commutes.

We claim that G preserves
 limit cone.

Let $\eta: D \Rightarrow G\bar{K} = K$ be a cone in DM Graph.
 Then, $\bar{\eta}: D \Rightarrow \bar{K}$ is a cone in Set^A

Since the functor $\bar{\cdot}$ does not change component
 of natural transformations $\eta_i \forall i \in J$, as a
 sub category. Thus, $\exists ! \eta$ s.t. above diagram holds.

By applying G ,



left diagram
commutes from
functoriality of G .

Moreover, such t is unique!

If there is another!

$$g': D \rightarrow G \lim_{\leftarrow} \bar{a}_i$$

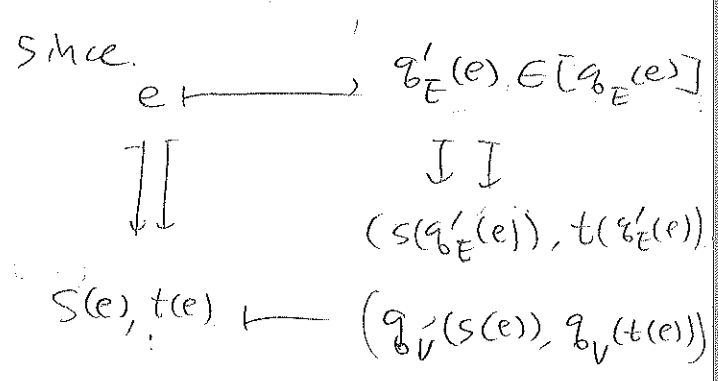
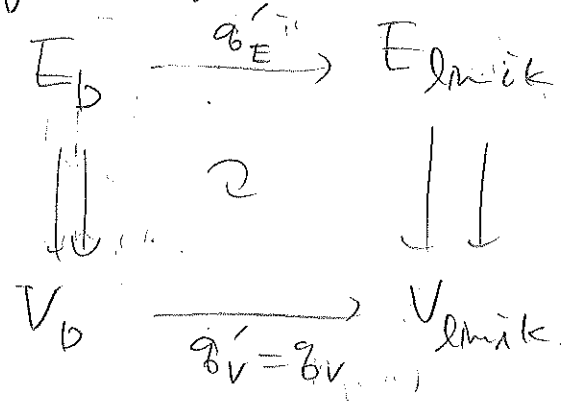
s.t. left diagram commutes.

then, define $g': D \rightarrow \lim_{\leftarrow} \bar{a}_i$ as a section;

i.e., $g'_E(x) = e'$ for some fixed representation $e' \in [t(x)]_E$.

$$g'_V(x) = g_V(x).$$

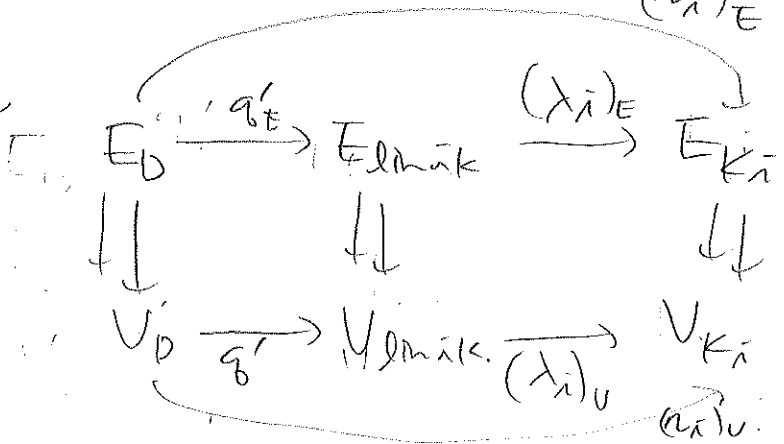
Then,



but $g'_E(e) \in [g_E(e)] \Rightarrow s(g'_E(e)) = s(g_E(e)) = g_V(s(e))$
 $t(g'_E(e)) = t(g_E(e)) = g_V(t(e))$

gives commutative property. Hence g' is well-defined natural transformation.

Now,



it suffices to show
that left diagram
commutes.

For $e \in E_P$,

$$e \longmapsto q'_E(e) \in [q_E(e)] \longrightarrow (\lambda_{\bar{n}})_E(q'_E(e))$$

Then, $s(q'_E(e)) \equiv s(q_E(e))$, $t(q'_E(e)) = t(q_E(e))$

implies that $(\lambda_{\bar{n}})_E(q_E(e)) = (\lambda_{\bar{n}})_E(q'_E(e))$

Since $K_{\bar{n}}$ is not a multigraph, so for any two ordered vertices, $\exists$ at most 1 arrow.

By universal property, $(\lambda_{\bar{n}})_E(q_E(e)) = (\kappa_{\bar{n}})_E(e)$

$$\Rightarrow q'_E \circ (\lambda_{\bar{n}})_E = (\kappa_{\bar{n}})_E$$

And by definition, $q'_U \circ (\lambda_{\bar{n}})_U = q_U \circ (\lambda_{\bar{n}})_U = (\kappa_{\bar{n}})_U$

$$\Rightarrow q' \circ \lambda = \kappa$$

Then, universal property shows that $q' = q$.

Thus, $q = Gq' = Gq$. $\therefore Gq$ is unique.

This shows that $G\lambda: G\lim \bar{K} \Rightarrow G\bar{K} = K$

is a limit cone in $\mathbf{DirGraph}$.

$\Rightarrow G$ preserves limit cone.

$$\text{Thus, } G(\lim \bar{K}) \cong \lim (G\bar{K}) = \lim K$$

$\Rightarrow \mathbf{DirGraph}$ admits limits if it is admittedly $\mathbf{Set}^A$

$\Rightarrow$ " is complete since $\mathbf{Set}^A$ is complete.

By the duality, doing the all argument dually,

we can get $\mathbf{DirGraph}$ admits colimit if it is in $\mathbf{Set}^A$

$\Rightarrow \mathbf{DirGraph}$ is cocomplete.

□

Ex. 3.5. N) Since we proved Cat is complete (co)category

using its set structure so we can make

J^Δ concretely: Define $i_0: J \rightarrow J \times 2$ by Ob_J
 $J \mapsto (J, 0)$

$\text{Ob} J \xrightarrow{!} \text{Ob} 1$

$i_0 \downarrow \quad \quad \downarrow$ is a pushout in Set

$\text{Ob} J \times 2 \longrightarrow \text{Ob} J^\Delta \Rightarrow \text{Ob} J^\Delta = \text{Ob} J \times 2 \amalg 1 / \sim$
 where $(J, 0) \sim 1 \quad \forall J \in J$

$\Rightarrow \text{Ob} J^\Delta \cong \{(J, 1) : J \in \text{Ob} J\} \cup \{1\}$

Now by Mor forgetful functor,

$\text{Mor} J \xrightarrow{!} \text{Mor} 1$ induces that

$i_0 \downarrow \quad \quad \downarrow$ $\text{Mor} J^\Delta = \text{Mor}(J \times 2) / \sim$

$\text{Mor}(J \times 2) \longrightarrow \text{Mor} J^\Delta$ where $(f, 1_0) = 1_1$

Thus $\text{Mor} J^\Delta \cong \{1 \rightarrow (J, 1) : \forall J \in \text{Ob} J\}$
 $\cup \{(f, 0 \rightarrow 1) : f \in \text{Mor} J\}$
 $\cup \{(f, 1_1) : f \in \text{Mor} J\}$

Now, we claim that J^Δ is a cone over J with
 summit 1 , by identifying $\{(J, 1)\}$ as image of J .

To see that, let $f: i \rightarrow j \in \text{Hom } J$. Then its image in J^Δ is $(i, 1) \xrightarrow{(f, 1_2)} (j, 1)$. It suffices to show that

$$\begin{array}{ccc} & 1_1 & \\ a \swarrow & & \searrow b \\ (i, 1) & \xrightarrow{(f, 1_2)} & (j, 1) \end{array} \quad \text{commutes.} \quad \text{where}$$

a is induced by $(i, 0) \rightarrow (i, 1)$
 b " $(j, 0) \rightarrow (j, 1)$.

Since $(i, 0) \xrightarrow{(1_i, 0 \rightarrow 1)} (i, 1)$ in $J \times \mathbb{Z}$,

$$\begin{array}{ccc} (f, 1_2) \downarrow & \curvearrowright & \downarrow (f, 1_2) \\ (j, 0) & \xrightarrow{(1_j, 0 \rightarrow 1)} & (j, 1) \end{array} \quad \text{we have.}$$

$$\begin{aligned} (f, 1_2) \circ a &= (f, 1_2) \circ (1_i, 0 \rightarrow 1) = (1_j, 0 \rightarrow 1) \circ (f, 1_0) \\ &= b \circ 1_1 = b. \end{aligned} \quad \square$$

Ex. 3.5.4) Let $k: J \rightarrow (N, \leq)$ be a small diagram. If $\lambda \vdash \text{Im } k \Rightarrow k$ and $\text{Im } k$ exists,

then $\forall i \rightarrow j \in J$, for any $\alpha \in \mathbb{Z} \mid \alpha \Rightarrow k$

$$\begin{array}{ccc} & X & \\ \lambda_i \swarrow & \downarrow & \searrow \lambda_j \\ & \text{Im } k & \\ \lambda_i \swarrow & & \searrow \lambda_j \\ & F_i & \\ F_i & \xrightarrow{\quad} & F_j \end{array}$$

From this, notes that

① If $X \Rightarrow k$ is a cone,

then $X \mid F_i \quad \forall i \in J$.

② If $\text{Im } k$ exists, then $X \mid \text{Im } k$.

Thus, $\text{Im } k$ is $\text{g.c.d.}(F_i; i \in \text{Ob } J)$.

Conversely, Suppose $Y = \text{g.c.d.}(F_i; i \in \text{Ob } J)$

Then, $Y \mid F_i \quad \forall i \Rightarrow Y \xrightarrow{\lambda_i} F_i$ exists.

Since $(N, \leq)$ is a poset as a category
 these $\{\lambda_i : i \in \text{Obj}\}$ form a core, since every
 division commutes?

If $X \Rightarrow F$ is a core, then $X(F_i \vee i \Rightarrow X) \mid Y$
 since Y is gcd thus the above limit diagram
 holds $\Rightarrow Y$ is limit.

In case of colimit, every arrows are reversed,
 so, by duality argument, colim k is l.c.m. $(F_i, i \in J)$

Ex. 3.5. vi)

$F_{\text{non}}^{\text{op}} : \text{Obj} \text{ (finite set)}$

Fix $X \in \text{Top}$.

Morphisms $\rightarrow$ injections.

$F_{\text{non}}^{\text{op}} \rightarrow \text{Top}$.

$[n] \mapsto \text{PConf}_n(X) : \{(x_1, \dots, x_n) \in X^n : x_i \neq x_j, \forall i \neq j\}$
 $(x_{f^{-1}(1)}, \dots, x_{f^{-1}(n)})$ partial function

$f \downarrow \mapsto \uparrow$
 $[m] \mapsto \text{PConf}_m(X) : (x_1, \dots, x_m)$
 $f^{-1} = [m] \rightarrow [n]$
 $\downarrow \mapsto f^{-1}(1)$

this is well-defined since f^{-1} is just a
 projection map. (with permutation of coordinate.)

However, $[n] \mapsto \text{Conf}_n(X)$ cannot be a functor!

Since $[(x_1, \dots, x_n)] \mapsto [(x_{f^{-1}(1)}, \dots, x_{f^{-1}(n)})]$ is

not well-defined. For example $f: \{1, 2\} \hookrightarrow \{1, 2, 3\}$ with

canonical injection cannot determine $[(4, 5, 6)] \rightarrow [(4, 5)]$

Since $[(4, 5, 6)] = [(5, 6, 4)] = [(4, 6, 5)] \dots$
 $\searrow \rightarrow [(5, 6)]$
 $\searrow \rightarrow [(4, 6)]$

Ex 3.5. vii) a morphism.

Fiber Space : $p: E \rightarrow B \in \text{Top}$

Map between fiber spaces: a commutative square, i.e., $(p: E \rightarrow B) \rightarrow (p': E' \rightarrow B')$

is $(g: E \rightarrow E', f: B \rightarrow B')$ s.t.

$$\begin{array}{ccc} E & \xrightarrow{g} & E' \\ p \downarrow & \wr & \downarrow p' \\ B & \xrightarrow{f} & B' \end{array}$$

Thus, Category of fiber space is $\subseteq$ diagram category Top^2 .

And one of its non-full subcategory is

Top/B , a slice category over B .

i.e. objects are $p: E \rightarrow B$ fiber space over B

and morphisms are $\begin{array}{ccc} & B & \\ E \nearrow & \wr & \nwarrow E' \\ & g & \end{array}$ identified as $(g, 1_B)$ in Top^2 .

1) In a morphism,

$$\begin{array}{ccc} E' & \xrightarrow{g} & E \\ p' \downarrow & & \downarrow p \\ B' & \xrightarrow{f} & B \end{array}$$

let $p^{-1}(b)$ a fiber of $b \in B$
 Then, $g^{-1}(p^{-1}(b)) = (p')^{-1}(f^{-1}(b))$
 by commutative diagram

$$\Rightarrow g^{-1}: p^{-1}(b) \rightarrow f^{-1}(b)$$

is a canonical map.

2) W.T.

fiber of product of fiber space = product of fibers.

First of all we need to figure out product of fiber spaces in Top^2

$$\text{Let } p: E \rightarrow B, \quad p': E' \rightarrow B'$$

Then product is $q: E'' \rightarrow B''$ with projection,

$$\pi_p: q \mapsto p, \quad \pi_{p'}: q \mapsto p' \text{ s.t.}$$

$$\begin{array}{ccc} & s & \\ \swarrow & \downarrow & \searrow \\ p & q & p' \end{array}$$

$$\forall s \in \text{Top}^2$$

$$\text{with } s \rightarrow p, s \rightarrow p'$$

Likewise. Ex 3.5 iv, we can generate q via Set.

$$\begin{array}{ccccc} & E_s & & & \\ & \downarrow s & & & \\ & B_s & & & \\ \swarrow & & \searrow & & \\ E & \xleftarrow{(\pi_p)_E} & E'' & \xrightarrow{(\pi_{p'})_E} & E' \\ p \downarrow & & \downarrow \pi_p & & \downarrow \pi_{p'} \\ B & \xleftarrow{(\pi_p)_B} & B'' & \xrightarrow{(\pi_{p'})_{B'}} & B' \end{array}$$

From left picture

which holds a set.

$$\text{we know } E'' = E \times E'$$

$$B'' = B \times B'$$

$$\text{Then } q = (p, p')$$

Thus, the unique map $s \mapsto q$ is

$$(E_s \rightarrow E'', B_s \rightarrow B'')$$

induced by E'' and B'' as a product.

Then it is just trivial to see that above diagram commutes.

Now, take a fiber in q over $(b, b') \in B''$.

Then it is $p^{-1}(b) \times (p')^{-1}(b')$ which is product of fiber

(iii) Let $\pi: B \times F \rightarrow B$. The fiber of π over $b \in B$ is $\pi^{-1}(b) = \{b\} \times F$.

As a subspace $\{b\} \times F$ is homeomorphic to F .

Since in finite product, box topology = product topology.

So any basic open set in $B \times F$ is $O_B \times O_F$ form,

thus $(\{b\} \times F) \cap (O_B \times O_F) = \{b\} \times O_F$ is basic open set

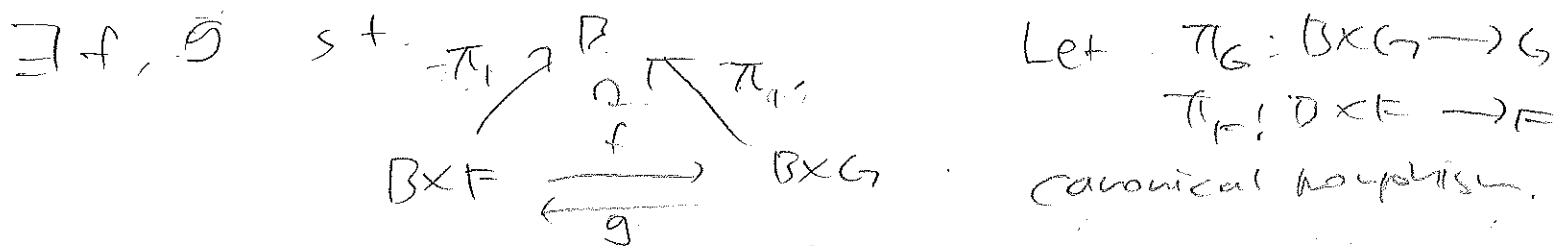
in $\{b\} \times F$, thus $f: \{b\} \times F \rightarrow F$ induces homeo-

$f^{-1}(O_F) = \{b\} \times O_F$ $\forall$ all open set O_F in F .

$\Rightarrow \{b\} \times F \cong F$ in Top.

(iv) Let $\pi_F: B \times F \rightarrow B$, $\pi_G: B \times G \rightarrow B$ be

two trivial fiber space. Then, if H is iso.



Then, $f(b, f) = (b, \gamma)$ for some $\gamma \in G$
 $g(b, g) = (b, \gamma')$ $\gamma' \in F$

$\Rightarrow f' = \pi_G \circ f$, $g' = \pi_F \circ g$ are cts.

Conversely, we claim that given $f': B \times F \rightarrow G$

and $g': B \times G \rightarrow F$ s.t. $g'(b, f'(b, x)) = x$

and $f'(b, g'(b, y)) = y$ this induces a map

f and g satisfying the above diagram.

Notes that

$$f := B \times F \xrightarrow{f'} D \times G$$

$$(b, x) \longmapsto (b, f'(b, x))$$

$$g := B \times G \longrightarrow B \times F$$

$$(b, y) \longmapsto (b, g'(b, y))$$

are cts; notes that for open set $O_B \times O_G \subset B \times G$,

$$f^{-1}(O_B \times O_G) = (f')^{-1}(O_G) \cap (O_B \times F)$$

Since if $f(b, x) \in O_B \times O_G$, then

$$f'(b, x) \in O_G \text{ and } b \in O_B$$

$$\Rightarrow (b, x) \in f^{-1}(O_G) \cap (O_B \times F)$$

Conversely, if $(b, x) \in (f')^{-1}(O_G) \cap (O_B \times F)$

then $f(b, x) = (b, f'(b, x)) \in O_B \times O_G$.

So the equality holds.

Moreover, $g(f(b, x)) = g(b, f'(b, x)) = (b, g'(b, f'(b, x)))$

$$= (b, x)$$

and $f(g(b, y)) = f(b, g'(b, y)) = (b, f'(b, g'(b, y))) = (b, y)$

$\Rightarrow f, g$ satisfy the diagram

(v) Let $p: E \rightarrow B$ a fiber space.

Then a section $s: B \rightarrow E$ is a cts map

$$\text{s.t. } p \circ s = 1_B.$$

Now define $\text{Sect}: \text{Top}/B \longrightarrow \text{Set}$

$$\begin{array}{ccc} p: E \rightarrow B & \longmapsto & \{ \text{sections of } p \} \\ f: E \rightarrow E' \downarrow & \longmapsto & \downarrow f_* \text{ post comp} \\ p': E' \rightarrow B' & \longmapsto & \{ \text{sections of } p' \} \end{array}$$

Then ① $\forall s: B \rightarrow E$ a fiber of $p: E \rightarrow B$,

$f \circ s: B \rightarrow E \rightarrow E'$ is a fiber of $p': E' \rightarrow B'$

$$\text{Since } p' \circ f \circ s(x) = p \circ s(x) = x, \forall x \in B$$

Thus, as a set map, f_* is well-defined.

② Functoriality ultimately holds by argument in ①

$\Rightarrow$ Sect is well-defined functor.

(vi). Notes that

$$\begin{array}{ccc} \text{Top}_{pb}^2 : \text{Obj} : p: E \rightarrow B & \text{any fiber space} & \\ \text{Mor} : (p: E \rightarrow B) & \xrightarrow{(f, g)} & (p': E' \rightarrow B') \end{array}$$

If the commutative square is actually a pull back, i.e.

$$\begin{array}{ccc} E & \xrightarrow{f} & E' \\ p \downarrow & \lrcorner & \downarrow p' \\ B & \xrightarrow{g} & B' \end{array}$$

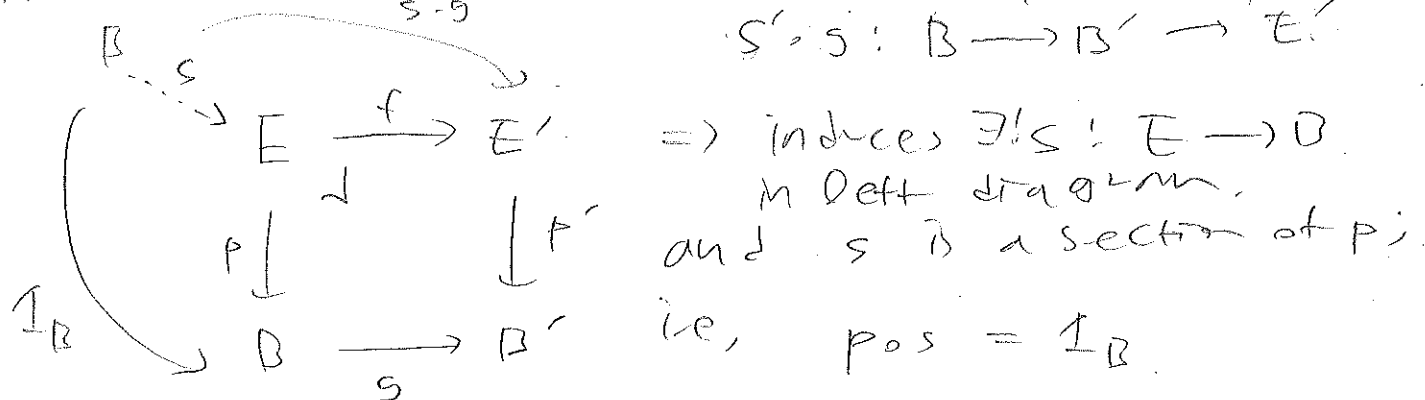
Now, $\text{Sect} : (\text{Topb})^{\text{ob}} \longrightarrow \text{Set}$

$p: E \longrightarrow B \longmapsto \text{Section of } p$

$(f, g) \downarrow \quad \quad \quad \uparrow \emptyset$

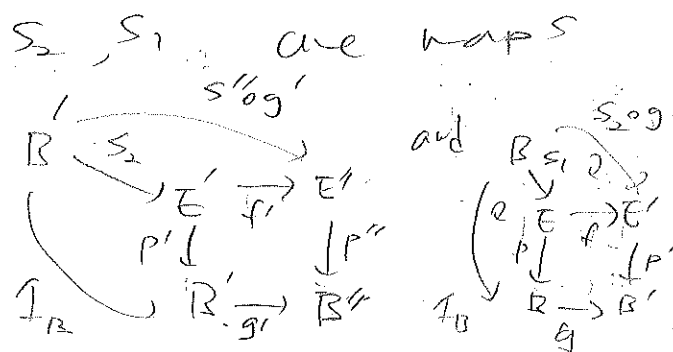
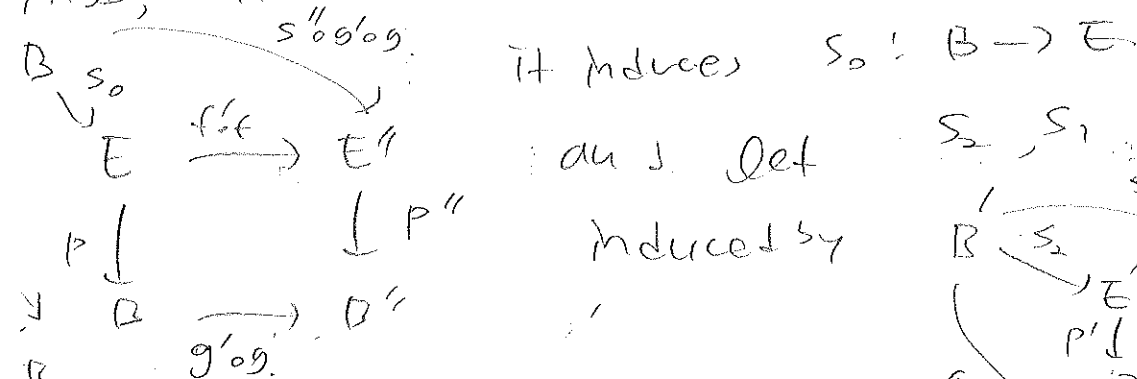
$p': E' \longrightarrow B' \longmapsto \text{Section of } p'$

from the fact that for any $s': B' \longrightarrow E'$ of Section of p' ,
 $s' \circ s: B \longrightarrow B' \longrightarrow E'$



thus $\emptyset$ is well-defined as the unique map induced by id_B and $(-) \circ g$ on the pullback of f and p .

Also, if $B \xrightarrow{(f, g)} p' \xrightarrow{(f', g')} p''$, then,



Now notes that

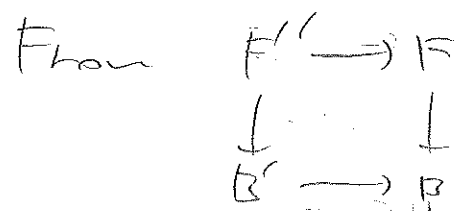
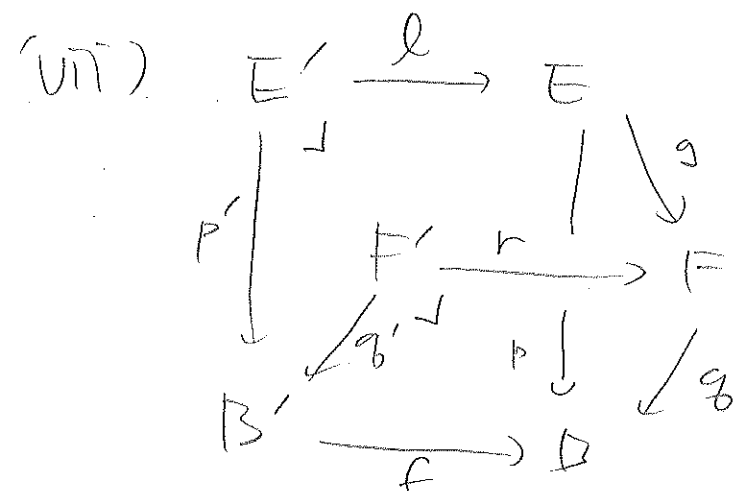
$$s'' \circ g' = f' \circ s_1, \quad s_2 \circ g = f \circ s_1$$

$$\Rightarrow s'' \circ s' \circ g = f' \circ s_2 \circ g = (f' \circ f) \circ s_1 \quad \text{and} \quad p \circ s_0 = \text{id}_B = p \circ s_1$$

thus universal property of pull back $E \rightarrow E''$
 shows that $S_0 = S_2'$ $\begin{array}{ccc} \downarrow & & \downarrow \\ B & \rightarrow & B'' \end{array}$

This implies functoriality of Sect .

($I_P \mapsto I_{\text{Sect}(P)}$ is trivial)



with $gl: E' \rightarrow F$
 and $p': E' \rightarrow B'$

Let unique map $g': E' \rightarrow F'$

st. $g' \circ g' = p'$ and $rg' = lg$, from the universal property of pull back.

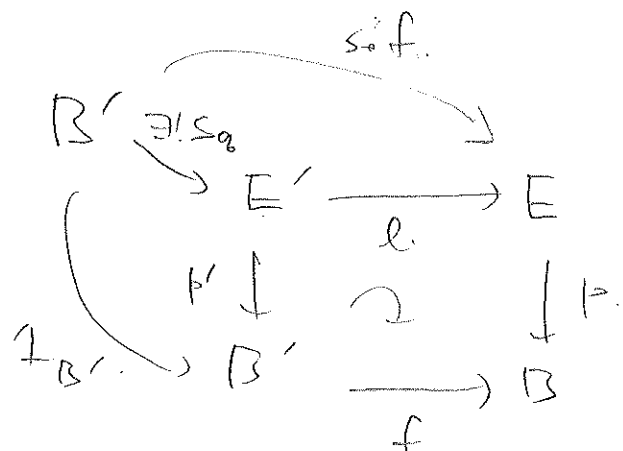
So, g and g' can be identified as a morphism in $\text{Top}/_B$ and $\text{Top}/_{B'}$.

Also, (r, f) and (l, f) are morphisms in Top^2_{pb} . Now it suffices to show that.

$$\begin{array}{ccc}
 \text{Sect}(p') & \xrightarrow{\text{Sect}_{pb}(l, f)} & \text{Sect}(p') \\
 \downarrow g_* & \curvearrowright & \downarrow g'_* \\
 \text{Sect}(g) & \xrightarrow{\text{Sect}_{pb}(r, f)} & \text{Sect}(g')
 \end{array}$$

Let $s \in \text{Sect}(A)$. Then, $\text{Seq}_{pl}(l, f)(C_{\Phi}^s) = s_q$.

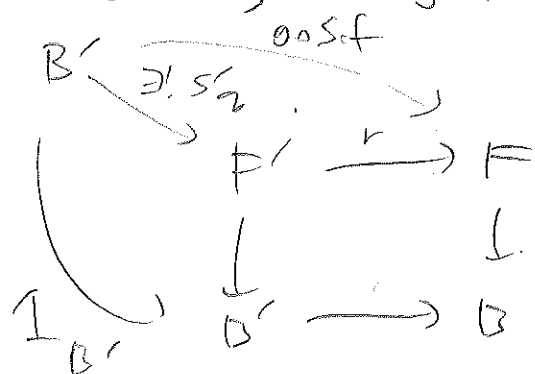
in the left diagram



So, $g' \circ s_q : B' \rightarrow F'$

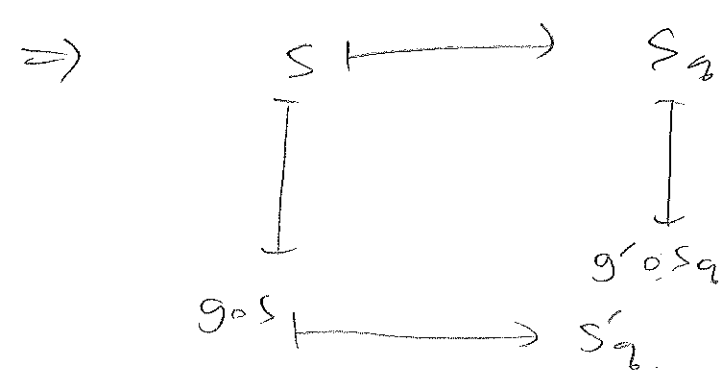
is in $\text{Sect}(q')$

Conversely, $g \circ s \in \text{Sect}(q)$, thus

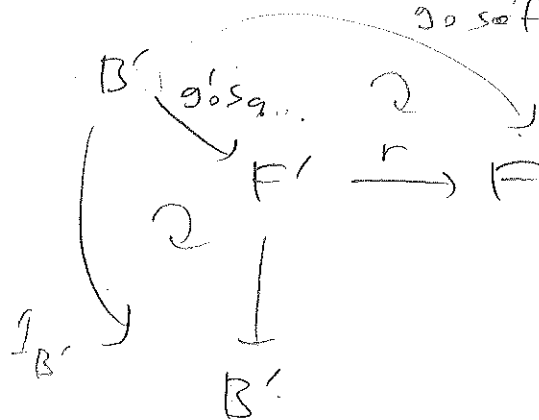


$s'_q \in \text{Sect}(q')$

by left diagram

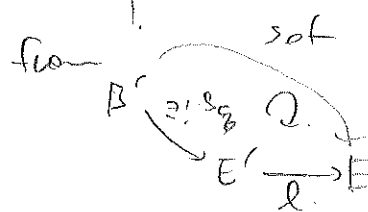
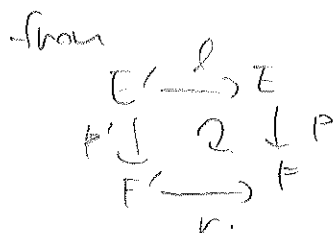


Now notes that $g \circ s \circ f$



Since $p' \circ g' \circ s_q = \text{id}_{B'}$ because $g' \circ s_q$ is a section.

And $r \circ g' \circ s_q = g \circ l \circ s_q = g \circ s \circ f$. Thus



Universal property says that $s'_q = g' \circ s_q$.

3.6. Functoriality of limit and colimits

Functoriality $\neq$ Canonicity

Lots of isomorphic objects can define limit.

Prop 3.6.1. Suppose $\mathcal{C}$ has all J -shaped limits.

Then choice of limit for each diagram

defines $\lim_J : \mathcal{C}^J \rightarrow \mathcal{C}$ as a functor.

2.1k) $\lim_J$ is not canonically defined.

p.f) By choice of limit and limit cone, we define $\lim_J$.

on $\text{Ob } \mathcal{C}$. To define $\lim_J$ on morphisms,

Suppose $F, G \in \mathcal{C}^J$ and $\lambda : \lim_J F \Rightarrow F$ are
chosen limit cone, $\eta : \lim_J G \Rightarrow G$.

Then if $\alpha : F \Rightarrow G \in \text{Hom}(F, G)$, then $\forall i \rightarrow j \in J$,

$$\begin{array}{ccc} \lambda_i \lim_J F & & \\ \downarrow \alpha_i & \searrow \lambda_j & \\ F_i & \longrightarrow & F_j \\ \downarrow \alpha_i & \searrow \alpha_j & \\ G_i & \longrightarrow & G_j \end{array} \Rightarrow \alpha \circ \lambda \text{ is a cone over } G,$$

thus $\exists ! k : \lim_J F \rightarrow \lim_J G$ st. $\alpha \circ \lambda = \eta \circ k$.

Thus, define $\lim_J(\alpha) = k$

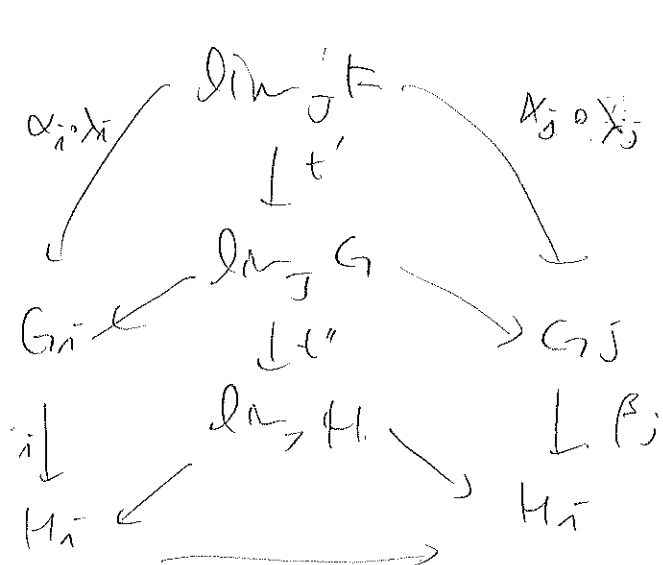
The uniqueness of k implies functoriality.

Notes that $1_F : F \Rightarrow F$ is sent to $1_{\lim_J F}$.

And for $F \xrightarrow{\alpha} G \xrightarrow{\beta} H$ we get to

$$\begin{array}{ccc} \beta_i \alpha_i & \searrow & \beta_j \alpha_j \\ \downarrow & \searrow & \downarrow \\ \lim_J F & \xrightarrow{\lim_J \alpha} & \lim_J G \\ \downarrow & \searrow & \downarrow \\ H_i & \longrightarrow & H_j \end{array} \text{ by universal property}$$

And if $t': \lim_j F \rightarrow \lim_j G$, $t'': \lim_j G \rightarrow \lim_j H$ derived from universal property of limit,



$t'' \circ t'$ gives commutative (left) diagram. $\Rightarrow t'' \circ t' = t$.

So composition is preserved. $\square$

Ex 3.6.2 Path: $\text{Top} \rightarrow \text{Set} \cong \text{Top}(I, -)$
 $X \mapsto \text{Top}(I, X)$ with $I = [0, 1]$

Let $0, 1, * \in I$ endpoints. Define

$$P: \text{Top} \rightarrow \text{Set} \xrightarrow{\cong} \text{Set}$$

$$X \mapsto \left(\text{Top}(I, X) \xrightleftharpoons[\text{ev}_1]{\text{ev}_0} \text{Top}(*, X) \right)$$

$$\parallel \qquad \qquad \qquad \parallel$$

$$\text{Path}(X) \qquad \qquad \qquad P(X)$$

This is a functor. And its coequalizer is

$$\text{Path}(X) \xrightleftharpoons[\text{ev}_1]{\text{ev}_0} P(X) \longrightarrow \pi_0(X)$$

the set of path components of X , since $\pi_0(X) =$ quotient of $P(X)$ by equivalence relation $\sim$.
 $x \sim y$ if x, y are path connected.

$\Rightarrow \pi_0: \text{Top} \xrightarrow{P} \text{Set} \xrightarrow{\cong} \text{Set}$ is well-defined by Prop 3.6.1. (We just choose specific coequalizer $\pi_0(X)$)

(Notes that any set with the same cardinality with $\mathcal{T}_0(Y)$ can be chosen. But we choose it, and it defines $\mathcal{T}_0$ as a functor.

(Other choice gives another functor.)

Rmk) Prop 3.6.1 shows any natural transformation $\eta: C \rightarrow J$ are sent to morphism between limits (or colimits) regardless of complete/cocompleteness of $\mathcal{C}$ domain category.

Corollary 3.6.3) A natural iso between diagram induces a naturally defined iso between their limits or colimits whenever they exist.

pf) $\eta_{\mathcal{I}}$. Prompt of 3.6.1 can be derived from the full subcategory of $\mathcal{C}^J$ spanned by functors having limits in $\mathcal{C}$. Now apply Lem 1.3.8. (all functor preserve iso.)

Rmk) Cor 3.6.3 is actually notation of Eilenberg and MacLane.

Ex 3.6.4) G ! gr. $X, Y: BG \Rightarrow \text{Set}$

X and Y are objectwise isomorphic $\Rightarrow X \cong Y$ as a set.

" naturally " $\Rightarrow$ There exists

an equivariant bijection $t: X \xrightarrow{\cong} Y$ st.

$$\begin{array}{ccc} X & \xrightarrow{t} & Y \\ g \downarrow & \cong & \downarrow g \\ X & \xrightarrow{t} & Y \end{array}$$

Let $\{a, b\}$ has the nontrivial $\mathbb{Z}/2$ action by sending $a \mapsto b, b \mapsto a$.

and $\{c, d\}$ has $\mathbb{Z}/2$ action by sending $c \mapsto d, d \mapsto c$.

Then $\{a, b\} : B(\mathbb{Z}/2) \rightarrow \text{Set}$ and $\{c, d\} : B(\mathbb{Z}/2) \rightarrow \text{Set}$ are objects in $\mathcal{B}$ but not natural isos.

Since there is no function $f : \{a, b\} \rightarrow \{c, d\}$ satisfying natural iso.

(p1) If $f(b) = c$, then $\begin{array}{ccc} b & \xrightarrow{\quad} & c \\ \downarrow & \times & \downarrow \\ b & \xrightarrow{\quad} & d \end{array}$

If $f(b) = d$, then $\begin{array}{ccc} b & \xrightarrow{\quad} & d \\ \downarrow & \times & \downarrow \\ b & \xrightarrow{\quad} & c \end{array}$

On invariance of limit, $\lim B \hookrightarrow X$ is X^G .
 $\lim \{a, b\} = \{a\}$ $\text{colim } B \hookrightarrow X$ is G -orbit sets.
 $\text{colim } \{a, b\} = \{a, b\}$
 $\lim \{c, d\} = \emptyset$ $\text{colim } \{c, d\} = \{*\}$

Ex. 3.6.5. X : finite set.

$\text{Sym}(X)$: all permutations of X
 $\text{Ord}(X)$: all total ordering on X . $\Rightarrow |X|!$ elements

But $\text{Sym} : \text{Fin} \rightarrow \text{Fin}$ acts by conjugation.

$\text{Ord} : \text{Fin} \rightarrow \text{Fin}$ acts by translation.

ex). $(132) \in \text{Sym}(\{1, 2, 3\})$
 $\downarrow$
 $(bca) \in \text{Sym}(\{b, a, c\})$

$(12322) \in \text{Sym}(\{1, 2, 3, 2, 2\})$
 $\downarrow$
 $(12322) \in \text{Sym}(\{1, 2, 3, 2, 2\})$

$\text{Ord}(\{1, 2, 3\})$
 $\downarrow$
 $\text{Ord}(\{1, 2, 3\})$

But this is not natural. (even objectwise iso.)
 $\Rightarrow$ Limit / colimit need not be iso.

Species := a functor $F: \mathbf{Fin}_{iso} \rightarrow \mathbf{FM}$.

labeled F -structures on $n := F(n)$.

unlabeled " := colimit of.

$$\mathbf{BS}_n \hookrightarrow \mathbf{Fin}_{iso} \xrightarrow{F} \mathbf{FM}$$

$\bullet \longmapsto [n] \longrightarrow F(n)$

objectwise iso $\Rightarrow$ labeled Sym structure
= labeled Ord "

But unlabeled Sym-structure: set of conjugacy
classes of permutations
of n -element.

(Since any bijection is permutation)

unlabeled Ord-structure: trivial.

since all linear orders on n are isomorphic.

Remark ① Even limit is unique, there might be distinct
limit cones.

② Choice of limit cone is usually not compatible.

ex) Freyd shows that $\mathbf{C}$ with pullback
has a "canonical pullback" with "horizontal
composite". But it seems not possible to
find such canonical one for "vertical composite".

③ Even when limit objects are equal, natural iso might
not be identity.

Lemma 3.6.6 For any triple $X, Y, Z \in \mathcal{C}$
 when $\mathcal{C}$ has binary product,

$\exists! \alpha: X \times (Y \times Z) \longrightarrow (X \times Y) \times Z$ natural
 Iso. commuting with projections to X, Y, Z .

pf) $X \times (Y \times Z) \xrightarrow{\pi_{1,X}} X$ The universal property of
 $X \times Y$ induces the unique
 $t: X \times (Y \times Z) \longrightarrow X \times Y$
 $\pi_{1,Y \times Z} \downarrow \quad \pi_{1,Y} \searrow$
 $Y \times Z \longrightarrow Y$
 $\pi_{1,Z} \downarrow \quad \text{, s.t.}$
 Z
 $\pi_{2,X} \swarrow \quad \pi_{2,Y} \searrow$
 $X \quad Y$
 $\pi_{1,X} = \pi_{2,X} \circ t$
 $\pi_{1,Y} \circ \pi_{1,Y \times Z} = \pi_{2,Y} \circ t$

(Now, t and $\pi_{1,Z} \circ \pi_{1,Y \times Z}$ induce)

$\alpha: X \times (Y \times Z) \longrightarrow (X \times Y) \times Z$
 s.t.
 $\pi_{3,X \times Y} \swarrow \quad \pi_{3,Z} \searrow$
 $X \times Y \quad Z$
 $t = \pi_{3,X \times Y} \circ \alpha$

$$\pi_{1,Z} \circ \pi_{1,Y \times Z} = \pi_{3,Z} \circ \alpha$$

Conversely, $\pi_{2,Y} \circ \pi_{3,X \times Y}$ and $\pi_{3,Z}$ induce

$s: (X \times Y) \times Z \longrightarrow Y \times Z$
 $\pi_{1,Y} \swarrow \quad \pi_{1,Z} \searrow$
 $Y \quad Z$
 s.t. $\pi_{2,Y} \circ \pi_{3,X \times Y} = \pi_{1,Y} \circ s$
 $\pi_{3,Z} = \pi_{1,Z} \circ s$

Also, s and $\pi_{2,X} \circ \pi_{3,X \times Y}$ induce

$\beta: (X \times Y) \times Z \longrightarrow X \times (Y \times Z)$ s.t. $\pi_{1,X} \circ \beta = \pi_{2,X} \circ \pi_{3,X \times Y}$
 $\pi_{1,Y \times Z} \circ \beta = s$

Then, $\pi_{3, X \times Y}(\alpha \circ \beta) = t \circ \beta : (X \times Y) \times Z \rightarrow (X \times Y)$

Since $\pi_{2, X}(t \circ \beta) = \pi_{1, X}(\beta) = \pi_{2, X} \circ \pi_{3, X \times Y}$

$$\begin{aligned}\pi_{2, Y}(t \circ \beta) &= \pi_{1, Y} \circ \pi_{1, Y \times Z} \circ \beta = \pi_{1, Y} \circ s \circ \pi_{3, X \times Y} \\ &= \pi_{2, Y} \circ \pi_{3, X \times Y}\end{aligned}$$

$\Rightarrow t \circ \beta = \pi_{3, X \times Y}$ by the universal property of $X \times Y$

$\Rightarrow \pi_{3, X \times Y}(\alpha \circ \beta) = \pi_{3, X \times Y}$

Also, $\pi_{3, Z}(\alpha \circ \beta) = \pi_{1, Z} \circ \pi_{1, Y \times Z} \circ \beta$
 $= \pi_{1, Z} \circ s = \pi_{3, Z}$

$\Rightarrow \alpha \circ \beta = 1_{(X \times Y) \times Z}$ by the univ. property of $(X \times Y) \times Z$

Conversely,

$$\pi_{1, X}(\beta \circ \alpha) = \pi_{2, X} \circ \pi_{3, X \times Y} \circ \alpha = \pi_{2, X} \circ t = \pi_{1, X}$$

$$\pi_{1, Y \times Z}(\beta \circ \alpha) = s \circ \alpha \quad \text{and}$$

$$\pi_{1, Y}(\beta \circ \alpha) = \pi_{2, Y} \circ \pi_{3, X \times Y} \circ \alpha = \pi_{2, Y} \circ t = \pi_{1, Y} \circ \pi_{1, Y \times Z}$$

$$\pi_{1, Z}(\beta \circ \alpha) = \pi_{3, Z} \circ \alpha = \pi_{1, Z} \circ \pi_{1, Y \times Z}$$

$\Rightarrow \pi_{1, Y \times Z} = s \circ \alpha$ by universal property

thus, $\pi_{1, Y \times Z} \circ (\beta \circ \alpha) = \pi_{1, Y \times Z}$

$\Rightarrow \beta \circ \alpha = 1_{X \times (Y \times Z)}$ by the universal property

$\Rightarrow \alpha, \beta$ are isomorphisms in $\mathcal{C}$.

This is natural! Iso. to see this, let $A = \{ - , - \}$

$$\text{Let } (X, \gamma) : C \rightarrow C^A$$

$$\begin{array}{ccc} Y \xrightarrow{\quad} (X, \gamma) : A \rightarrow C & & \text{Categories without nontrivial morphisms.} \\ \downarrow \xrightarrow{\quad} \downarrow & & \\ Z \xrightarrow{\quad} (X, \gamma) : A \rightarrow C & & \end{array}$$

Then, $Y \rightarrow Z$ induces natural transformation between (X, γ) and (Y, γ) since A has no nontrivial morphisms.

Then, $(X, \lim_A(Y, -)) \xrightarrow{\alpha} (\lim_A(X, Y), -)$
by our construction of α, β .

$\therefore$ Natural! □

Ex. 3.6.17. $\text{sk}(\text{Set})$ a skeletal category of Set .
Then actually for each cardinal k , there is only one object in $\text{sk}(\text{Set})$.

$\text{sk}(\text{Set})$ is complete; (just use isomorphisms for limits in Set)

$\Rightarrow$ It has a product.

Let C : cble infinite. $\Rightarrow CXC = C$.

as $\pi_1, \pi_2 : C \rightarrow C$ are epi.

If $CX(CXC) \xrightarrow{i} (CXC) \times C$ is identity, then.

$$\begin{array}{ccc} CX(CXC) & = & (CXC) \times C \\ \downarrow f(x \times h) & & \downarrow (fx) \times h \\ CX(C \times C) & = & (CXC) \times C \end{array} \quad \text{for any } f, g, h : C \rightarrow C.$$

Now, we set below commutative diagram

$$\begin{array}{ccc}
 C & \xrightarrow{g \times h} & C \\
 \uparrow \pi_2 & & \uparrow \pi_1 \\
 C \times C & \xrightarrow[\overline{(f \times g) \times h}]{f \times (g \times h)} & C \\
 \downarrow \pi_1 & & \downarrow \pi_1 \\
 C & \xrightarrow[\overline{f \times g}]{f} & C
 \end{array}$$

$\Rightarrow$ Since π_1, π_2 are epi,
 $f = f \times g, \quad g \times h = h$

$\Rightarrow f, g : C \Rightarrow C$ equal.

Since

$$\begin{array}{ccc}
 C \times C & \xrightarrow{f \times f} & C \\
 \parallel & & \parallel \\
 C & \xrightarrow{f \times g} & C
 \end{array}$$

$\Rightarrow f = g, f = f$

But this is absurd! since $\text{sk}(\text{Set})(C, C) = \text{Set}(C, C)$

Ex 3.6.1) Special case of prop 3.6.1

((ii) shown in the above.

3.7. Size Matters

Lemma 3.7.1. C : Category. $I_C : C \rightarrow C$ admits a limit $\iff C$ has initial object.

Pf) $(\lambda_c : I \rightarrow C)_{c \in C}$ a limit cone.

We can say that I is "weakly initial object"

in the sense that $\forall c \in C, \exists \lambda_c : I \rightarrow c$.

To see λ_c is only morphism, let $f : I \rightarrow c$.

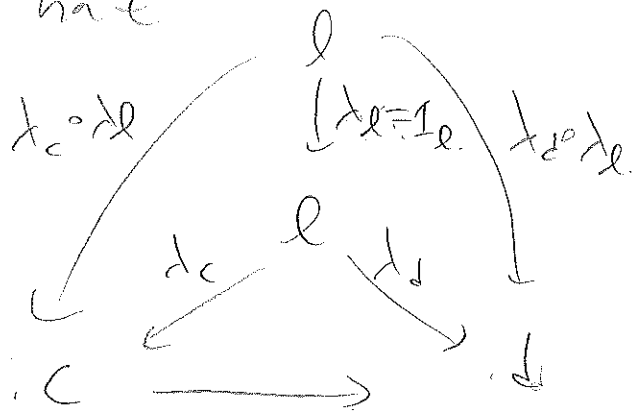
Then, $I \Rightarrow f \lambda_I = \lambda_c$ so WTS λ_I is identity.

$$\begin{array}{ccc}
 & I & \\
 \lambda_I \swarrow & & \searrow \lambda_c \\
 I & \xrightarrow{f} & c
 \end{array}$$

To see this, we have $\lambda_l \circ \lambda_c = \lambda_c \circ \lambda_l = \lambda_c$

Thus, $\lambda \circ \lambda_l : (\lambda_c \circ \lambda_l : l \rightarrow c)_{c \in C}$

is another cone. By the universal property of λ , we have



$\lambda_l = I_l$ since both λ_l and I_l make the left diagram commutes.

$$\Rightarrow f = \lambda_c$$

Conversely, if C has initial element I , then, $\lambda_c : l \rightarrow c$ form a cone over I_c , and it is limit cone. Let $\eta : I \rightarrow I_c$.

Then, $\forall f : d \rightarrow c \in C$, $\eta_d \circ \eta_c = f$. This implies $\Rightarrow \eta_l : I \rightarrow I_c$. $\Rightarrow \eta_l : I \rightarrow I_c$ exists.

So, λ is a limit cone. □

Def 3.2.7. Cardinality of a small category.
the set of its morphisms

a k -small category = a category whose cardinality is less than k .

k -small diagram: a diagram whose indexing category is k -small.

Prop 3.17.3 (Freyd). Any k -small category admits all k -small limits or k -small colimits is a preorder.

pf). Let C be a k -small category with $|C| = \lambda$.
 $f, g: B \rightrightarrows A \in \text{Mor } C$ with $f \neq g$.

By Ex 3.4.4, $B \rightarrow A^k$ is determined by
 λ indexed morphisms $B \rightarrow A$.

$\Rightarrow$ The collection of all $B \rightarrow A^k$ whose component is either f or g has cardinality 2^λ .

$$|C(B, A^k)| \geq 2^\lambda > \lambda = |\text{Mor } C|.$$

$\uparrow$ from Cantor's diagonalization argument.

But this contradicts the fact that $C(B, A^k) \subseteq \text{Mor } C$.

$\Rightarrow$ every hom set in C is singleton.

$\Rightarrow C$ is preorder.

Do the same thing with cotensor for k -complete case.

2nd) Prop 3.5.9 shows that there is a preorder
 $\mathcal{P}$ to have all limit/colimit of unrestricted size

Since $\mathcal{P}$ poset has this property

Ex 3.11.1) done in the above. $(\mathcal{P})^{\mathcal{P}} = \text{Complete lattice}$.

8. Interaction between limits and colimits.

Let $F: I \times J \rightarrow \mathcal{C}$. By Ex 1.7. VII), F can be regarded as $F: I \rightarrow \mathcal{C}^J$ or $F: J \rightarrow \mathcal{C}^I$.

If $\forall i \in I, \exists \lim_{j \in J} F(i, j)$, then prop 3.3.9

gives $\lim_{j \in J} F(-, j): I \rightarrow \mathcal{C}$. Moreover,

Thm 3.8.1. If $\lim_{i \in I} \lim_{j \in J} F(i, j)$, $\lim_{j \in J} \lim_{i \in I} F(i, j)$ exist.

then they are iso and define the $\lim_{I \times J} F$.

Pf) From Yoneda Lemma, it suff to prove.

$$\mathcal{C}(X, \lim_{i \in I} \lim_{j \in J} F(i, j)) \cong \mathcal{C}(X, \lim_{(i, j) \in I \times J} F(i, j)) \cong \mathcal{C}(Y, \lim_{j \in J} \lim_{i \in I} F(i, j))$$

By Thm 3.4.6(i), covariant representable functor preserves limit.

$$\Rightarrow \mathcal{C}(X, \lim_I \lim_J F(i, j)) \cong \lim_I \mathcal{C}(X, \lim_J F(i, j)) \cong \lim_I \lim_J \mathcal{C}(X, F(i, j))$$

$$\mathcal{C}(X, \lim_{(i, j)} F(i, j)) \cong \lim_{(i, j)} \mathcal{C}(X, F(i, j))$$

$$\mathcal{C}(X, \lim_J \lim_I F(i, j)) \cong \lim_J \lim_I \mathcal{C}(X, F(i, j))$$

thus, it suffices to show that $\forall H: I \times J \rightarrow \text{Set}$ a set valued functor, the limits.

$$\lim_I \lim_J H \cong \lim_{I \times J} H \cong \lim_J \lim_I H \text{ is o.}$$

Isms is o of category $I \times J \cong J \times I$,
 only left iso is needed to show.

By 3.2.3, let $\lambda: f \Rightarrow \lim_{i \in I} H(i, j) \in \lim_I \lim_J H$.

Then,

$$\begin{array}{ccc} & 1 & \lambda_{i'} \\ \lambda_i \swarrow & \circlearrowleft & \searrow \\ \lim_J H(i, j) & \longrightarrow & \lim_J H(i', j) \end{array} \quad \forall i \rightarrow i' \in I$$

By prop 3.6.1, since we already chosen limit cone
 at $\lim_J H(i, j)$, $\forall i \in I$, $i \rightarrow i'$ also a natural
 transformation of $H(i, -) \rightarrow H(i', -)$.

determines $\lim_J H(i, j) \rightarrow \lim_J H(i', j)$

Denote $(\eta_i): \lim_J H(i, j) \Rightarrow H(i, -)$.

Then, $\forall i \in I, \phi_{i, j}: 1 \xrightarrow{\lambda_i} \lim_J H(i, -) \xrightarrow{\eta_{i, j}} H(i, j)$

is a cone over $F: I \times J \rightarrow C$. To see this

notes that $\forall f: i \rightarrow i',$

$$\begin{array}{ccc} & 1 & \\ \lambda_i \swarrow & \circlearrowleft & \searrow \lambda_{i'} \\ \lim_J H(i, j) & \longrightarrow & \lim_J H(i', j) \\ \eta_{i, j} \downarrow & \circlearrowleft & \downarrow \eta_{i', j} \\ H(i, j) & \xrightarrow{H(f, -)} & H(i', j) \end{array}$$

left rectangular
 commutes by proof of
 prop 3.6.1.

(Actually, $\lim_J H(i, j) \rightarrow \lim_J H(i', j)$

Thus, over is given by the universal property of

$\Rightarrow \phi_{i, j} \xrightarrow{1} \phi_{i', j} \xrightarrow{\phi_{i', j'}} H(i', j')$ (cone $H(i', -) \Rightarrow H(f, -) \circ \eta_{i, j} = k \circ \eta_{i', j}$)

$\Rightarrow H(i, j) \rightarrow H(i', j) \rightarrow H(i', j')$ (Last triangle is from η)

This shows that $\phi: I \Rightarrow F \in \lim_{I \times J} F$.

Conversely, if $\phi \in \lim_{I \times J} F$, then, for fixed i ,

$\phi_i: I \Rightarrow H(i, -)$, thus, $\phi_i \in \lim_J H(i, J)$

So we get $\lambda_i: I \rightarrow \lim_J (H(i, J))$
 $I \mapsto \phi_i$

Hence, this induces $\lambda: I \Rightarrow \lim_J (H(i, -))$ (element, colimit)

This shows that $\lim_I \lim_J H \cong \lim_{I \times J} H$.

By duality, colim case also holds. $\square$

Exk). This shows that $\exists$ natural iso

$$\begin{array}{ccc} \lim_I \lim_J & \xrightarrow{\cong} & \lim_I \lim_J \\ \downarrow & & \downarrow \\ C^I & \xrightarrow{\lim_J} & C^I \end{array} \quad \begin{array}{ccc} \lim_I \lim_J & \xrightarrow{\cong} & \lim_I \lim_J \\ \downarrow & & \downarrow \\ C^J & \xrightarrow{\lim_I} & C^J \end{array}$$

for any choice of $\lim_I, \lim_J$ or $\lim_I, \lim_J$.

Lemma 3.8.3. $F: I \times J \rightarrow C$ bifunctor s.t.

$\lim F, \text{colim} F$ exist.

$\Rightarrow \exists k: \lim_{i \in I} \lim_{j \in J} F(i, j) \rightarrow \lim_{j \in J} \lim_{i \in I} F(i, j)$

a canonical map exists.

pt) By the univ. prop. of colimit, if k is canonical, then $\exists$

$$k_i : \left(\lim_{j \in J} F(i, j) \longrightarrow \lim_{\substack{j \in J \\ i' \in I}} \text{colim}_{i' \in I} F(i, j) \right)_{i \in I}$$

that defines a cone.

By the univ. prop. of limit, k_i is defined by

$$(k_i, : \lim_{j \in J} F(i, j) \longrightarrow \text{colim}_{i' \in I} F(i, j))_{j \in J}$$

that defines a cone over $\text{colim}_{i' \in I} F(i, -)$.

So, define $k_{i,j} : \lim_{j' \in J} F(i, j') \xrightarrow{\pi_{i,j,j'}} F(i, j) \xrightarrow{\bar{c}_{i,j,j}} \text{colim}_{i' \in I} F(i, j)$

Where $\pi_{i,j}$, $\bar{c}_{i,j}$ are limit and colimit cone over $F(i, -)$ and $F(-, j)$ respectively.

Then, $k_{i,j}$ is a cone over $\text{colim}_{i' \in I} F(i', j)$.

thus we have k_i ; thus k by above argument. $\square$

Corollary 3.8.4 $f : X \times Y \rightarrow \mathbb{R}$ a function

$$\Rightarrow \sup_{x \in X} \inf_{y \in Y} f(x, y) \leq \inf_{y \in Y} \sup_{x \in X} f(x, y)$$

pt) In a poset Category, $\text{colim}_X = \sup_X$, $\lim_Y = \inf_Y$. $\square$

Ex: 3.8.5. Direct proof of the Gallai, in RMI analysis is actually categorical.

Pf) $\sup_x \inf_y f$ is $\text{bdl}_{\text{by } M}^{\text{above}} \Leftarrow \inf_Y f(x, -) \text{ bdl above by } M. \forall x \in X$

$\inf_Y \sup_x f$ is $\text{bdl}_{\text{below } N}^{\text{below}} \Leftarrow \sup_x f(-, y) \text{ bdl below by } N. \forall y \in Y$

Then we have $\inf_Y \sup_x f(x, -) \leq f(x, y) \leq \sup_x \inf_Y f(-, y)$

Ex: 3.8.6 Let $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ with $-\infty \leq x \leq \infty$ poset.

$\Rightarrow \overline{\mathbb{R}}$ is complete and cocomplete as a category for poset.

any function gives diagram, and its limit = infimum
co-lim = supremum.

Now let $x: \mathbb{N} \rightarrow \mathbb{R}$ is a sequence.

Then, we have: $\mathbb{N} \times \mathbb{N} \xrightarrow{+} \mathbb{N} \xrightarrow{x} \mathbb{R}$
(m, n) $\mapsto$ m+n $\mapsto$ x_{m+n} .

So. $\liminf x_n = \sup_{n \geq 0} \inf_{m \geq n} x_m = \sup_{n \geq 0} \inf_{m \geq n} x_{n+m} = \sup_n \inf_m x_{n+m}$

By Lem. 3.8.3 $\Rightarrow \liminf x_n \leq \limsup x_n$.

In Set and some related category there exists an invariant K .

Def: 3.8.7. J is filtered if there is a cone.

$J' \rightarrow J$ $\forall J'$ finite index category

Ex) $W, \circ \rightarrow \circ$: filtered, $\circ \leftarrow \circ \rightarrow \circ$, $\circ \rightrightarrows \circ$, discrete : not filtered category

Thm 3.8.9. Filtered colimit commute with finite limit in Set

Observation: $F: J \rightarrow \text{Set}$, J is small filtered category. Then, from diagram in p. 9/7.

$$\text{colim}_{j \in J} F_j = \coprod_{j \in J} F_j / \sim \quad \text{where} \quad x \sim y \text{ for } x \in F_j, y \in F_k$$

$$\Leftrightarrow \exists f: J \rightarrow t \quad \text{s.t. } Ff(j) = Ff(k) \\ g: k \rightarrow t$$

Thus, let A be a finite subset of $\coprod_{j \in J} F_j$ s.t.

$A/\sim := \{\lambda_i\}_{i \in I}$ where I is finite.

Then, $\exists t \in J$ s.t. $A' \subseteq F_t$ and $A'/\sim = A/\sim$.

*): A is a finite subset of $\coprod_{j \in J} F_j$, so

$\{j \in J: F_j \cap A \neq \emptyset\}$ is a finite set. Let $\bar{B}$ be

a full subcategory spanned by $\bar{B}$. By definition,

$\bar{B} \hookrightarrow J$ has a cone with vertex, say t .

thus, by sending $F_j \cap A \rightarrow F_t$ using core map, we get such A' .

(Proof of Thm) WTS $k: \lim_{\substack{\longrightarrow \\ J}} \lim_{\substack{\longrightarrow \\ I}} F \rightarrow \lim_{\substack{\longrightarrow \\ I}} \lim_{\substack{\longrightarrow \\ J}} F$

is a bijection when I is finite, J is small filtered category

By def 3.2.3, as a set,

$$\lim_{\substack{\longrightarrow \\ I}} \lim_{\substack{\longrightarrow \\ J}} F = \{\lambda: I \Rightarrow \lim_{\substack{\longrightarrow \\ J}} F(-, J)\}$$

Since I is finite, $\lambda = \{\lambda_i\}_{i \in I}$ is finite.

Also, only finitely many compatibility conditions λ satisfy.

Then, each $\lambda_{\bar{i}} : 1 \rightarrow \text{colim}_{\mathcal{J}} F(\bar{i}, \mathcal{J})$ ^{can be viewed} as a element of $\text{colim}_{\mathcal{J}} F(\bar{i}, \mathcal{J})$.

[Now fix $\bar{i} \in \mathcal{I}$, $t \in \mathcal{J}$. Choose $\lambda'_{\bar{i}} \in F(\bar{i}, t)$ s.t. $[\lambda'_{\bar{i}}] = \lambda_{\bar{i}}$ as equivalence class. Then, for any $f: \bar{i} \rightarrow \bar{i}'$,

Since $\text{colim}_{\mathcal{J}} F(f, \mathcal{J})(\lambda_{\bar{i}}) = \lambda_{\bar{i}'}$;

$F(f, \mathcal{J})(\lambda'_{\bar{i}}) \in \lambda_{\bar{i}'}$ as an equivalence class.

Thus we can choose $\lambda'_{\bar{i}} \in F(\bar{i}, t)$ s.t.

$$\begin{array}{ccc} \lambda'_{\bar{i}} & \xrightarrow{1} & \lambda'_{\bar{i}'} \\ \searrow \eta & & \searrow \eta' \\ F(\bar{i}, t) & \xrightarrow{F(f, t)} & F(\bar{i}', t) \end{array}$$

So if $\mathcal{I}$ is connected, then we have a cone $\{\lambda'_{\bar{i}} : 1 \Rightarrow F(\cdot, t)\}$.

Otherwise, if $\bar{i}''$ is not connected with $\bar{i}$ in $\mathcal{I}$, then still $\lambda'_{\bar{i}''} \in F(\bar{i}'', t')$ for some t' s.t.

$[\lambda'_{\bar{i}''}] = \lambda_{\bar{i}''}$. Since $\mathcal{J}$ is filtered, $\exists$ a cone $\begin{array}{ccc} t & \xleftrightarrow{\quad} & t' \\ \eta_1 \searrow & & \swarrow \eta_2 \\ & t'' & \end{array}$

Under t and t' , say t'' , i.e.

$\Rightarrow$ We may regard both $\lambda'_{\bar{i}}$ and its related $\lambda'_{\bar{i}'}$ and $\lambda'_{\bar{i}''}$ as an element of $F(\bar{i}, t'')$, $F(\bar{i}', t'')$ and $F(\bar{i}'', t'')$ respectively via the map η_1, η_2 .

Thus, we can say that $\exists t \in \mathcal{J}$ s.t.

$\lambda'_{\bar{i}} : 1 \Rightarrow F(\bar{i}, t)$ and $[\lambda'_{\bar{i}}] = \lambda_{\bar{i}} \quad \forall \bar{i} \in \mathcal{I}$.

Now this λ' is inside of $\lim_{i \in I} F(-, t)$.

By p. 97 diagram, $\text{den } k$ is

$$\text{colim}_{j \in J} \lim_I F(\bar{i}, j) = \frac{(\lim_{j \in J} \lim_I F(\bar{i}, j))}{\sim}.$$

thus, $[\lambda']$ is in $\text{den } k$ and by

$$\text{construction, } k \text{ is induced by } K_{j, \bar{i}}: \lim_{i \in I} F(\bar{i}, j) \rightarrow F(\bar{i}, j)$$

$$\downarrow$$

$$\text{colim}_{j' \in J} F(\bar{i}, j')$$

and by setting $j = t$, we get $K_{j, \bar{i}}(\lambda') = [\lambda']$ in $\text{cod } k$.

$\Rightarrow k$ is surjective.

Now, to see k is injective,

$$\text{let } \left(1 \xrightarrow{\alpha_{\bar{i}}} F(\bar{i}, j)\right)_{\bar{i} \in I}, \left(1 \xrightarrow{\beta_{\bar{i}}} F(\bar{i}, k)\right)_{\bar{i} \in I}$$

$$\text{a cone } \alpha: I \rightrightarrows \lim_I F(-, j), \beta = 1 \Rightarrow \lim_I F(-, k)$$

for some $j, k \in \text{Ob } J$.

If $k([\alpha]) = k([\beta])$, then for each $\bar{i} \in I$,

$$\exists t_{\bar{i}} \in J \text{ s.t. } \forall \bar{i}, \alpha_{\bar{i}} \text{ and } \beta_{\bar{i}} \text{ have a common cone}$$

in $F(\bar{i}, t_{\bar{i}})$. Since I is finite and J is filtered,

by the same argument used for λ , $\{t_{\bar{i}}\}$ is finite.

$$\exists t \in J \text{ s.t. } \forall \bar{i}, \alpha_{\bar{i}} \text{ and } \beta_{\bar{i}} \text{ have}$$

a common cone in $F(\bar{i}, t)$. Then, as an element of $F(\bar{i}, j)$ or $F(\bar{i}, k)$

~~$\alpha_{\bar{A}}$~~ Image of $\alpha_{\bar{A}}$ in $F(\bar{A}, t) =$ Image of $\beta_{\bar{A}}$ in $F(\bar{A}, t)$

$$\Rightarrow [\alpha] = [\beta] \text{ in } \text{dom } k$$

$\therefore k$ is injective.

Rank) Proof is not categorical, since choosing $\lambda_{\bar{A}}$ is arbitrary. In large class of category (Ch 5) these are created in Set.

Ex. 3.8.i) Let s be a coequalizer map.

$$\begin{array}{ccc} A & \xrightleftharpoons[e]{1} A & \xrightarrow{s} A/e \\ & \searrow e & \downarrow r \\ & & A \end{array}$$

Then the universal property (left diagram) gives unique r

$$\text{s.t. } rs = e$$

Thus we can set below diagram since

$$\begin{array}{ccc} A & \xrightleftharpoons[e]{1} A & \xrightarrow{s} A/e \\ & \searrow s & \downarrow sr \\ & & A/e \end{array}$$

$$sr s = se = s$$

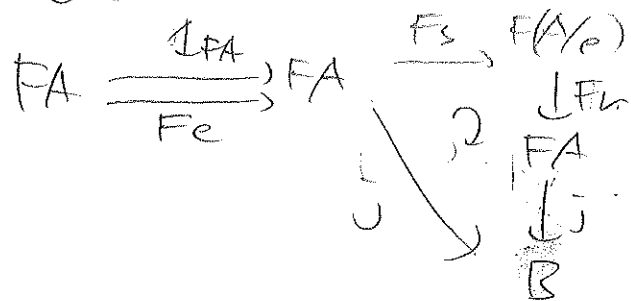
The universal property gives $sr = 1_{A/e}$.

Thus, A/e is a retract of A . Conversely, any retract of the form $B \xrightarrow{r} A \xrightarrow{s} B$ gives an idempotent sr which is split by B .

To see any functor preserves idempotent, if F is a functor, then $Fe \cdot Fe = F(e \cdot e) = Fe$.

Now suppose that $j: FA \rightarrow B \in \text{cod } F$.
 $\text{s.t. } \bar{j} = j \circ Fe$

Then we have a diagram.



We claim that

left triangle is commutative

since

$$j \circ F_r \circ F_s = j \circ F_e = \bar{j}.$$

Moreover, since F_s is still split epi,

(since functor preserves split epi)

thus epi, therefore $\exists b: F(A/e) \rightarrow B$

s.t. $b \circ F_s = j$, then $(j \circ F_r) \circ F_s = j$

implies $b = j \circ F_r$.

Thus, $F(A/e)$ is a coequalizer in $\text{cod } F$.

Now, by duality, idempotent as a limit also preserved by any functor.

Thus, by Thm 3.8.1, they commute with

both limits and colimits of any shape.

Ex. 3.8.ii) It suff to show that $\text{colim}_H (\lim_G X)$

$$= \lim_G (\text{colim}_H X).$$

And from previous lecture, we know that

LHS = The set of H -orbits on X^G .

RHS = The set of // consists of G -fixed elements.

Suppose $\exists$ $BG \times BH \longrightarrow X$ exists.

$$(g) \quad \begin{array}{ccc} & & x \\ (g, h) \downarrow & \longrightarrow & \downarrow \\ & & x \end{array}$$

I_G

(at least $X = G \times H$ exists in Set)

Then, by identifying $gx := (g, I_H)x$
 $hx := (I_G, h)x$,

X is G -set and H -set.

Moreover, if x, y are in the same H -orbit,
 then $x = (I, h)y$ for some $h \in H$, thus

$$\forall g \in G, \quad gx = (g, I_H)x = (g, I_H)(I, h)y = (g, h)x = (I, h)(g, I)x.$$

$\Rightarrow gx$ and gy are in the same orbit.

Thus, $\exists G$ action on the set of H -orbits.

Let A be an H -orbit. s.t. $gA = A \quad \forall g$.

Then, ~~for any~~ $\forall x \in A, \quad \forall g \in G, \quad \exists h \in H$ s.t.

$$gx \in A \Rightarrow gx = hx \text{ for some } h.$$

But this implies $g^n x = h^n x \quad \forall n \Rightarrow \exists n > 0$ s.t.

$$g^n x = x \quad \text{and} \quad n \mid |g| \quad \text{and} \quad n \mid |h|.$$

$\Rightarrow n = 1$ since $|g|$ and $|h|$ are coprime.

$\therefore$ all elements in A are in X^G .

Conversely, if we take an H -orbit on X^G , then these are orbits of X consisting of X^G elements. done